

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\HTS EFF. ,STEAM GEN103-C1-NVS.pvdb

Date of Analysis : Apr 10,2026 11:13am

PV Elite 28, January 2026



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FileName : HTS EFF

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Vessel Design Summary:

Step: 39 11:13am Apr 10,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2520.000 mm ID
Vessel Design Length, Tangent to Tangent 8.74 m
Specified Datum Line Distance 0.00 m
Shell Side Design Temperature 343 °C
Channel Side Design Temperature 475 °C
Shell Side Design Pressure 14.281 N/mm²
Channel Side Design Pressure 4.950 N/mm²
Shop Shell Side Test Pressure 19.748 N/mm²
Field Shell Side Test Pressure 18.565 N/mm²
Shop Channel Side Test Pressure 10.001 N/mm²
Field Channel Side Test Pressure 6.661 N/mm²
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 22	2	...	K21590	
Shell	SA-387 11	2	...	K11789	
Head	SA-387 22	2	...	K21590	
Head	SA-387 11	2	...	K11789	
Nozzle	SA-336 F22	3	...	K21590	+IT
Nozzle	SA-336 F11	3	...	K11572	+IT
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-336 F11	2	...	K11572	
Tubes	SA-213 T22	...	...	K21590	
Tubesheet	SA-336 F22	3	...	K21590	+IT
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (N/mm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head	4.950	0.00	5.490	3.0000	N/A	No
Channel shell	4.951	0.00	5.790	3.0000	N/A	No
Shell	14.306	0.10	14.800	3.0000	N/A	No
Shell head	14.281	0.10	15.500	3.0000	N/A	No

Liquid Level: 2.52 m Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev m	Element Length m	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Vessel Design Summary:

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Sphere	0.0	0.0	40.0	33.0	30.1	...	1.00	1.00
Cylinder	2.5	2.5	68.0	68.0	58.3	...	1.00	1.00
Cylinder	8.7	5.7	138.0	138.0	132.6	...	1.00	1.00
Sphere	8.7	0.0	85.0	70.0	64.5	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	33	No Calc	...	4.98
20	Tsht	68	No Calc	...	9.1
Tsht	40	68	No Calc	...	No Calc
40	50	138	14.7003	0.10131	15.2
50	60	70	6.51545	0.10131	13

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2.91997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	6.12003	25400	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Tube inlet	WRC-297	0.903	Shell	Passed
N2- Tube outle	WRC-297	0.903	Shell	Passed
Shell inlet-4	WRC-297	0.985	Shell	Passed
Shell outlet-3	WRC-297	0.983	Shell	Passed
Shell outlet-4	WRC-297	0.983	Shell	Passed
Shell inlet-1	WRC-297	0.985	Shell	Passed
Shell outlet-1	WRC-297	0.983	Shell	Passed
Shell outlet-2	WRC-297	0.983	Shell	Passed
shell inlet-2	WRC-297	0.985	Shell	Passed
Shell inlet-3	WRC-297	0.985	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	230.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1800.000 mm
Wear Pad Width	300.000 mm
Wear Pad Thickness	30.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1150.000 mm

Baseplate Length	2600.000 mm
Baseplate Thickness	35.000 mm
Baseplate Width	254.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	15.000 mm
Web Thickness	15.000 mm



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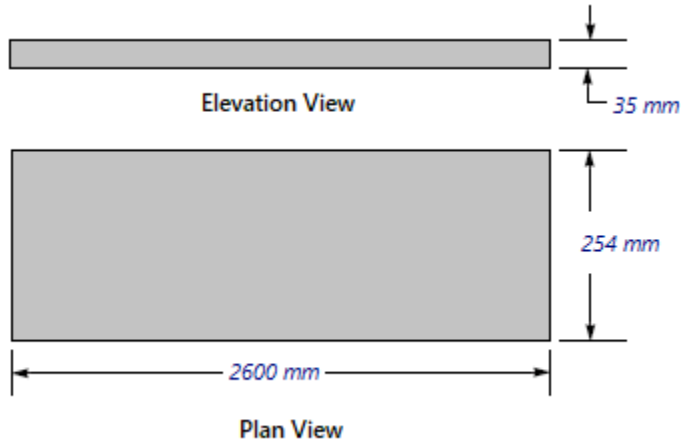
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Vessel Design Summary:

Step: 39 11:13am Apr 10,2026

Height of Center Web	306.000 mm
Number of Bolts in Baseplate	8

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	1498766.00 N
Maximum Transverse Saddle Shear Load	48770.65 N
Maximum Longitudinal Saddle Shear Load	97541.30 N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	1780833.75 N
Maximum Transverse Saddle Shear Load	3443.32 N
Maximum Longitudinal Saddle Shear Load	3725.15 N

Weights:

Fabricated - Bare w/o removable internals	150049.6 kgm
Shop Test - Fabricated + water (full)	197435.4 kgm
Shipping - Fab. + rem. intls.+ shipping app.	150049.6 kgm
Erected - Fab. + rem. intls.+ insul. (etc)	154112.5 kgm
Empty - Fab. + intls. + details + wghts.	157643.8 kgm
Operating - Empty + operating liquid (no ca)	170837.5 kgm
Field Test - Empty weight + test medium (full)	200966.7 kgm

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Nozzle Summary:

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Nozzle Calculation Summary (N/mm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Manhole	5.49	...	...	OK	17.32	OK	Passed
N1- Tube inlet	5.79	...	...	OK	24.41	OK	Passed
N2- Tube outlet	5.47	...	...	OK	26.26	OK	Passed
Shell inlet-4	14.8	OK	...	OK	11.40	OK	Passed
Shell outlet-3	14.8	OK	...	OK	14.07	OK	Passed
Shell outlet-4	14.8	OK	...	OK	14.07	OK	Passed
Shell inlet-1	14.8	OK	...	OK	11.40	OK	Passed
Shell outlet-1	14.8	OK	...	OK	14.07	OK	Passed
Shell outlet-2	14.8	OK	...	OK	14.07	OK	Passed
BD1	14.8	...	...	OK	8.26	OK	No Calc[*]
BD2	14.8	...	...	OK	8.26	OK	No Calc[*]
shell inlet-2	14.8	OK	...	OK	11.40	OK	Passed
Shell inlet-3	14.8	OK	...	OK	11.40	OK	Passed
BD3	14.8	...	...	OK	8.26	OK	No Calc[*]
BD5	14.8	...	...	OK	8.26	OK	No Calc[*]
BD4	14.8	...	...	OK	8.26	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 14.8 Nozzle : Shell inlet-4 [Shellside]
 Minimum MAWP Nozzles : 5.47 Nozzle : N2- Tube outlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Manhole	0.000	0.000	1231.200
20	N1- Tube inlet	1100.000	0.000	1840.800
20	N2- Tube outlet	1100.000	180.000	1840.800
40	Shell inlet-4	6740.000	180.000	458.920
40	Shell outlet-3	5740.000	0.000	519.225
40	Shell outlet-4	6740.000	0.000	519.225
40	Shell inlet-1	3694.005	180.000	458.920
40	Shell outlet-1	3705.833	0.000	519.225
40	Shell outlet-2	4732.080	0.000	519.225
40	BD1	3240.000	180.000	150.000
40	BD2	4585.070	180.000	150.000
40	shell inlet-2	4978.561	180.000	458.920
40	Shell inlet-3	5714.621	180.000	458.920
40	BD3	5370.980	180.000	150.000
40	BD5	6040.000	180.000	150.000
40	BD4	7163.050	180.000	150.000

The nozzle spacing is computed by the following:

$$= \sqrt{I_l^2 + I_c^2} \text{ where}$$

I_l - Arc length along the inside vessel surface in the long. direction.

I_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.



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Nozzle Summary:

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No interference violations have been detected!

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Nozzle Schedule:

Step: 36 11:13am Apr 10,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
BD1	2.000 in	Actual	LWN	4.120	26.924	...	...	329.59	...
BD2	2.000 in	Actual	LWN	4.120	26.924	...	...	329.59	...
BD3	2.000 in	Actual	LWN	4.120	26.924	...	...	329.59	...
BD5	2.000 in	Actual	LWN	4.120	26.924	...	...	329.59	...
BD4	2.000 in	Actual	LWN	4.120	26.924	...	...	329.59	...
Shell inlet-4	6.000 in	Actual	None	7.438	18.260	...	...	399.01	...
Shell inlet-1	6.000 in	Actual	None	7.438	18.260	...	...	399.01	...
shell inlet-2	6.000 in	Actual	None	7.438	18.260	...	...	399.01	...
Shell inlet-3	6.000 in	Actual	None	7.438	18.260	...	...	399.01	...
Shell outlet-3	8.000 in	Actual	None	9.812	23.012	...	...	404.23	...
Shell outlet-4	8.000 in	Actual	None	9.812	23.012	...	...	404.23	...
Shell outlet-1	8.000 in	Actual	None	9.812	23.012	...	...	404.23	...
Shell outlet-2	8.000 in	Actual	None	9.812	23.012	...	...	404.23	...
Manhole	24.000 in	Actual	WNF	25.938	24.610	...	...	514.84	900
N1- Tube inlet	36.000 in	Actual	None	38.750	34.925	...	...	871.90	...
N2- Tube outlet	36.000 in	Actual	None	38.750	34.925	...	...	871.90	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
BD1	SA-336 F11	138.000	10.000	...	...	...
BD2	SA-336 F11	138.000	10.000	...	...	...
BD3	SA-336 F11	138.000	10.000	...	...	...
BD5	SA-336 F11	138.000	10.000	...	...	...
BD4	SA-336 F11	138.000	10.000	...	...	...
Shell inlet	SA-336 F11	138.000	10.000	...	...	...
Shell inlet	SA-336 F11	138.000	10.000	...	...	...
shell inlet	SA-336 F11	138.000	10.000	...	...	...
Shell inlet	SA-336 F11	138.000	10.000	...	...	...
Shell outle	SA-336 F11	138.000	10.000	...	...	...
Shell outle	SA-336 F11	138.000	10.000	...	...	...
Shell outle	SA-336 F11	138.000	10.000	...	...	...
Shell outle	SA-336 F11	138.000	10.000	...	...	...
Manhole	SA-336 F22	59.000	10.000	...	...	...
N1- Tube in	SA-336 F22	68.000	15.000	...	...	...
N2- Tube ou	SA-336 F11	68.000	15.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum m	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
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Nozzle Schedule:

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BD1	3.240	180.0	190.50	0.00	Shell
BD2	4.585	180.0	190.50	0.00	Shell
BD3	5.371	180.0	190.50	0.00	Shell
BD5	6.040	180.0	190.50	0.00	Shell
BD4	7.163	180.0	190.50	0.00	Shell
Shell inlet-4	6.740	180.0	250.00	0.00	Shell
Shell inlet-1	3.694	180.0	250.00	0.00	Shell
shell inlet-2	4.979	180.0	250.00	0.00	Shell
Shell inlet-3	5.715	180.0	250.00	0.00	Shell
Shell outlet-3	5.740	0.0	250.00	0.00	Shell
Shell outlet-4	6.740	0.0	250.00	0.00	Shell
Shell outlet-1	3.706	0.0	250.00	0.00	Shell
Shell outlet-2	4.732	0.0	250.00	0.00	Shell
Manhole	...	0.0	396.90	0.00	Channel head
N1- Tube inlet	1.100	0.0	600.00	0.00	Channel shell
N2- Tube outlet	1.100	180.0	600.00	0.00	Channel shell

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Bill of Material:

Step: 35 11:13am Apr 10,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 40.0mm THK X 2520.0mm ID X 0.0m	SA-387 22
1	SPHERICAL HEAD: 85.0mm THK X 2520.0mm ID X 0.0m	SA-387 11
9	BAFFLES: 2509.0mm DIA X 13.0mm THK	SA-516 70
20	TIE RODS: 16mm DIA X 4800mm	SA-193 B7M
20	SPACERS: 24.0mm DIA X 3.0mm THK X 4800.0mm	SA-179
1	TUBESHEET: 2796.0mm X 540.0mm THK	SA-336 F22
4952	TUBES: 4800.0mm X 25.4mm DIA X 2.1mm THK	SA-213 T22
2	SADDLE: 230.0mm X 120 DEG	SA-516 70
1	INSULATION: 5700mm X 150mm THK	
1	INSULATION: 680mm X 150mm THK	
1	CLASS 900 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
2	CYLINDRICAL SEGMENT 68.0mm THK X 3144.7mm X 2.5m	SA-387 22
1	CYLINDRICAL SEGMENT 68.0mm THK X 1627.3mm X 2.5m	SA-387 22
2	CYLINDRICAL SEGMENT 138.0mm THK X 3048.0mm X 5.7m	SA-387 11
1	CYLINDRICAL SEGMENT 138.0mm THK X 1820.8mm X 5.7m	SA-387 11
1	NAMEPLATE	...

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MDMT Summary:

Step: 38 11:13am Apr 10,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell head [9]	A	43	40		0.947	70.000	1.00	Yes
Left Saddle [24]	A	7						No
Right side sa[24]	A	7						No
Shell inlet-4 [1]	A	47	-104		0.097	90.000	1.00	No
Shell outlet-3[1]	A	48	-104		0.114	100.000	1.00	No
Shell outlet-4[1]	A	48	-104		0.114	100.000	1.00	No
Shell inlet-1 [1]	A	47	-104		0.097	90.000	1.00	No
Shell outlet-1[1]	A	48	-104		0.114	100.000	1.00	No
Shell outlet-2[1]	A	48	-104		0.114	100.000	1.00	No
BD1 [1]	A	21	-104		0.124	26.924	1.00	No
BD2 [1]	A	21	-104		0.124	26.924	1.00	No
shell inlet-2 [1]	A	47	-104		0.097	90.000	1.00	No
Shell inlet-3 [1]	A	47	-104		0.097	90.000	1.00	No
BD3 [1]	A	21	-104		0.124	26.924	1.00	No
BD5 [1]	A	21	-104		0.124	26.924	1.00	No
BD4 [1]	A	21	-104		0.124	26.924	1.00	No
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Channel head [9]	A	26	12		0.750	33.000	1.00	Yes
Channel shell [8]	A	42	26		0.704	68.000	1.00	Yes
Manhole [1]	A	26	12		0.750	33.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
N1- Tube inlet[1]	A	42	26		0.704	68.000	1.00	No
N2- Tube outle[1]	A	42	26		0.704	68.000	1.00	No
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:		42	26					

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



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MDMT Summary:

Step: 38 11:13am Apr 10, 2026

UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 11:13am Apr 10, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 11:13am Apr 10,2026

Units used in this Analysis (SI):

Name	System Unit	Constant	User Unit
Length	Feet	0.3048	m
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	0.0006	m ²
Moment	ft. lbs.	1.3563	N-m
Stress	lbs./sq.in.	6.8946	kPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	N/mm ²
Elast. Modulus	lbs./sq.in.	0.0069	N/mm ²
Pipe Density	lbs./cu.in.	27680.0000	kg/m ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	0.3048	m
Volume	in.3	0.1600E-04	m ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	14.281	N/mm ²
Channel Side Design Pressure	4.9503	N/mm ²
Shell Side Design Temperature	343.0	°C
Channel Side Design Temperature	475.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	N/mm ²
User defined MAWP, Channel Side	0	N/mm ²
User defined MAPnc, Shell Side	0	N/mm ²
User defined MAPnc, Channel Side	0	N/mm ²
User defined Test Pres., Shell Side	0	N/mm ²
User defined Test Pres., Channel Side	0	N/mm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

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Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	Yes
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Channel head
Distance "FROM" to "TO"	0 m
Inside Diameter	2520 mm
Element Thickness	33 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	40 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.9503 N/mm ²
Design Temperature Internal Pressure	475 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	24 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 22
Allowable Stress, Ambient	147989 kPa
Allowable Stress, Operating	116013 kPa
Allowable Stress, Hydrotest	278979 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	0 mm
Yield Stress, Operating	224994 kPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K21590
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Nozzle
Detail ID	Manhole
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	Y



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Input Echo:

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Weight of Nozzle (Used if > 0)	22320	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-336 F22	+IT

Element From Node	10	
Detail Type	Weight	
Detail ID	DAVIT	WEIGHT
Dist. from "FROM" Node / Offset dist	-0.65	m
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	300	mm

Element#	2/4	
Element From Node	20	
Element To Node	40	
Element Type	Cylinder	
Description	Channel shell	
Distance "FROM" to "TO"	2.5	m
Inside Diameter	2520	mm
Element Thickness	68	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	68	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.9503	N/mm ²
Design Temperature Internal Pressure	475	°C
Design External Pressure	0	N/mm ²
Design Temperature External Pressure	24	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 22	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	2.52	m
Liquid Density	18.491	kg/m ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1- Tube inlet	
Dist. from "FROM" Node / Offset dist	1.1	m
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15959	N
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	+IT

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N2- Tube outlet	
Dist. from "FROM" Node / Offset dist	1.1	m
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	16185	N
Grade of Attached Flange	None	



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Nozzle Matl SA-336 F11 +IT

Element From Node	20
Detail Type	Weight
Detail ID	Pass partition
Dist. from "FROM" Node / Offset dist	1.25 m
Miscellaneous Weight	19249 N
Offset from Element Centerline	0 mm

Element#	3/4
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	5.7 m
Inside Diameter	2520 mm
Element Thickness	138 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	138 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	14.281 N/mm ²
Design Temperature Internal Pressure	343 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	147989 kPa
Allowable Stress, Operating	147989 kPa
Allowable Stress, Hydrotest	278979 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	16.002 mm
Yield Stress, Operating	239106 kPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1.15 m
Width of Saddle	230 mm
Height of Saddle at Bottom	1800 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	300 mm
Thickness of Wear Plate	30 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 m

Element From Node	40
Detail Type	Saddle
Detail ID	Right side sadd
Dist. from "FROM" Node / Offset dist	4.55 m



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Width of Saddle	230	mm
Height of Saddle at Bottom	1800	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	300	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	m

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	2.52	m
Liquid Density	999.54	kg/m³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Insulation	5.7	m
Thickness of Insulation	150	mm
Density	352.4	kg/m³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell inlet-4	
Dist. from "FROM" Node / Offset dist	3.7	m
Nozzle Diameter	6	in.
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1478	N
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell outlet-3	
Dist. from "FROM" Node / Offset dist	2.7	m
Nozzle Diameter	8	in.
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2261.6	N
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell outlet-4	
Dist. from "FROM" Node / Offset dist	3.7	m
Nozzle Diameter	8	in.
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2261.6	N



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Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	Shell inlet-1
Dist. from "FROM" Node / Offset dist	0.654 m
Nozzle Diameter	6 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1478 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	Shell outlet-1
Dist. from "FROM" Node / Offset dist	0.6658 m
Nozzle Diameter	8 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	2261.6 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	Shell outlet-2
Dist. from "FROM" Node / Offset dist	1.6921 m
Nozzle Diameter	8 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	2261.6 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	BD1
Dist. from "FROM" Node / Offset dist	0.2 m
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	286.54 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	BD2
Dist. from "FROM" Node / Offset dist	1.5451 m
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	286.54 N



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Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	shell inlet-2
Dist. from "FROM" Node / Offset dist	1.9386 m
Nozzle Diameter	6 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1478 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	Shell inlet-3
Dist. from "FROM" Node / Offset dist	2.6746 m
Nozzle Diameter	6 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1478 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	BD3
Dist. from "FROM" Node / Offset dist	2.331 m
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	286.54 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	BD5
Dist. from "FROM" Node / Offset dist	3 m
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	286.54 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Nozzle
Detail ID	BD4
Dist. from "FROM" Node / Offset dist	4.1231 m
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	286.54 N



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Input Echo:

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Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	40
Detail Type	Weight
Detail ID	TOP HEADER-16"
Dist. from "FROM" Node / Offset dist	2.2 m
Miscellaneous Weight	13728 N
Offset from Element Centerline	2000 mm
Element From Node	40
Detail Type	Weight
Detail ID	BTM. HEADER-10"
Dist. from "FROM" Node / Offset dist	2.2 m
Miscellaneous Weight	6864.2 N
Offset from Element Centerline	2000 mm

Element#	4/4
Element From Node	50
Element To Node	60
Element Type	Spherical
Description	Shell head
Distance "FROM" to "TO"	0 m
Inside Diameter	2520 mm
Element Thickness	70 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	85 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	14.281 N/mm ²
Design Temperature Internal Pressure	343 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Insulation
Detail ID	insl
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	0.68 m
Thickness of Insulation	150 mm
Density	352.4 kg/m ³



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XY Coordinate Calculations: Step: 2 11:13am Apr 10,2026

XY Coordinate Calculations:

From	To	X (Horiz.) m	Y (Vert.) m	DX (Horiz.) m	DY (Vert.) m
Channel head		...	...	...	...
Channel shell		2.5	...	2.5	...
Shell		8.74	...	5.7	...
Shell head		8.74	...	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Calculations: Step: 3 11:13am Apr 10,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd N/mm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kPa	Element#
Channel head		4.9503	40	3	2520	116013	1
Channel shell		4.9507	68	3	2520	116013	2
Shell		14.306	138	3	2520	147989	3
Shell head		14.281	85	3	2520	147989	4

Element Required Thickness and MAWP :

From	To	Design Pressure N/mm ²	M.A.W.P. Corroded N/mm ²	M.A.P. New & Cold N/mm ²	Minimum Thickness mm	Required Thickness mm	Element#
Channel head		4.95029	5.49	7.71	33	30.06	1
Channel shell		4.95029	5.79	7.74	68	58.3102	2
Shell		14.2808	14.8	15.2	138	132.599	3
Shell head		14.2808	15.5	16.3	70	64.5296	4

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP	=	14.7 N/mm ²
Shell Side MAPnc	=	15.2 N/mm ²
Channel Side MAWP	=	5.25 N/mm ²
Channel Side MAPnc	=	7.71 N/mm ²

Elements Suitable for Design Internal Pressure.

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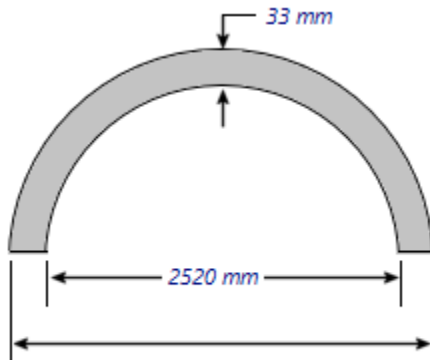
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Element: Channel head

Step: 4 11:13am Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel head

Spherical Head Design Information:			
Design Pressure P	4.950 N/mm ²	Int. Design Temperature	475 °C
Material	SA-387 22	Ext. Design Temperature	24 °C
Length	0.000 m	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	116013.2 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2520.000 mm	Actual Stress	104692.5 kPa
Spec. Min. Thk t _e , or s _a	33.0000 mm	Spec. Nominal Thk.	40.000 mm
Surface Area	10.6 m ²	SG of Contents	0.000
Element Volume	4.1 m ³	Weight of Contents	0.0 N
Empty Weight	31298.0 N	Operating Weight	31298.0 N
		Comp. All. Ope. Stress	66542.5 kPa
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (4.95 \cdot 1263) / (2 \cdot 116013 \cdot 1 - 0.2 \cdot 4.95)$$

$$= 27.0600 + 3.0000 = 30.0600 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 116013 \cdot 1 \cdot 30) / (1263 + 0.2 \cdot 30)$$

$$= 5.49 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 147989 \cdot 1 \cdot 33) / (1260 + 0.2 \cdot 33)$$



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Element: Channel head

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= 7.71 N/mm²

Actual stress at given pressure and thickness, corroded [Sact]:

= $(P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$
= $(4.95 \cdot (1263 + 0.2 \cdot 30)) / (2 \cdot 1 \cdot 30)$
= 104692.523 kPa

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.344 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 33, tr = 22.5, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 26 °C
Min Metal Temp. at Required thickness (UCS 66.1) 12 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	5.251 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	33.000 mm	Basic MDMT	26 °C
Stress/Thk. Ratio	0.750	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	12 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.750 mm

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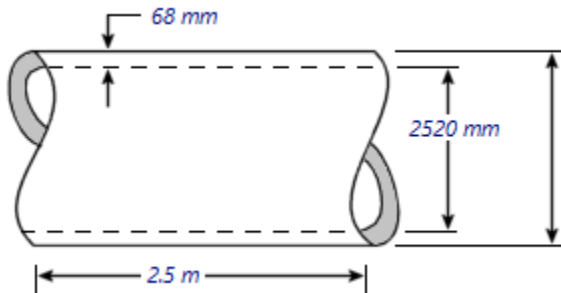
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Element: Channel shell

Step: 5 11:13am Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	4.951 N/mm ²	Int. Design Temperature	475 °C
Material	SA-387 22	Ext. Design Temperature	24 °C
Length	2.500 m	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	116013.2 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2520.000 mm	Actual Stress	99161.5 kPa
Spec. Min. Thk t,e, or sa	68.0000 mm	Spec. Nominal Thk.	68.000 mm
Surface Area	20.9 m ²	SG of Contents	0.018
Element Volume	12.2 m ³	Weight of Contents	2261.0 N
Empty Weight	105047.2 N	Operating Weight	107308.2 N
		Comp. All. Ope. Stress	76068.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4.95 \cdot 1263) / (116013 \cdot 1 - 0.6 \cdot 4.95) \\
 &= 55.3102 + 3.0000 = 58.3102 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.000 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (116013 \cdot 1 \cdot 65) / (1263 + 0.6 \cdot 65) \\
 &= 5.79 - 0.00046 = 5.79 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147989 \cdot 1 \cdot 68) / (1260 + 0.6 \cdot 68) \\
 &= 7.74 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4.95 \cdot (1263 + 0.6 \cdot 65)) / (1 \cdot 65) \\
 &= 99161.477 \text{ kPa}
 \end{aligned}$$



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Element: Channel shell

Step: 5 11:13am Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.628 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 68$, $t_r = 45.8$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.7$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 42 °C

Min Metal Temp. at Required thickness (UCS 66.1) 26 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	5.251 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	68.000 mm	Basic MDMT	42 °C
Stress/Thk. Ratio	0.704	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	26 °C
		Joint Efficiency E^*	100 %

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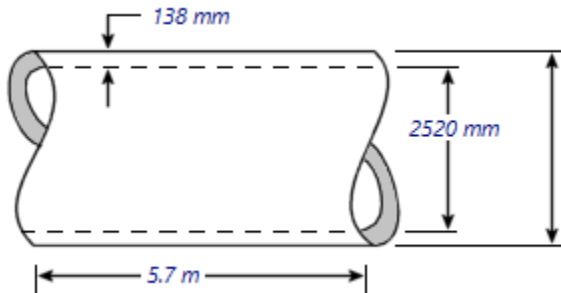
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Element: Shell

Step: 6 11:13am Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	14.306 N/mm ²	Int. Design Temperature	343 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	5.700 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2520.000 mm	Actual Stress	142411.5 kPa
Spec. Min. Thk t _e , or sa	138.0000 mm	Spec. Nominal Thk.	138.000 mm
Surface Area	50.1 m ²	SG of Contents	1.000
Element Volume	12.4 m ³	Weight of Contents	124973.0 N
Empty Weight	499206.8 N	Operating Weight	624179.8 N
		Comp. All. Ope. Stress	109728.9 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (14.3 \cdot 1263) / (147989 \cdot 1 - 0.6 \cdot 14.3) \\
 &= 129.5991 + 3.0000 = 132.5991 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.025 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147989 \cdot 1 \cdot 135) / (1263 + 0.6 \cdot 135) \\
 &= 14.9 - 0.025 = 14.8 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147989 \cdot 1 \cdot 138) / (1260 + 0.6 \cdot 138) \\
 &= 15.2 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (14.3 \cdot (1263 + 0.6 \cdot 135)) / (1 \cdot 135) \\
 &= 142411.547 \text{ kPa}
 \end{aligned}$$



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Element: Shell

Step: 6 11:13am Apr 10,2026

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 5.192 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.727 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	138.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 6120 / \sqrt{(1398 \cdot 135)}$$

$$= 14.087$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 13 \text{ and } Mx < 2 \cdot (Do/t)^{0.94}, Ch = 1.12 \cdot Mx - 1.058$$

$$= 1.12 \cdot 14.1 - 1.058$$

$$= 0.0682$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.068 \cdot 193337 (135/2796)$$

$$= 1018494.562 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 1018494.6/261995.9$$

$$= 3.887$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = Fy$$

$$= 261995.875 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [261996] \geq Fy [261996], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = Fy/FS$$

$$= 261995.875/1.667$$

$$= 157166.078 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:



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Element: Shell

Step: 6 11:13am Apr 10,2026

= 2 • Fha(t / Do)
= 2 • 157 (135/2796)
= 15.178 N/mm²

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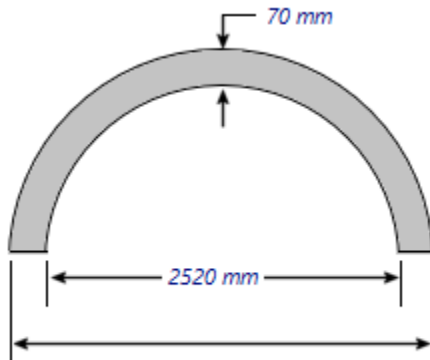
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Element: Shell head

Step: 7 11:13am Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell head

Spherical Head Design Information:			
Design Pressure P	14.281 N/mm ²	Int. Design Temperature	343 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	0.000 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2520.000 mm	Actual Stress	136022.3 kPa
Spec. Min. Thk t _e , or s _a	70.0000 mm	Spec. Nominal Thk.	85.000 mm
Surface Area	11.4 m ²	SG of Contents	0.000
Element Volume	4.1 m ³	Weight of Contents	0.0 N
Empty Weight	68885.9 N	Operating Weight	68885.9 N
		Comp. All. Ope. Stress	109728.9 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (14.3 \cdot 1263) / (2 \cdot 147989 \cdot 1 - 0.2 \cdot 14.3)$$

$$= 61.5296 + 3.0000 = 64.5296 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 147989 \cdot 1 \cdot 67) / (1263 + 0.2 \cdot 67)$$

$$= 15.5 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 147989 \cdot 1 \cdot 70) / (1260 + 0.2 \cdot 70)$$



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Element: Shell head

Step: 7 11:13am Apr 10,2026

= 16.3 N/mm²

Actual stress at given pressure and thickness, corroded [Sact]:

= $(P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$
 = $(14.3 \cdot (1263 + 0.2 \cdot 67)) / (2 \cdot 1 \cdot 67)$
 = 136022.328 kPa

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.894 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 70, tr = 63.5, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.95$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 40 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.727 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	70.000 mm	Basic MDMT	43 °C
Stress/Thk. Ratio	0.947	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	40 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.750 mm

EXTERNAL PRESSURE RESULTS for: Shell head

Spherical Head From 50 to 60 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per Code Case 2286:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $0.075 \cdot E_y (t / R_o)$
 = $0.075 \cdot 193337 (67 / 1330)$
 = 730422.188 kPa

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Sy
 = $730422.2 / 261995.9$
 = 2.788

Predicted Buckling Stress [Fic]:

Since $Fr > 1.6$ and $Fr < 6.25$, $Fic = 1.31 \cdot Fy / (1.15 + Sy / Fhe)$
 = $1.31 \cdot 261996 / (1.15 + 261996 / 730422)$
 = 227491.641 kPa

Factor of Safety [FS]:

Since $Fic > 0.55 Fy$ and $Fic < Fy$
 = $2.407 - 0.741 \cdot Fic / Fy$
 = $2.407 - 0.741 \cdot 227492 / 261996$
 = 1.764



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Element: Shell head

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Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 1.6$ and $Fr < 6.25$, $Fha = 1.31 \cdot Fy / (FS \cdot (1.15 + Sy/Fhe))$
 $= 1.31 \cdot 261996 / (1.76 \cdot (1.15 + 261996/730422))$
 $= 128993.656 \text{ kPa}$

Allowable External Pressure at the given Thickness [Pa]:

$= 2 \cdot Fha[t/Ro]$
 $= 2 \cdot 128994 [67/1330]$
 $= 12.997 \text{ N/mm}^2$

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Shop Pressure Test Results: Step: 8 11:13am Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	19.145 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	18.565 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	19.748 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	16.200 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	20.422 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.422 N/mm ²
App 27-4	= M.A.W.P.	14.727 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.066 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6.661 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	10.001 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	5.979 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	7.079 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7.079 N/mm ²
App 27-4	= M.A.W.P.	5.251 N/mm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel head	192385.4	278978.8	0.690	10.03
Channel shell	191768.9	278978.8	0.687	10.03
Shell	192385.4	278978.8	0.690	19.77
Shell head	179920.7	278978.8	0.645	19.77

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	147991.42	115993.27	1.276
N1- Tube inlet	Nozzle	147991.42	115993.27	1.276
N2- Tube outlet	Nozzle	147988.41	107048.78	1.382
Shell inlet-4	Nozzle	147988.73	147988.73	1.000
Shell outlet-3	Nozzle	147988.73	147988.73	1.000
Shell outlet-4	Nozzle	147988.73	147988.73	1.000
Shell inlet-1	Nozzle	147988.73	147988.73	1.000
Shell outlet-1	Nozzle	147988.73	147988.73	1.000
Shell outlet-2	Nozzle	147988.73	147988.73	1.000
BD1	Nozzle	147988.73	147988.73	1.000
BD2	Nozzle	147988.73	147988.73	1.000
shell inlet-2	Nozzle	147988.73	147988.73	1.000
Shell inlet-3	Nozzle	147988.73	147988.73	1.000
BD3	Nozzle	147988.73	147988.73	1.000
BD5	Nozzle	147988.73	147988.73	1.000
BD4	Nozzle	147988.73	147988.73	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
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Shop Pressure Test Results:

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Channel head	147988.73	116013.23	1.276
Channel shell	147988.73	116013.23	1.276
Shell	147988.73	147988.73	1.000
Shell head	147988.73	147988.73	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
Tube Material	117991.02	113991.31	1.035
Tubesheet Material	147988.41	116013.27	1.276

Minimum 1.035

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Manhole	130174.36	191988.86	0.678
N1- Tube inlet	137248.88	191988.86	0.715
N2- Tube outlet	137248.88	192385.84	0.713
Shell inlet-4	94370.51	192385.34	0.491
Shell outlet-3	99154.33	192385.34	0.515
Shell outlet-4	99154.33	192385.34	0.515
Shell inlet-1	94370.51	192385.34	0.491
Shell outlet-1	99154.33	192385.34	0.515
Shell outlet-2	99154.33	192385.34	0.515
BD1	30515.27	192385.34	0.159
BD2	30515.27	192385.34	0.159
shell inlet-2	94370.51	192385.34	0.491
Shell inlet-3	94370.51	192385.34	0.491
BD3	30515.27	192385.34	0.159
BD5	30515.27	192385.34	0.159
BD4	30515.27	192385.34	0.159

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 9 11:13am Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	19.145 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	18.565 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	19.748 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	16.200 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	20.422 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.422 N/mm ²
App 27-4	= M.A.W.P.	14.727 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.066 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6.661 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	10.001 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	5.979 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	7.079 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7.079 N/mm ²
App 27-4	= M.A.W.P.	5.251 N/mm ²

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 14.3 • 1

= 18.565 N/mm²

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4.95 • 1.04

= 6.661 N/mm²

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel head	141398.2	278978.8	0.507	6.69
Channel shell	133915.5	278978.8	0.480	6.69
Shell	185060.8	278978.8	0.663	18.59
Shell head	177064.3	278978.8	0.635	18.59

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	147991.42	115993.27	1.276
N1- Tube inlet	Nozzle	147991.42	115993.27	1.276
N2- Tube outlet	Nozzle	147988.41	107048.78	1.382
Shell inlet-4	Nozzle	147988.73	147988.73	1.000
Shell outlet-3	Nozzle	147988.73	147988.73	1.000
Shell outlet-4	Nozzle	147988.73	147988.73	1.000
Shell inlet-1	Nozzle	147988.73	147988.73	1.000
Shell outlet-1	Nozzle	147988.73	147988.73	1.000
Shell outlet-2	Nozzle	147988.73	147988.73	1.000
BD1	Nozzle	147988.73	147988.73	1.000
BD2	Nozzle	147988.73	147988.73	1.000
shell inlet-2	Nozzle	147988.73	147988.73	1.000



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Field Pressure Test Results: Step: 9 11:13am Apr 10,2026

Shell inlet-3	Nozzle	147988.73	147988.73	1.000
BD3	Nozzle	147988.73	147988.73	1.000
BD5	Nozzle	147988.73	147988.73	1.000
BD4	Nozzle	147988.73	147988.73	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
Channel head	147988.73	116013.23	1.276
Channel shell	147988.73	116013.23	1.276
Shell	147988.73	147988.73	1.000
Shell head	147988.73	147988.73	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
Tube Material	117991.02	113991.31	1.035
Tubesheet Material	147988.41	116013.27	1.276

Minimum 1.035

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Manhole	86812.51	191988.86	0.452
N1- Tube inlet	91530.47	191988.86	0.477
N2- Tube outlet	91530.47	192385.84	0.476
Shell inlet-4	88724.91	192385.34	0.461
Shell outlet-3	93222.54	192385.34	0.485
Shell outlet-4	93222.54	192385.34	0.485
Shell inlet-1	88724.91	192385.34	0.461
Shell outlet-1	93222.54	192385.34	0.485
Shell outlet-2	93222.54	192385.34	0.485
BD1	28689.73	192385.34	0.149
BD2	28689.73	192385.34	0.149
shell inlet-2	88724.91	192385.34	0.461
Shell inlet-3	88724.91	192385.34	0.461
BD3	28689.73	192385.34	0.149
BD5	28689.73	192385.34	0.149
BD4	28689.73	192385.34	0.149

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 11:13a Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		18.57 N/mm ²
Shell Thickness	ts	138.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2520.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	147988.73 kPa
Shell Allowable Stress at Ambient		147988.73 kPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Test Pressure		6.66 N/mm ²
Channel Thickness	tc	68.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2520.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	147988.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	4952
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	4800.00 mm
Straight Tube Length (bet. inner tubsht faces) L		4260.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		117991.02 kPa
Tube Allowable Stress At Ambient		117991.02 kPa
Tube Yield Stress At design Temperature	Syt	204984.41 kPa
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1215.220 mm
Largest Center-to-Center Tube Distance	U1	101.6000 mm
Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	4.4 m ²
Perimeter of Tube Layout	Cp	7805.0 mm



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ASME TS Calc: Pressure Test

Case: 1 11:13a Apr 10,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F22

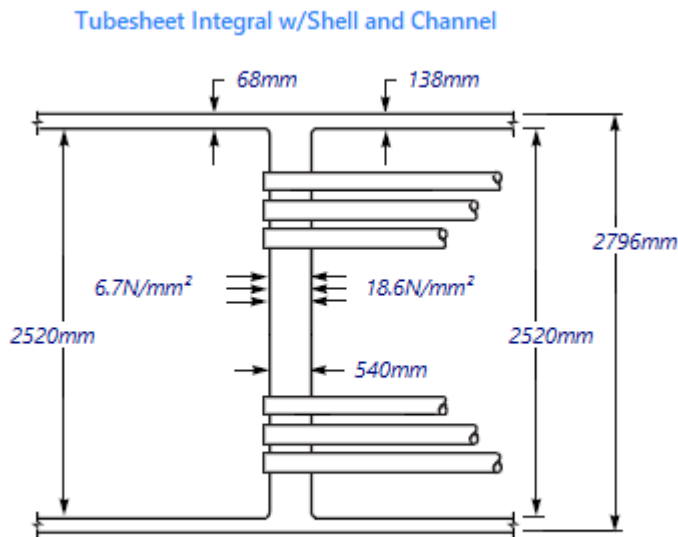
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K21590
Tubesheet Allowable Stress at Temperature S 147988.41 kPa
Tubesheet Allowable Stress at Ambient Tt 147988.41 kPa
Thickness of Tubesheet h 540.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 2796.000 mm

Area of the Untubed Lanes AL 0.192 m²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes

Tubesheet Integral with Both
Tubesheet Extended as Flange No



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature 21.1 °C 0.20409E+06 N/mm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20409E+06 N/mm²

Channel - TM-1 Material Group D



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ASME TS Calc: Pressure Test

Case: 1 11:13a Apr 10, 2026

Elastic Mod. at Design Temperature	21.1 °C	0.21099E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tubes - TM-1 Material Group D

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.21099E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 °C	0.21099E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(6.66 \cdot (25.4/2 - 0)) / (117991 \cdot 1 + 0.4 \cdot 6.66)$

= $0.7011 + 0.0000 = 0.7011$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$

= $5475 / \sqrt{(12.7 \cdot 2.11)}$

= 1057.690

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)0.94$, $Ch = 0.55 / (Do/t)$

= $0.55 / (25.4/2.11)$

= 0.0457

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey (t / Do)$

= $1.6 \cdot 0.046 \cdot 210987 (2.11/25.4)$

= 1281181.625 kPa

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy

= $1281181.6/204984.4$

= 6.250

Predicted Buckling Stress [Fic]:

Since $Fr \geq 2.439$, $Fic = Fy$

= 204984.406 kPa

Factor of Safety [FS]:

Since $Fic [204984] \geq Fy [204984]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 2.439$, $Fha = Fy/FS$

= $204984.406/1.667$

= 122966.039 kPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha (t / Do)$

= $2 \cdot 123 (2.11/25.4)$

= 20.431 N/mm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.9173 mm
Allowable Internal Pressure at Corroded thickness	21.00 N/mm ²
Required Internal Design Pressure	6.66 N/mm ²
Allowable External Pressure at Corroded thickness	20.43 N/mm ²



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ASME TS Calc: Pressure Test

Case: 1 11:13a Apr 10,2026

Required External Design Pressure 18.57 N/mm²
Required Thickness due to Shell Side pressure 1.9173 mm

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	232.071	...	107335	588955	34843	235582	Ps+Pt D1	Ok
D2uc	380.324	...	282702	588955	97109	235582	Ps+Pt D2	Ok
D3uc	310.055	...	188453	588955	62266	235582	Ps+Pt D3	Ok
Max:	380.3244	...	mm	0.480		0.412	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Reqd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
	Shell	Channel					
D1uc	...	...	...	441716	...	441716	Ok
D2uc	...	...	...	441716	...	441716	Ok
D3uc	...	...	...	441716	...	441716	Ok
Max	...	...		...		...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	380.324	540.000	Ok
Tube thickness :	1.917	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	3.000	Ok
Tube-tubesheet groove weld leg :	1.333	3.000	Ok

Given Shell Thickness: 138.0000 mm

Given Channel Thickness: 68.0000 mm

Min Shell length of thk, (138.000) adj. to tubesheet: 1061.481 mm

Min Channel length of thk, (68.000) adj. to tubesheet: 745.122 mm

Note: This is a full strength Tube to Tubesheet Joint.



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ASME TS Calc: Pressure Test

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External Pressure Calculations: Step: 10 11:13am Apr 10,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length m	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kPa
10	20	No Calc	2586	30	No Calc	No Calc
20	Tsht	2.91997	2656	65	No Calc	No Calc
Tsht	40	No Calc	...	65	No Calc	No Calc
40	50	6.12003	2796	135	No Calc	No Calc
50	60	No Calc	2660	67	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	33	No Calc	...	4.98
20	Tsht	68	No Calc	...	9.1
Tsht	40	68	No Calc	...	No Calc
40	50	138	14.7003	0.10131	15.2
50	60	70	6.51545	0.10131	13

Minimum

4.98

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2.91997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	6.12003	25400	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 11 11:13am Apr 10,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume m ³	Corroded Metal Wgt. kgm	Corroded ID Volume m ³	Extra due Misc % kgm
10	20	3191.72	4.09062	2959.23	4.11991	319.172
20	40	10712.5	12.1745	10251.8	12.2325	1071.25
40	50	50908.3	12.4489	49857.8	12.5812	5090.83
50	60	7024.87	4.09062	6792.38	4.11991	702.487
Total		71837	32.80	69861	33.05	7183



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. m	Y Offset, Dtl. Cent. m	Z Offset, Dtl. Cent. m	Description
10	Nozl	2503.8	-1.26	...	...	Manhole
10	Wght	200	-1.26	0.3	...	DAVIT WEIGHT
20	Liqd	230.572	1.25	...	...	Liquid: 20
20	Nozl	1790.25	1.1	1.7172	...	N1- Tube inlet
20	Nozl	1815.55	1.1	-1.7172	...	N2- Tube outlet
20	Wght	1963	1.25	...	...	Pass partition
40	Sadl	834.527	1.15	1.446	...	Left Saddle
40	Sadl	834.527	4.55	1.446	...	Right side sadd
40	Liqd	12744.5	2.85	...	...	Liquid: 40
40	Insl	2788.69	2.85	...	...	Insulation
40	Nozl	165.796	3.7	-1.3362	...	Shell inlet-4
40	Nozl	253.694	2.7	1.3616	...	Shell outlet-3
40	Nozl	253.694	3.7	1.3616	...	Shell outlet-4
40	Nozl	165.796	0.65401	-1.3362	...	Shell inlet-1
40	Nozl	253.694	0.66583	1.3616	...	Shell outlet-1
40	Nozl	253.694	1.69208	1.3616	...	Shell outlet-2
40	Nozl	32.1429	0.2	-1.2854	...	BD1
40	Nozl	32.1429	1.54507	-1.2854	...	BD2
40	Nozl	165.796	1.93856	-1.3362	...	shell inlet-2
40	Nozl	165.796	2.67462	-1.3362	...	Shell inlet-3
40	Nozl	32.1429	2.33098	-1.2854	...	BD3
40	Nozl	32.1429	3	-1.2854	...	BD5
40	Nozl	32.1429	4.12305	-1.2854	...	BD4
40	Wght	1400	2.2	2	...	TOP HEADER-16"
40	Wght	700	2.2	-2	...	BTM. HEADER-10"
50	Insl	742.607	0.34	...	...	insl
20	FTsh	16715.1	2.77	...	...	Tubesheet
20	Tube	41286.3	4.9	...	...	Tubes
40	Baf1	356.506	0.364	...	...	Baffles
40	Baf1	356.506	0.864	...	...	Baffles
40	Baf1	356.506	1.364	...	...	Baffles
40	Baf1	356.506	1.864	...	...	Baffles
40	Baf1	356.506	2.364	...	...	Baffles
40	Baf1	356.506	2.864	...	...	Baffles
40	Baf1	356.506	3.364	...	...	Baffles
40	Baf1	356.506	3.864	...	...	Baffles
40	Baf1	357.567	4.364	...	...	Baffles

Total Weight of Each Detail Type:



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Element and Detail Weights: Step: 11 11:13am Apr 10,2026

Saddles	1669.1
Liquid	12975.1
Insulation	3531.3
Nozzles	7948.3
Weights	4263.0
Exchanger Components	61211.0
Liquid in Tubes	218.5

Sum of the Detail Weights 91816.3 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	79021.2	79021.2	79021.2	79021.2	79021.2	79021.2
Saddles	1669.1	1669.1	1669.1	1669.1	1669.1	1669.1
Nozzles	7948.3	7948.3	7948.3	7948.3	7948.3	7948.3
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0
Exchanger	61211.0	61211.0	61211.0	61211.0	61211.0	61211.0
Insulation	...	...	...	...	3531.3	3531.3 *
Empty Weights	...	...	...	4063.0	4063.0	...
Test Weights	...	1963.0	...	...	...	...
Ope Weights	...	...	...	...	...	4063.0
Ope. Liquid	...	...	...	...	...	12975.1
Tube Ope Lqd	...	...	...	...	...	218.5
Test Liquid	...	33583.7	...	...	...	...
Tube Tst Lqd	...	11839.1	...	...	...	...
Totals	150049.6	197435.4	150049.6	154112.5	157643.8	170837.5

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	150049.6 kgm
Shop Test Wt.	- Fabricated weight + water (full)	197435.4 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	150049.6 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	154112.5 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	157643.8 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	170837.5 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	168663.7 kgm
Field Test Wt.	- Empty weight + test medium (full)	200966.7 kgm

Exchanger Tube Data

Volume of Exchanger tubes :	11.6 m ³
Weight of Ope Liq in tubes :	218.5 kgm
Weight of Water in tubes :	11839.1 kgm

Surface Areas of Elements:

From	To	Outside Surface Area m ²	Inside Surface Area m ²
10	20	10.616	9.97271
20	40	20.855	19.7871
40	50	50.0558	45.1146



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Element and Detail Weights: Step: 11 11:13am Apr 10, 2026

50	60	11.3636	9.97271
----	----	---------	---------

Total	92.9	84.8 m ²
Total	1000.1	913.5 ft ²

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Nozzle Flange MAWP:

Step: 12 11:13am Apr 10,2026

Nozzle Flange MAWP Results: (N/mm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Manhole	5.25	15.31	475	900	GR 1.1	...	...	...	...

Shellside Flange Rating**Channelside Flange Rating**Lowest Flange Pressure Rating was (Ope)[TubeSide]: 5.251 N/mm²Lowest Flange Pressure Rating was (Amb)[TubeSide]: 15.311 N/mm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 13 11:13am Apr 10,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97)/(1 + 1.7 \cdot 3.4 \cdot 0.23))) \\
 &= \min(0.85, 0.91) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.546
Structure Height to Diameter ratio	3.767

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274 \text{ m}$

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 1.8 + 0.3 = 2.1 \text{ m} \\
 &= 6.89 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (6.89 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900)^{2/9.5} \\
 &= 0.85
 \end{aligned}$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 13 11:13am Apr 10,2026

Wind Directionality Factor [Kd]:
= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
= (1 + K1 • K2 • K3)²
= (1 + 0 • 0 • 0)²
= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)
= 16 psf [0.77] kPa

Force on the first element [F]:
= qz • G • Cf • WindArea
= 16 • 0.85 • 0.55 • 28.3
= 210 lbs. [934] N

Element	Hgt (z) m	K1	K2	K3	Kz	Kzt	qz kPa
Channel head	2.1	0.000	0.000	0.000	0.849	1.000	0.766
Channel shell	2.1	0.000	0.000	0.000	0.849	1.000	0.766
Shell	2.1	0.000	0.000	0.000	0.849	1.000	0.766
Shell head	2.1	0.000	0.000	0.000	0.849	1.000	0.766

Wind Loads on Masses/Equipment/Piping

ID	Wind Area m ²	Elevation m	Pressure kPa	Force N
DAVIT WEIGHT	0.00	2.40	0.77	0.00
Pass partitio	0.00	2.10	0.77	0.00
TOP HEADER-16	0.00	4.10	0.77	0.00
BTM. HEADER-1	0.00	0.30	0.77	0.00

Wind Load Calculation:

From	To	Wind Height m	Wind Diameter m	Wind Area m ²	Wind Pressure kPa	Element Wind Load N
10	20	2.1	3.1032	2.62558	0.76606	1027.27
20	40	2.1	3.1872	7.96602	0.76606	3116.73
40	50	2.1	3.7152	21.1714	0.76606	8283.38
50	60	2.1	3.552	3.43993	0.76606	1345.88

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 14 11:13am Apr 10,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		No
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(100.75)$
 $= 0.113$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.11, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 1658865 \cdot 1.25$
 $= 97541.297$ N

Earthquake Load Calculation:

From	To	Earthquake Height m	Earthquake Weight N	Element Ope Load N
10	20	1.26	276478	16256.9
20	40	1.26	276478	16256.9
40	Sad1	1.26	276478	16256.9
Sad1	50	1.26	276478	16256.9
40	50	1.26	276478	16256.9
50	60	1.26	276478	16256.9



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Earthquake Load Calculation: Step: 14 11:13am Apr 10, 2026

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 15 11:13am Apr 10,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5.890 m
Center of Gravity of the Liquid	5.808 m
Center of Gravity of the Insulation	6.561 m
Center of Gravity of the Nozzles	1.317 m
Center of Gravity of the Added Weights (Operating)	3.098 m
Center of Gravity of the Added Weights (Empty)	3.098 m
Center of Gravity of the Tubesheet(s)	2.770 m
Center of Gravity of the Tubes	4.900 m
Center of Gravity of the Baffles	5.405 m
Center of Gravity of Bare Shell New and Cold	5.250 m
Center of Gravity of Bare Shell Corroded	5.273 m
Vessel CG in the Operating Condition	4.768 m
Vessel CG in the Fabricated (Shop/Empty) Condition	4.635 m
Vessel CG in the Test Condition	4.509 m

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Saddle Calcs: Operating Case: Step: 16 11:13am Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

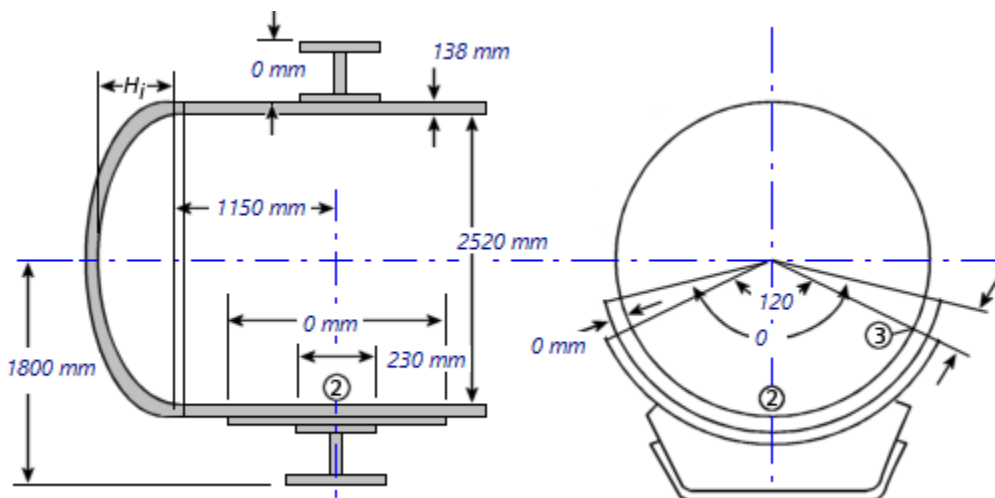
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(230 + 1.56 \cdot \sqrt{1330 \cdot 135}, 2 \cdot 1150)$$

$$= 891.1483 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1330.50	mm
Shell Thickness used in this Case	t	138.000	mm
Stiffened Vessel Length per 4.15.6	L	5.70	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	230.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.28	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		116013.23	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		1490582.62	N
Pressure used in Analysis	P	14.293	N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top of Midspan		70516.29	147988.73
Long. Stress at Bottom of Midspan		70342.76	147988.73



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Saddle Calcs: Operating Case: Step: 16 11:13am Apr 10,2026

Long. Stress at Top of Saddles	78512.65	147988.73
Long. Stress at Bottom of Saddles	65949.36	147988.73
Tangential Shear in Shell	4461.46	118390.98
Circ. Stress at Horn of Saddle	12741.81	184985.91
Circ. Compressive Stress in Shell	941.99	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (13773/2 + 0) \cdot 1800/2473$$

$$= 15035.3 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(7450, 0) \cdot 1800/3400$$

$$= 3944.3 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(97541, 0) \cdot 1800/3400$$

$$= 51639.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (97541/2 + 0) \cdot 1800/2473$$

$$= 106478.9 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1384103 + \max(3944, 15035, 51640, 106479)$$

$$= 1490582.6 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.8 \cdot 1.2)^2)$$

$$= 7450.694 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 1498766.00 N

Transverse Shear Load 48770.65 N

Longitudinal Shear Load 97541.30 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2520/2 + 138 + 30) \sin(120/2), 2 \cdot 1330)$$

$$= 2473.369 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0421 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4 \text{hm}) / (3L))]$$

$$= -1490582 \cdot 1.15 [1 - (1 - 1.15/5.7 + (1.33^2 - 1.28^2) / (2 \cdot 1.15 \cdot 5.7)) / (1 + (4 \cdot 1.28) / (3 \cdot 5.7))]$$

$$= -647231.1 \text{ N-m}$$



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Saddle Calcs: Operating Case: Step: 16 11:13am Apr 10, 2026

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 1490582 \cdot 5.7 / 4 \cdot ((1 + 2(1.33^2 - 1.28^2) / (5.7^2)) / (1 + (4 \cdot 1.28) / (3 \cdot 5.7)) - 4 \cdot 1.15 / 5.7)$$

$$= -65168.7 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1330 / (2 \cdot 135) - 65169 / (\pi \cdot 1330^2 \cdot 135)$$

$$= 70516.29 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1330 / (2 \cdot 135) + 65169 / (\pi \cdot 1330^2 \cdot 135)$$

$$= 70342.76 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1330 / (2 \cdot 135) - 647231 / (0.11 \cdot \pi \cdot 1330^2 \cdot 135)$$

$$= 78512.65 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 14.3 \cdot 1330 / (2 \cdot 135) + 647231 / (0.19 \cdot \pi \cdot 1330^2 \cdot 135)$$

$$= 65949.36 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 1490582 \cdot (5.7 - 2 \cdot 1.15) / (5.7 + (4 \cdot 1.28 / 3))$$

$$= 684495.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 684496 / (1330 \cdot 135)$$

$$= 4461.46 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{1330 \cdot 135}$$

$$= 330.574 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 1490582 \cdot 0.1 / (135(230 + 331 + 331))$$

$$= -941.99 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -1490582 / (4 \cdot 135(230 + 331 + 331)) - 12 \cdot 0.042 \cdot 1490582 \cdot 1330 / (5.7 \cdot 135^2)$$

$$= -12741.81 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(230 + 1.56 \cdot \sqrt{1330 \cdot 135}, 2 \cdot 1.15)$$

$$= 891.15 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 3400.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13497E-04 \cdot 3400(343 - 21.1)$$

$$= 14.771 \text{ mm}$$



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Saddle Calcs: Operating Case: Step: 16 11:13am Apr 10,2026

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2600.0002	mm
Baseplate Thickness	Bpthk	35.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1263 \cdot 135)} + 300) 135^3) / 12$$

$$= 193581952.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 193581952 + 0.13 (108 - 67.5)^2$$

$$= 403608928.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	944.2	135.0	67.5	0.1	8603650.0	0.404E+09
Wearplate	300.0	30.0	150.0	0.0	1350000.0	0.165E+08
Web	15.0	337.0	333.5	0.0	1685842.1	0.305E+09
BasePlate	254.0	35.0	519.5	0.0	4618354.5	0.151E+10
Totals	...	...	...	0.2	16257846.0	0.223E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16257846 / 0.15$$

$$= 108.093 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1490582$$

$$= 303365.9062 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 13221.7959 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1800 - 1263 \cdot \sin(120 / 2) / 1.05$$

$$= 755.507 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 229288.0938 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 11108.0703 \text{ kPa}$$



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Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(1490582 + 8183) \cdot 254}{4 \cdot 2600 \cdot 147989}}$$

$$= 27.241 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) \cdot (1260 + 138 + 30)$$

$$= 2473.369 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2473 / (5 - 1)$$

$$= 618.342 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 618 \cdot 254 / 2$$

$$= 0.079 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1498766 / (2600 \cdot 254)$$

$$= 2270044.750 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.079 \cdot 2270044$$

$$= 178221.156 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0032 + 0.0046$$

$$= 0.008 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 178221 / 0.0079$$

$$= 22667.658 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	230.0	...	...	...	0.152E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 230 - 15$$

$$= 215.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.0079$$

$$= 0.000 \text{ mm}$$



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230/2$$

$$= 115.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{15208737/0.0079}$$

$$= 43.981 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1800 - \cos(120/2) (1260 + 138 + 30) - 35$$

$$= 1051.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955/194672}$$

$$= 142.386$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1051/44$$

$$= 23.897$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 97541 / (2 \cdot 2600) \cdot 618 \cdot 1051/2$$

$$= 6097.654 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($23.9 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.9)^2 / (2 \cdot 142^2)) 194672 /$$

$$(5/3 + 3 \cdot (23.9) / (8 \cdot 142) - (23.9)^3 / (8 \cdot 142^3))$$

$$= 111006 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 22668/111006 + (6098 \cdot 115/15208751) / 129781$$

$$= 0.559$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	215.0	0.0	0.0	0.0	0.152E+08
Web	618.3	15.0	0.0	0.0	0.0	0.174E+06
Totals	...	...	...	...	...	0.154E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/0.012$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230/2$$

$$= 115.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1800 - \cos(1428/2) (35 + 0 + 0) - 0$$

$$= 337.000 \text{ mm}$$



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Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(15378426 / 0.012)}$$

$$= 35.075 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 337 / 35.1$$

$$= 9.608$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 97541 / (2 \cdot 2600)$$

$$= 18.758 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 18.8 \cdot 618 \cdot 337$$

$$= 3910.389 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3910 / 133726$$

$$= 29230.832 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($9.61 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.61)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (9.61) / (8 \cdot 142) - (9.61^3) / (8 \cdot 142^3))$$

$$= 114797 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 28564 / 114797 + (3910 \cdot 115 / 15378426) / 129781$$

$$= 0.474$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	137991.98 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
Nominal Bolt Diameter	Bnd	31.7500 mm
Thread Series	Series	UNC
BasePlate Allowable Stress	S	129114.48 kPa
Area Available in a Single Bolt	BltArea	0.0005 m²
Saddle Load Q0 (Weight)	Q0	1392287.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	51639.5 N
Maximum Transverse Force	Ft	48770.6 N
Maximum Longitudinal Force	F1	97541.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 97541 / (0.00045 \cdot 8)$$

$$= 26859.7 \text{ kPa. Must be less than } 82795.2 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 48771 / (0.00045 \cdot 8)$$

$$= 13429.9 \text{ kPa. Must be less than } 82795.2 \text{ kPa.}$$



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Saddle Calcs: Operating Case: Step: 16 11:13am Apr 10, 2026

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 97541/(82795 • 8)
= 0.0001 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1800 • 48771 + 0
= 87822.76 N-m

Eccentricity (e):
= Rmom / QO
= 87823/1392287
= 63.05 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.00015, 0]
= 0.0001 m²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: A

Govern. thk, tg = 30, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = -1, Temp. Reduction = 0 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
= $\min(230 + 1.56 \cdot \sqrt{1330 \cdot 135}, 2 \cdot 1150)$
= 891.1483 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1330.50	mm
Shell Thickness used in this Case	t	138.000	mm
Stiffened Vessel Length per 4.15.6	L	5.70	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	230.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.30	m



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Shell Allowable Stress used in Calculation	147988.73	kPa
Head Allowable Stress used in Calculation	147988.73	kPa
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	1800.00 mm
Coefficient of Friction	mu	0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	381240.66	N
Pressure used in Analysis	P	14.293 N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	70455.21	147988.73
Long. Stress at Bottom of Midspan	70403.84	147988.73
Long. Stress at Top of Saddles	72523.50	147988.73
Long. Stress at Bottom of Saddles	69268.91	147988.73
Tangential Shear in Shell	1137.30	118390.98
Circ. Stress at Horn of Saddle	3258.92	184985.91
Circ. Compressive Stress in Shell	240.93	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (13773/2 + 0) \cdot 1800/2473$
 $= 15035.3 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(7450, 0) \cdot 1800/3400$
 $= 3944.3 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(97541, 0) \cdot 1800/3400$
 $= 51639.5 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (97541/2 + 0) \cdot 1800/2473$
 $= 106478.9 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 274762 + \max(3944, 15035, 51640, 106479)$
 $= 381240.7 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.8 \cdot 1.2)^2)$
 $= 7450.694 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	389424.03	N
Transverse Shear Load	48770.65	N
Longitudinal Shear Load	97541.30	N

Note: Wear Plate is Welded to the Shell, k = 0.1



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Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2520/2 + 138 + 30)\sin(120/2), 2 \cdot 1330)$$

$$= 2473.369 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0421	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -381241 \cdot 1.15 [1 - (1 - 1.15/5.7 + (1.33^2 - 1.3^2)/(2 \cdot 1.15 \cdot 5.7))/(1 + (4 \cdot 1.3)/(3 \cdot 5.7))]$$

$$= -167669.2 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 381241 \cdot 5.7/4 [(1 + 2(1.33^2 - 1.3^2)/(5.7^2))/(1 + (4 \cdot 1.3)/(3 \cdot 5.7)) - 4 \cdot 1.15/5.7]$$

$$= -19291.6 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 14.3 \cdot 1330/(2 \cdot 135) - -19292/(\pi \cdot 1330^2 \cdot 135)$$

$$= 70455.21 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 14.3 \cdot 1330/(2 \cdot 135) + -19292/(\pi \cdot 1330^2 \cdot 135)$$

$$= 70403.84 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 14.3 \cdot 1330/(2 \cdot 135) - 167669/(0.11 \cdot \pi \cdot 1330^2 \cdot 135)$$

$$= 72523.50 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 14.3 \cdot 1330/(2 \cdot 135) + -167669/(0.19 \cdot \pi \cdot 1330^2 \cdot 135)$$

$$= 69268.91 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 381241 (5.7 - 2 \cdot 1.15)/(5.7 + (4 \cdot 1.3/3))$$

$$= 174489.5 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 174490 / (1330 \cdot 135)$$

$$= 1137.30 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1330 \cdot 135)}$$

$$= 330.574 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 381241 \cdot 0.1 / (135(230 + 331 + 331))$$

$$= -240.93 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:



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$$\begin{aligned}
 &= -Q/(4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2) \\
 &= -381241 / (4 \cdot 135 (230 + 331 + 331)) - \\
 &\quad 12 \cdot 0.042 \cdot 381241 \cdot 1330 / (5.7 \cdot 135^2) \\
 &= -3258.92 \text{ kPa}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\
 &= \min(230 + 1.56 \cdot \sqrt{1330 \cdot 135}, 2 \cdot 1.15) \\
 &= 891.15 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2600.0002	mm
Baseplate Thickness	Bpthk	35.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3)/12 \\
 &= ((1.56 \cdot \sqrt{1263 \cdot 135} + 300)135^3)/12 \\
 &= 193581952.000 \text{ mm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\
 &= 193581952 + 0.13 (108 - 67.5)^2 \\
 &= 403608928.000 \text{ mm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	944.2	135.0	67.5	0.1	8603650.0	0.404E+09
Wearplate	300.0	30.0	150.0	0.0	1350000.0	0.165E+08
Web	15.0	337.0	333.5	0.0	1685842.1	0.305E+09
BasePlate	254.0	35.0	519.5	0.0	4618354.5	0.151E+10
Totals	...	...	...	0.2	16257846.0	0.223E+10

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 16257846 / 0.15 \\
 &= 108.093 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 381241 \\
 &= 77590.7344 \text{ N}
 \end{aligned}$$



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Tension Stress, $St = (Fh/As) = 3381.6882 \text{ kPa}$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 116803.2656 \text{ kPa}$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1800 - 1263 \cdot \sin(120/2) / 1.05$
 $= 755.507 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 58644.1406 \text{ N-m}$

Bending Stress, $Sb = (M \cdot C1 / I) = 2841.0691 \text{ kPa}$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 129781.4062 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 * \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(381241 + 8183) \cdot 254 / (4 \cdot 2600 \cdot 147989))}$
 $= 13.886 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (1260 + 138 + 30)$
 $= 2473.369 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nrifs} - 1)$
 $= 2473 / (5 - 1)$
 $= 618.342 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 618 \cdot 254 / 2$
 $= 0.079 \text{ m}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 389424 / (2600 \cdot 254)$
 $= 589825.250 \text{ N/m}^2$

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 0.079 \cdot 589825$
 $= 46307.168 \text{ N}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 0.0032 + 0.0046$
 $= 0.008 \text{ m}^2$

Compressive Stress [Sc]:
 $= P / Ar$
 $= 46307 / 0.0079$
 $= 5889.733 \text{ kPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io
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Rib+Web	15.0	230.0	...	...	...	0.152E+08
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Rib dimension [D]:

= Saddle Width - Web Thickness
= 230 - 15
= 215.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/0.0079
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 230/2
= 115.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{15208737 / 0.0079}$
= 43.981 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1800 - cos(120/2)(1260 + 138 + 30) - 35
= 1051.000 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199955 / 194672)}$
= 142.386

Slenderness ratio [KL/r]:

= KL/r
= 1 • 1051/44
= 23.897

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 97541 / (2 • 2600) • 618 • 1051/2
= 6097.654 N-m

Compressive Allowable, KL/r < Cc (23.9 < 142) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (23.9)^2 / (2 \cdot 142^2)) 194672 /$
 $(5/3 + 3 \cdot (23.9) / (8 \cdot 142) - (23.9^3) / (8 \cdot 142^3))$
= 111006 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 5890/111006 + (6098 • 115/15208751) / 129781
= 0.408

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	215.0	0.0	0.0	0.0	0.152E+08
Web	618.3	15.0	0.0	0.0	0.0	0.174E+06
Totals	...	...	...	...	...	0.154E+08

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/0.012



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= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 230/2

= 115.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1800 - cos(1428/2)(35 + 0 + 0) - 0

= 337.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(15378426/0.012)}$

= 35.075 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 337/35.1

= 9.608

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 97541/(2 • 2600)

= 18.758 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 18.8 • 618 • 337

= 3910.389 N-m

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 3910/133726

= 29230.832 kPa

Compressive Allowable, KL/r < Cc (9.61 < 142) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (9.61)^2 / (2 \cdot 142^2)) 194672 /$

$(5/3 + 3 \cdot (9.61) / (8 \cdot 142) - (9.61^3) / (8 \cdot 142^3))$

= 114797 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 7422/114797 + (3910 • 115/15378426)/129781

= 0.290

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	137991.98 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
Nominal Bolt Diameter	Bnd	31.7500 mm
Thread Series	Series	UNC
BasePlate Allowable Stress	S	129114.48 kPa
Area Available in a Single Bolt	BltArea	0.0005 m²
Saddle Load Q0 (Weight)	Q0	282945.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	51639.5 N
Maximum Transverse Force	Ft	48770.6 N
Maximum Longitudinal Force	F1	97541.3 N



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Saddle Calcs: Operating Case: Step: 16 11:13am Apr 10, 2026

Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $97541 / (0.00045 \cdot 8)$

= 26859.7 kPa. Must be less than 82795.2 kPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $48771 / (0.00045 \cdot 8)$

= 13429.9 kPa. Must be less than 82795.2 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$

= $97541 / (82795 \cdot 8)$

= 0.0001 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1800 \cdot 48771 + 0$

= 87822.76 N-m

Eccentricity (e):

= R_{mom} / QO

= $87823 / 282945$

= 310.26 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0, 0.00015, 0]$

= 0.0001 m²

MDMT Calculations for: Right side sadd, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 30, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = -1$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit 7 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 17 11:13am Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

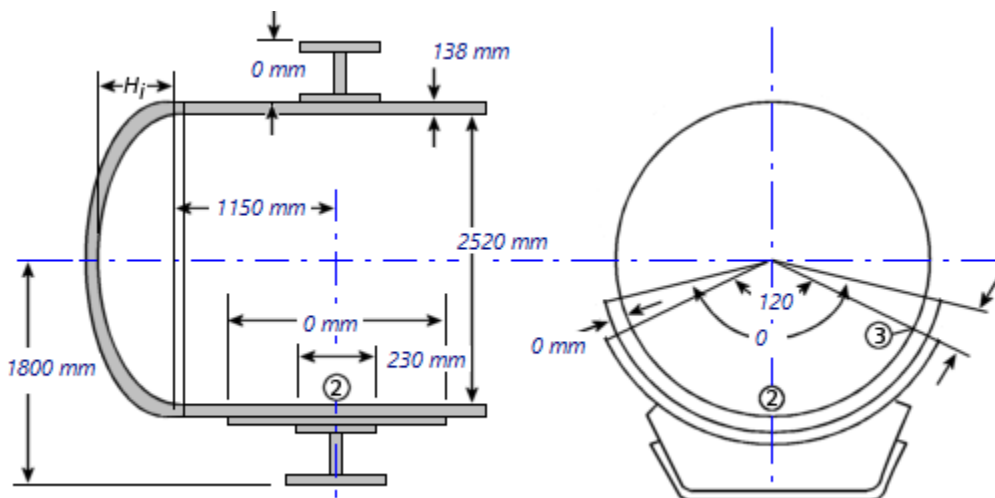
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(230 + 1.56 \cdot \sqrt{1329 \cdot 138}, 2 \cdot 1150)$$

$$= 898.0771 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1329.00	mm
Shell Thickness used in this Case	t	138.000	mm
Stiffened Vessel Length per 4.15.6	L	5.70	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	230.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.28	m
Shell Allowable Stress used in Calculation		278978.75	kPa
Head Allowable Stress used in Calculation		278978.75	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1772650.50	N
Pressure used in Analysis	P	18.578	N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top of Midspan		89552.93	278978.75
Long. Stress at Bottom of Midspan		89351.92	278978.75



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Long. Stress at Top of Saddles	98873.44	278978.75
Long. Stress at Bottom of Saddles	84230.71	278978.75
Tangential Shear in Shell	5197.64	165217.41
Circ. Stress at Horn of Saddle	14559.62	418468.09
Circ. Compressive Stress in Shell	1087.43	278978.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (6887/2 + 0) \cdot 1800/2473$$

$$= 7517.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(3725, 0) \cdot 1800/3400$$

$$= 1972.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1765132 + \max(1972, 7518, 0, 0)$$

$$= 1772650.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.8 \cdot 1.2)^2)$$

$$= 7450.694 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 1780833.75 N

Transverse Shear Load 3443.32 N

Longitudinal Shear Load 3725.15 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2520/2 + 138 + 30)\sin(120/2), 2 \cdot 1329)$$

$$= 2473.369 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0422 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$$

$$= -1772650 \cdot 1.15 [1 - (1 - 1.15/5.7 + (1.33^2 - 1.28^2)/$$

$$(2 \cdot 1.15 \cdot 5.7)/(1 + (4 \cdot 1.28)/(3 \cdot 5.7))]$$

$$= -769384.6 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$$

$$= 1772650 \cdot 5.7/4((1 + 2(1.33^2 - 1.28^2)/(5.7^2))/(1 + (4 \cdot 1.28)/$$

$$(3 \cdot 5.7)) - 4 \cdot 1.15/5.7)$$

$$= -76989.6 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 t)$$

$$= 18.6 \cdot 1329/(2 \cdot 138) - -76990/(\pi \cdot 1329^2 \cdot 138)$$

$$= 89552.93 \text{ kPa}$$



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1329 / (2 \cdot 138) + -76990 / (\pi \cdot 1329^2 \cdot 138)$$

$$= 89351.92 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1329 / (2 \cdot 138) - 769385 / (0.11 \cdot \pi \cdot 1329^2 \cdot 138)$$

$$= 98873.44 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1329 / (2 \cdot 138) + -769385 / (0.19 \cdot \pi \cdot 1329^2 \cdot 138)$$

$$= 84230.71 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1772650 (5.7 - 2 \cdot 1.15) / (5.7 + (4 \cdot 1.28 / 3))$$

$$= 814244.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 814245 / (1329 \cdot 138)$$

$$= 5197.64 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1329 \cdot 138}$$

$$= 334.039 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 1772650 \cdot 0.1 / (138 (230 + 334 + 334))$$

$$= -1087.43 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -1772650 / (4 \cdot 138 (230 + 334 + 334)) - 12 \cdot 0.042 \cdot 1772650 \cdot 1329 / (5.7 \cdot 138^2)$$

$$= -14559.62 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(230 + 1.56 \cdot \sqrt{1329 \cdot 138}, 2 \cdot 1.15)$$

$$= 898.08 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2600.0002	mm
Baseplate Thickness	Bpthk	35.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{1260 \cdot 138} + 300)^3) / 12$$

$$= 208165744.000 \text{ mm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 208165744 + 0.13 (109 - 69)^2$$

$$= 416354816.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	950.5	138.0	69.0	0.1	9050691.0	0.416E+09
Wearplate	300.0	30.0	153.0	0.0	1377000.1	0.182E+08
Web	15.0	337.0	336.5	0.0	1701007.2	0.310E+09
BasePlate	254.0	35.0	522.5	0.0	4645024.5	0.152E+10
Totals	...	...	...	0.2	16773723.0	0.227E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16773723 / 0.15$$

$$= 108.839 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1772650$$

$$= 360772.7812 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 15723.7979 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1800 - 1260 \cdot \sin(120/2) / 1.05$$

$$= 757.988 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 273572.4375 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13131.9971 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(1772650 + 8183)254) / (4 \cdot 2600 \cdot 173320)}$$

$$= 27.438 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1260 + 138 + 30)$$

$$= 2473.369 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2473 / (5 - 1)$$

$$= 618.342 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 618 \cdot 254 / 2$$

$$= 0.079 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 1780833 / (2600 \cdot 254)$$

$$= 2697267.250 \text{ N/m}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 0.079 \cdot 2697267$$

$$= 211762.391 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 0.0032 + 0.0046$$

$$= 0.008 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 211762 / 0.0079$$

$$= 26933.715 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	230.0	...	...	...	0.152E+08

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 230 - 15$$

$$= 215.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.0079$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 230 / 2$$

$$= 115.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (15208737 / 0.0079) }$$

$$= 43.981 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1800 - \cos(120 / 2) (1260 + 138 + 30) - 35$$

$$= 1051.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199955 / 259980) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1051/44$$

$$= 23.897$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 3725 / (2 \cdot 2600) \cdot 618 \cdot 1051/2$$

$$= 232.873 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($23.9 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.9)^2 / (2 \cdot 123^2)) 259980 /$$

$$(5/3 + 3 \cdot (23.9) / (8 \cdot 123) - (23.9^3) / (8 \cdot 123^3))$$

$$= 146731 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 26934/146731 + (233 \cdot 115/15208751) / 173320$$

$$= 0.194$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	215.0	0.0	0.0	0.0	0.152E+08
Web	618.3	15.0	0.0	0.0	0.0	0.174E+06
Totals	...	...	...	...	...	0.154E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/0.012$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230/2$$

$$= 115.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1800 - \cos(1428/2) (35 + 0 + 0) - 0$$

$$= 337.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (15378426/0.012) }$$

$$= 35.075 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 337/35.1$$

$$= 9.608$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3725 / (2 \cdot 2600)$$

$$= 0.716 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.72 \cdot 618 \cdot 337$$

$$= 149.340 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, base, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 149/133726
= 1116.341 kPa

Compressive Allowable, $KL/r < C_c$ ($9.61 < 123$) per AISC E2-1 [Sca]:

= $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
= $(1 - (9.61)^2 / (2 \cdot 123^2)) 259980 /$
 $(5/3 + 3 \cdot (9.61) / (8 \cdot 123) - (9.61^3) / (8 \cdot 123^3))$
= 152838 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
= $33940 / 152838 + (149 \cdot 115 / 15378426) / 173320$
= 0.229

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48296.32 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
Nominal Bolt Diameter	Bnd	31.7500 mm
Thread Series	Series	UNC
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0005 m ²
Saddle Load Q0 (Weight)	Q0	1773316.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	1972.1 N
Maximum Transverse Force	Ft	3443.3 N
Maximum Longitudinal Force	Fl	3725.2 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $3725 / (0.00045 \cdot 8)$
= 1025.8 kPa. Must be less than 28977.8 kPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $3443 / (0.00045 \cdot 8)$
= 948.2 kPa. Must be less than 28977.8 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
= $3725 / (28978 \cdot 8)$
= 0.0000 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
= $1800 \cdot 3443 + 0$
= 6200.48 N-m

Eccentricity (e):

= R_{mom} / QO
= $6200 / 1773316$
= 3.50 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:



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= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.16065E-04, 0]

= 0.0000 m²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= min($b + 1.56 \cdot \sqrt{Rm \cdot t}$, 2a)

= min(230 + $1.56 \cdot \sqrt{1329 \cdot 138}$, 2 • 1150)

= 898.0771 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1329.00	mm
Shell Thickness used in this Case	t	138.000	mm
Stiffened Vessel Length per 4.15.6	L	5.70	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	230.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.30	m
Shell Allowable Stress used in Calculation		278978.75	kPa
Head Allowable Stress used in Calculation		278978.75	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		182661.17	N
Pressure used in Analysis	P	18.578	N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top of Midspan		89464.42	278978.75
Long. Stress at Bottom of Midspan		89440.43	278978.75
Long. Stress at Top of Saddles		90435.70	278978.75
Long. Stress at Bottom of Saddles		88907.44	278978.75
Tangential Shear in Shell		533.81	165217.41
Circ. Stress at Horn of Saddle		1500.28	418468.09
Circ. Compressive Stress in Shell		112.05	278978.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (6887/2 + 0) • 1800/2473

= 7517.6 N

Longitudinal Saddle Reaction Force [Fwl]:



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Saddle Calcs: Test Case: Step: 17 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s \\
 &= \max(3725, 0) \cdot 1800/3400 \\
 &= 1972.1 \text{ N}
 \end{aligned}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st}) \\
 &= 175144 + \max(1972, 7518, 0, 0) \\
 &= 182661.2 \text{ N}
 \end{aligned}$$

Longitudinal Wind Force [F_l]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 766 (0 + \pi/4 (2.8 \cdot 1.2)^2) \\
 &= 7450.694 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	190844.53	N
Transverse Shear Load	3443.32	N
Longitudinal Shear Load	3725.15	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2 (2520/2 + 138 + 30) \sin(120/2), 2 \cdot 1329) \\
 &= 2473.369 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0422	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)] \\
 &= -182661 \cdot 1.15 [1 - (1 - 1.15/5.7 + (1.33^2 - 1.3^2) / (2 \cdot 1.15 \cdot 5.7)) / (1 + (4 \cdot 1.3)/(3 \cdot 5.7))] \\
 &= -80300.5 \text{ N-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/3L) - 4a/L) \\
 &= 182661 \cdot 5.7/4 ((1 + 2(1.33^2 - 1.3^2)/(5.7^2)) / (1 + (4 \cdot 1.3)/(3 \cdot 5.7)) - 4 \cdot 1.15/5.7) \\
 &= -9190.2 \text{ N-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t) \\
 &= 18.6 \cdot 1329 / (2 \cdot 138) - 9190 / (\pi \cdot 1329^2 \cdot 138) \\
 &= 89464.42 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 18.6 \cdot 1329 / (2 \cdot 138) + 9190 / (\pi \cdot 1329^2 \cdot 138) \\
 &= 89440.43 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t) \\
 &= 18.6 \cdot 1329 / (2 \cdot 138) - 80300 / (0.11 \cdot \pi \cdot 1329^2 \cdot 138) \\
 &= 90435.70 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 18.6 \cdot 1329 / (2 \cdot 138) + 80300 / (0.19 \cdot \pi \cdot 1329^2 \cdot 138) \\
 &= 88907.44 \text{ kPa}
 \end{aligned}$$



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Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 182661 (5.7 - 2 \cdot 1.15) / (5.7 + (4 \cdot 1.3/3))$$

$$= 83624.5 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 83624 / (1329 \cdot 138)$$

$$= 533.81 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (1329 \cdot 138) }$$

$$= 334.039 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t (b + X_1 + X_2))$$

$$= -0.76 \cdot 182661 \cdot 0.1 / (138 (230 + 334 + 334))$$

$$= -112.05 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t (b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -182661 / (4 \cdot 138 (230 + 334 + 334)) -$$

$$12 \cdot 0.042 \cdot 182661 \cdot 1329 / (5.7 \cdot 138^2)$$

$$= -1500.28 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(230 + 1.56 \cdot \sqrt{ (1329 \cdot 138) }, 2 \cdot 1.15)$$

$$= 898.08 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2600.0002	mm
Baseplate Thickness	Bpthk	35.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (1260 \cdot 138) } + 300) 138^3) / 12$$

$$= 208165744.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 208165744 + 0.13 (109 - 69)^2$$

$$= 416354816.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	950.5	138.0	69.0	0.1	9050691.0	0.416E+09
Wearplate	300.0	30.0	153.0	0.0	1377000.1	0.182E+08
Web	15.0	337.0	336.5	0.0	1701007.2	0.310E+09
BasePlate	254.0	35.0	522.5	0.0	4645024.5	0.152E+10
Totals	...	...	...	0.2	16773723.0	0.227E+10



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 16773723 / 0.15$$

$$= 108.839 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 182661$$

$$= 37175.5078 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 1620.2445 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta / 2) / (\theta / 2 \text{ in radians})$$

$$= 1800 - 1260 \cdot \sin(120 / 2) / 1.05$$

$$= 757.988 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 28190.0234 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1353.1748 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(182661 + 8183) \cdot 254) / (4 \cdot 2600 \cdot 173320)}$$

$$= 8.982 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (1260 + 138 + 30)$$

$$= 2473.369 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2473 / (5 - 1)$$

$$= 618.342 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 618 \cdot 254 / 2$$

$$= 0.079 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 190845 / (2600 \cdot 254)$$

$$= 289054.906 \text{ N/m}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 0.079 \cdot 289055$$

$$= 22693.693 \text{ N}$$



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Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 0.0032 + 0.0046 \\
 &= 0.008 \text{ m}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/Ar \\
 &= 22694/0.0079 \\
 &= 2886.374 \text{ kPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	230.0	...	...	...	0.152E+08

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 230 - 15 \\
 &= 215.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/0.0079 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 230/2 \\
 &= 115.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(15208737/0.0079)} \\
 &= 43.981 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1800 - \cos(120/2) (1260 + 138 + 30) - 35 \\
 &= 1051.000 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 199955/259980)} \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1051/44 \\
 &= 23.897
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2 \\
 &= 3725 / (2 \cdot 2600) \cdot 618 \cdot 1051/2 \\
 &= 232.873 \text{ N-m}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($23.9 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (23.9)^2 / (2 \cdot 123^2)) 259980 / \\
 &\quad (5/3 + 3 \cdot (23.9) / (8 \cdot 123) - (23.9)^3 / (8 \cdot 123^3)) \\
 &= 146731 \text{ kPa}
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)



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$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 2886/146731 + (233 \cdot 115/15208751) / 173320 \\
 &= 0.030
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	215.0	0.0	0.0	0.0	0.152E+08
Web	618.3	15.0	0.0	0.0	0.0	0.174E+06
Totals	...	...	...	...	...	0.154E+08

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/0.012 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 230/2 \\
 &= 115.000 \text{ mm}
 \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \text{Outside Radius} - Bpthk \\
 &= 1800 - \cos(1428/2)(35 + 0 + 0) - 0 \\
 &= 337.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\
 &= \sqrt{15378426/0.012} \\
 &= 35.075 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 337/35.1 \\
 &= 9.608
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 3725 / (2 \cdot 2600) \\
 &= 0.716 \text{ N/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.72 \cdot 618 \cdot 337 \\
 &= 149.340 \text{ N-m}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 149/133726 \\
 &= 1116.341 \text{ kPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($9.61 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (9.61)^2 / (2 \cdot 123^2)) 259980 / \\
 &\quad (5/3 + 3 \cdot (9.61) / (8 \cdot 123) - (9.61^3) / (8 \cdot 123^3)) \\
 &= 152838 \text{ kPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Mbase,c \cdot C1/I) / Sba \\
 &= 3637/152838 + (149 \cdot 115/15378426) / 173320 \\
 &= 0.030
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:



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Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48296.32 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
Nominal Bolt Diameter	Bnd	31.7500 mm
Thread Series	Series	UNC
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0005 m ²
Saddle Load Q0 (Weight)	Q0	183326.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	1972.1 N
Maximum Transverse Force	Ft	3443.3 N
Maximum Longitudinal Force	Fl	3725.2 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3725 / (0.00045 \cdot 8)$$

$$= 1025.8 \text{ kPa. Must be less than } 28977.8 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3443 / (0.00045 \cdot 8)$$

$$= 948.2 \text{ kPa. Must be less than } 28977.8 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3725 / (28978 \cdot 8)$$

$$= 0.0000 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1800 \cdot 3443 + 0$$

$$= 6200.48 \text{ N-m}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 6200 / 183327$$

$$= 33.81 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.16065E-04, 0]$$

$$= 0.0000 \text{ m}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Ext. Pressure: Step: 18 11:13am Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

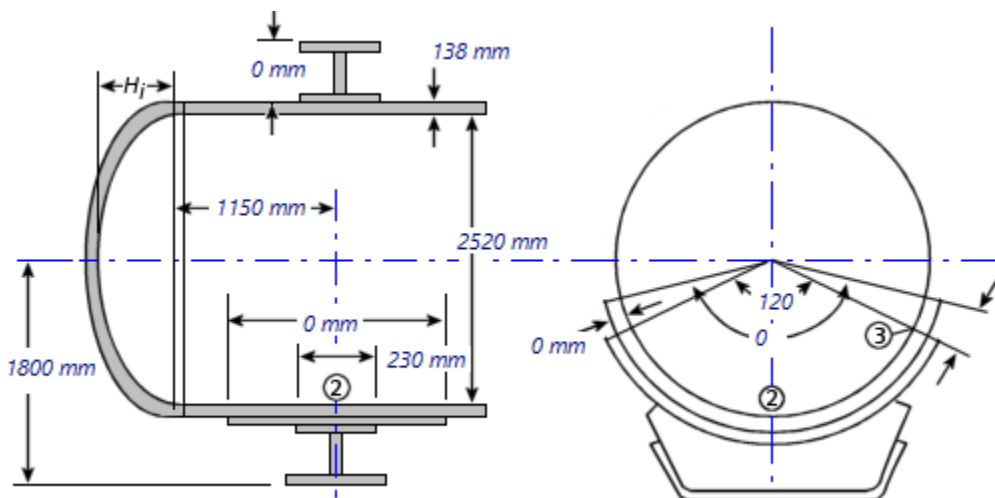
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(230 + 1.56 \cdot \sqrt{1330 \cdot 135}, 2 \cdot 1150)$$

$$= 891.1483 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1330.50	mm
Shell Thickness used in this Case	t	138.000	mm
Stiffened Vessel Length per 4.15.6	L	5.70	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	230.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.28	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		116013.23	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	1490582.62	N
Pressure used in Analysis	P	-0.101 N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top	of Midspan	-412.42	-215195.78
Long. Stress at Top	of Midspan	412.42	147988.73



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Long. Stress at Bottom of Midspan	-585.95	-215195.78
Long. Stress at Bottom of Midspan	585.95	147988.73
Long. Stress at Top of Saddles	7583.94	147988.73
Long. Stress at Bottom of Saddles	-4979.35	-215195.78
Long. Stress at Bottom of Saddles	4979.35	147988.73
Tangential Shear in Shell	4461.46	118390.98
Circ. Stress at Horn of Saddle	12741.81	184985.91
Circ. Compressive Stress in Shell	941.99	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (13773/2 + 0) \cdot 1800/2473$$

$$= 15035.3 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(7450, 0) \cdot 1800/3400$$

$$= 3944.3 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(97541, 0) \cdot 1800/3400$$

$$= 51639.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (97541/2 + 0) \cdot 1800/2473$$

$$= 106478.9 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1384103 + \max(3944, 15035, 51640, 106479)$$

$$= 1490582.6 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.8 \cdot 1.2)^2)$$

$$= 7450.694 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1498766.00	N
Transverse Shear Load	48770.65	N
Longitudinal Shear Load	97541.30	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2520/2 + 138 + 30)\sin(120/2), 2 \cdot 1330)$$

$$= 2473.369 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0421	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -1490582 \cdot 1.15 [1 - (1 - 1.15/5.7 + (1.33^2 - 1.28^2)/$$



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$$\begin{aligned} & (2 \cdot 1.15 \cdot 5.7) / (1 + (4 \cdot 1.28) / (3 \cdot 5.7)) \\ & = -647231.1 \text{ N-m} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} & = Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ & = 1490582 \cdot 5.7 / 4 \cdot ((1 + 2(1.33^2 - 1.28^2) / (5.7^2)) / (1 + (4 \cdot 1.28) / (3 \cdot 5.7)) - 4 \cdot 1.15 / 5.7) \\ & = -65168.7 \text{ N-m} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} & = P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ & = -0.1 \cdot 1330 / (2 \cdot 135) - 65169 / (\pi \cdot 1330^2 \cdot 135) \\ & = -412.42 \text{ kPa} \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} & = P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ & = -0.1 \cdot 1330 / (2 \cdot 135) + 65169 / (\pi \cdot 1330^2 \cdot 135) \\ & = -585.95 \text{ kPa} \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned} & = P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ & = -0.1 \cdot 1330 / (2 \cdot 135) - 647231 / (0.11 \cdot \pi \cdot 1330^2 \cdot 135) \\ & = 7583.94 \text{ kPa} \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned} & = P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ & = -0.1 \cdot 1330 / (2 \cdot 135) + 647231 / (0.19 \cdot \pi \cdot 1330^2 \cdot 135) \\ & = -4979.35 \text{ kPa} \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} & = Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ & = 1490582 \cdot (5.7 - 2 \cdot 1.15) / (5.7 + (4 \cdot 1.28 / 3)) \\ & = 684495.6 \text{ N} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} & = K2 \cdot T / (Rm \cdot t) \\ & = 1.17 \cdot 684496 / (1330 \cdot 135) \\ & = 4461.46 \text{ kPa} \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned} & = 0.78 \cdot \sqrt{Rm \cdot t} \\ & = 0.78 \cdot \sqrt{1330 \cdot 135} \\ & = 330.574 \text{ mm} \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned} & = -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ & = -0.76 \cdot 1490582 \cdot 0.1 / (135(230 + 331 + 331)) \\ & = -941.99 \text{ kPa} \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{7*}]:

$$\begin{aligned} & = -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2) \\ & = -1490582 / (4 \cdot 135(230 + 331 + 331)) - 12 \cdot 0.042 \cdot 1490582 \cdot 1330 / (5.7 \cdot 135^2) \\ & = -12741.81 \text{ kPa} \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} & = \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\ & = \min(230 + 1.56 \cdot \sqrt{1330 \cdot 135}, 2 \cdot 1.15) \\ & = 891.15 \text{ mm} \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2600.0002	mm
Baseplate Thickness	Bpthk	35.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	



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Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1263 \cdot 135) + 300})^3) / 12$$

$$= 193581952.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 193581952 + 0.13 (108 - 67.5)^2$$

$$= 403608928.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	944.2	135.0	67.5	0.1	8603650.0	0.404E+09
Wearplate	300.0	30.0	150.0	0.0	1350000.0	0.165E+08
Web	15.0	337.0	333.5	0.0	1685842.1	0.305E+09
BasePlate	254.0	35.0	519.5	0.0	4618354.5	0.151E+10
Totals	...	...	...	0.2	16257846.0	0.223E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16257846 / 0.15$$

$$= 108.093 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1490582$$

$$= 303365.9062 \text{ N}$$

Tension Stress, St = (Fh/As)	=	13221.7959	kPa
Allowed Stress, Sa = 0.6 * Yield Str	=	116803.2656	kPa

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1800 - 1263 \cdot \sin(120/2) / 1.05$$

$$= 755.507 \text{ mm}$$

Bending Moment, M = Fh * d	=	229288.0938	N-m
----------------------------	---	-------------	-----

Bending Stress, Sb = (M * C1 / I)	=	11108.0703	kPa
Allowed Stress, Sa = 2/3 * Yield Str	=	129781.4062	kPa

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used



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to determine the baseplate thickness.

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(1490582 + 8183)254}{4 \cdot 2600 \cdot 147989}}$$

$$= 27.241 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1260 + 138 + 30)$$

$$= 2473.369 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2473 / (5 - 1)$$

$$= 618.342 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 618 \cdot 254 / 2$$

$$= 0.079 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1498766 / (2600 \cdot 254)$$

$$= 2270044.750 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.079 \cdot 2270044$$

$$= 178221.156 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0032 + 0.0046$$

$$= 0.008 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 178221 / 0.0079$$

$$= 22667.658 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	230.0	...	...	...	0.152E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 230 - 15$$

$$= 215.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 0.0079$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230 / 2$$

$$= 115.000 \text{ mm}$$



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Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{15208737 / 0.0079}$$

$$= 43.981 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1800 - \cos(120/2) (1260 + 138 + 30) - 35$$

$$= 1051.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955 / 194672}$$

$$= 142.386$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1051 / 44$$

$$= 23.897$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 97541 / (2 \cdot 2600) \cdot 618 \cdot 1051 / 2$$

$$= 6097.654 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($23.9 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.9)^2 / (2 \cdot 142^2)) 194672 /$$

$$(5/3 + 3 \cdot (23.9) / (8 \cdot 142) - (23.9)^3 / (8 \cdot 142^3))$$

$$= 111006 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 22668 / 111006 + (6098 \cdot 115 / 15208751) / 129781$$

$$= 0.559$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	215.0	0.0	0.0	0.0	0.152E+08
Web	618.3	15.0	0.0	0.0	0.0	0.174E+06
Totals	...	...	...	...	...	0.154E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.012$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230 / 2$$

$$= 115.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1800 - \cos(1428/2) (35 + 0 + 0) - 0$$

$$= 337.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{15378426 / 0.012}$$

$$= 35.075 \text{ mm}$$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 337/35.1$$

$$= 9.608$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 97541 / (2 \cdot 2600)$$

$$= 18.758 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 18.8 \cdot 618 \cdot 337$$

$$= 3910.389 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3910/133726$$

$$= 29230.832 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($9.61 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.61)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (9.61) / (8 \cdot 142) - (9.61^3) / (8 \cdot 142^3))$$

$$= 114797 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 28564/114797 + (3910 \cdot 115/15378426) / 129781$$

$$= 0.474$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(230 + 1.56 \cdot \sqrt{1330 \cdot 135}, 2 \cdot 1150)$$

$$= 891.1483 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1330.50	mm
Shell Thickness used in this Case	t	138.000	mm
Stiffened Vessel Length per 4.15.6	L	5.70	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	230.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.30	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		147988.73	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, External Pressure Case 381240.66 N
Pressure used in Analysis P -0.101 N/mm²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	-473.50	-215195.78
Long. Stress at Top of Midspan	473.50	147988.73
Long. Stress at Bottom of Midspan	-524.87	-215195.78
Long. Stress at Bottom of Midspan	524.87	147988.73
Long. Stress at Top of Saddles	1594.80	147988.73
Long. Stress at Bottom of Saddles	-1659.80	-215195.78
Long. Stress at Bottom of Saddles	1659.80	147988.73
Tangential Shear in Shell	1137.30	118390.98
Circ. Stress at Horn of Saddle	3258.92	184985.91
Circ. Compressive Stress in Shell	240.93	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (13773/2 + 0) \cdot 1800/2473$$

$$= 15035.3 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(7450, 0) \cdot 1800/3400$$

$$= 3944.3 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(97541, 0) \cdot 1800/3400$$

$$= 51639.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (97541/2 + 0) \cdot 1800/2473$$

$$= 106478.9 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 274762 + \max(3944, 15035, 51640, 106479)$$

$$= 381240.7 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.8 \cdot 1.2)^2)$$

$$= 7450.694 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	389424.03	N
Transverse Shear Load	48770.65	N
Longitudinal Shear Load	97541.30	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + \text{t} + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2520/2 + 138 + 30) \sin(120/2), 2 \cdot 1330)$$

$$= 2473.369 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011



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K5 = 0.7603 K6 = 0.0529 K7 = 0.0421 K8 = 0.3405
K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / (1 + (4hm)/3L) \right]$$

$$= -381241 \cdot 1.15 \left[1 - \left(1 - \frac{1.15}{5.7} + \frac{(1.33^2 - 1.3^2)}{(2 \cdot 1.15 \cdot 5.7)} \right) / (1 + (4 \cdot 1.3)/(3 \cdot 5.7)) \right]$$

$$= -167669.2 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 5.7)} \right) - 4 \cdot 1.15 / 5.7$$

$$= 381241 \cdot 5.7 / 4 \left(\frac{(1 + 2(1.33^2 - 1.3^2)/(5.7^2)) / (1 + (4 \cdot 1.3)/(3 \cdot 5.7)) - 4 \cdot 1.15 / 5.7}{(3 \cdot 5.7)} \right) - 4 \cdot 1.15 / 5.7$$

$$= -19291.6 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -0.1 \cdot 1330 / (2 \cdot 135) - 19292 / (\pi \cdot 1330^2 \cdot 135)$$

$$= -473.50 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1330 / (2 \cdot 135) + 19292 / (\pi \cdot 1330^2 \cdot 135)$$

$$= -524.87 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -0.1 \cdot 1330 / (2 \cdot 135) - 167669 / (0.11 \cdot \pi \cdot 1330^2 \cdot 135)$$

$$= 1594.80 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1330 / (2 \cdot 135) + 167669 / (0.19 \cdot \pi \cdot 1330^2 \cdot 135)$$

$$= -1659.80 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 381241 (5.7 - 2 \cdot 1.15) / (5.7 + (4 \cdot 1.3 / 3))$$

$$= 174489.5 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 174490 / (1330 \cdot 135)$$

$$= 1137.30 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1330 \cdot 135)}$$

$$= 330.574 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 381241 \cdot 0.1 / (135(230 + 331 + 331))$$

$$= -240.93 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, L < 8Rm (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -381241 / (4 \cdot 135(230 + 331 + 331)) - 12 \cdot 0.042 \cdot 381241 \cdot 1330 / (5.7 \cdot 135^2)$$

$$= -3258.92 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$



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$$= \min(230 + 1.56 \cdot \sqrt{(1330 \cdot 135)}, 2 \cdot 1.15)$$

$$= 891.15 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2600.0002	mm
Baseplate Thickness	Bpthk	35.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3)/12$$

$$= ((1.56 \cdot \sqrt{(1263 \cdot 135)} + 300)^3)/12$$

$$= 193581952.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 193581952 + 0.13(108 - 67.5)^2$$

$$= 403608928.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	944.2	135.0	67.5	0.1	8603650.0	0.404E+09
Wearplate	300.0	30.0	150.0	0.0	1350000.0	0.165E+08
Web	15.0	337.0	333.5	0.0	1685842.1	0.305E+09
BasePlate	254.0	35.0	519.5	0.0	4618354.5	0.151E+10
Totals	...	...	...	0.2	16257846.0	0.223E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16257846/0.15$$

$$= 108.093 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2)/(\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 381241$$

$$= 77590.7344 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 3381.6882 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians})$$

$$= 1800 - 1263 \cdot \sin(120/2)/1.05$$

$$= 755.507 \text{ mm}$$



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Saddle Calcs: Ext. Pressure: Step: 18 11:13am Apr 10,2026

$$\text{Bending Moment, } M = F_h \cdot d = 58644.1406 \text{ N-m}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 2841.0691 \text{ kPa}$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{(3(381241 + 8183) \cdot 254) / (4 \cdot 2600 \cdot 147989)} \\ &= 13.886 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (1260 + 138 + 30) \\ &= 2473.369 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 2473 / (5 - 1) \\ &= 618.342 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 618 \cdot 254 / 2 \\ &= 0.079 \text{ m}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 389424 / (2600 \cdot 254) \\ &= 589825.250 \text{ N/m}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= A_p \cdot B_p \\ &= 0.079 \cdot 589825 \\ &= 46307.168 \text{ N} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 0.0032 + 0.0046 \\ &= 0.008 \text{ m}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / A_r \\ &= 46307 / 0.0079 \\ &= 5889.733 \text{ kPa} \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	230.0	...	...	...	0.152E+08

Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 230 - 15 \\ &= 215.000 \text{ mm} \end{aligned}$$



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Saddle Calcs: Ext. Pressure: Step: 18 11:13am Apr 10,2026

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/0.0079$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230/2$$

$$= 115.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(15208737/0.0079)}$$

$$= 43.981 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1800 - \cos(120/2) (1260 + 138 + 30) - 35$$

$$= 1051.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/194672)}$$

$$= 142.386$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1051/44$$

$$= 23.897$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 97541 / (2 \cdot 2600) \cdot 618 \cdot 1051/2$$

$$= 6097.654 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($23.9 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.9)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (23.9) / (8 \cdot 142) - (23.9)^3 / (8 \cdot 142^3))$$

$$= 111006 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5890/111006 + (6098 \cdot 115/15208751) / 129781$$

$$= 0.408$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	215.0	0.0	0.0	0.0	0.152E+08
Web	618.3	15.0	0.0	0.0	0.0	0.174E+06
Totals	...	...	...	...	...	0.154E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/0.012$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230/2$$

$$= 115.000 \text{ mm}$$



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Saddle Calcs: Ext. Pressure: Step: 18 11:13am Apr 10, 2026

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1800 - \cos(1428/2)(35 + 0 + 0) - 0$$

$$= 337.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{15378426 / 0.012}$$

$$= 35.075 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 337 / 35.1$$

$$= 9.608$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 97541 / (2 \cdot 2600)$$

$$= 18.758 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 18.8 \cdot 618 \cdot 337$$

$$= 3910.389 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3910 / 133726$$

$$= 29230.832 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($9.61 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.61)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (9.61) / (8 \cdot 142) - (9.61^3) / (8 \cdot 142^3))$$

$$= 114797 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 7422 / 114797 + (3910 \cdot 115 / 15378426) / 129781$$

$$= 0.290$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: Manhole

Nozl: 33 11:13am Apr 10,2026

Input, Nozzle Desc: Manhole

From: 10

Pressure for Reinforcement Calculations	P	5.2510	N/mm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	116013.23	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Hemispherical Head	D	2520.00	mm
Head Finished (Minimum) Thickness	t	33.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22 +IT	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115993.27	kPa
Allowable Stress At Ambient	Sna	147991.42	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.6100	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	70.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	396.9000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		59.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



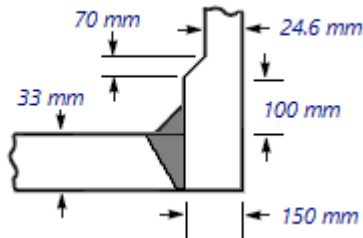
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Nozzle Calcs.: Manhole

Nozl: 33 11:13am Apr 10, 2026



Reinforcement CALCULATION, Description: Manhole

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.969 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 5.25 \cdot 1263 / (2 \cdot 116013 \cdot 1 - 0.2 \cdot 5.25)$
 $= 28.7111 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.25 \cdot 308 / (115993 \cdot 1 - 0.6 \cdot 5.25)$
 $= 14.3221 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 125 / 0.5773$
 $= 217.1819 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001 mm
Parallel to Vessel Wall, opening length	d	615.6000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	75.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115993 / 116013)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115993 / 116013)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



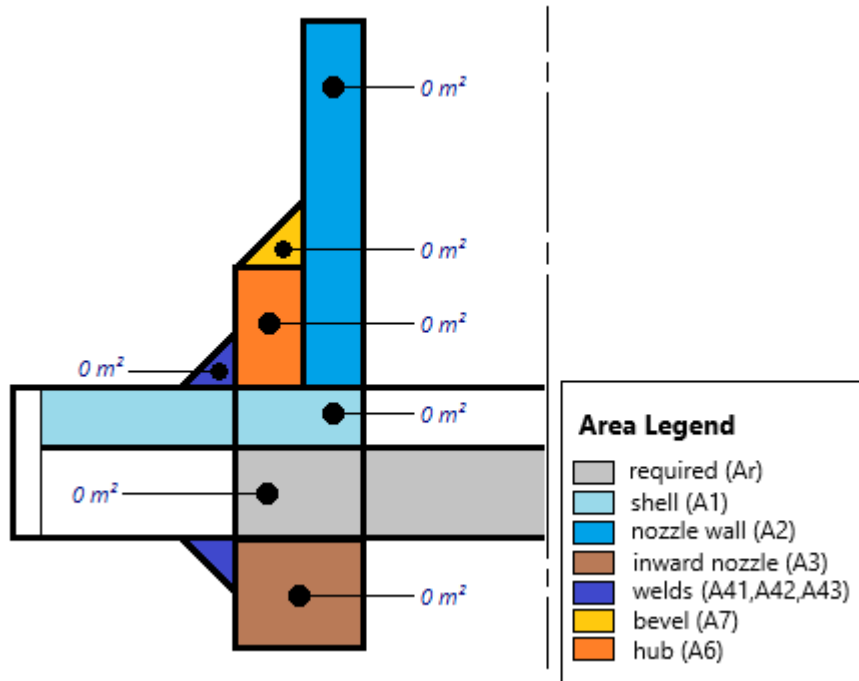
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Nozzle Calcs.: Manhole

Noz1: 33 11:13am Apr 10,2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.018	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	0.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.019	NA	NA
TOTAL AREA AVAILABLE	Atot	0.021	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (616 \cdot 28.7 \cdot 1 + 2 \cdot 147 \cdot 28.7 \cdot 1 \cdot (1 - 1))$$

$$= 0.018 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 616(1 \cdot 30 - 1 \cdot 28.7) - 2 \cdot 147$$

$$(1 \cdot 30 - 1 \cdot 28.7) \cdot (1 - 1)$$

$$= 0.001 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 75)(21.6 - 14.3)1$$

$$= 0.001 \text{ m}^2$$



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Nozzle Calcs.: Manhole

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(75, 397, 100)) \cdot (150 - 24.6) \cdot 1$$

$$= 0.019 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.3221 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 31.7111 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 31.7111 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(31.7, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.3, 11.3)$$

$$= 17.3221 \text{ mm}$$

Available Nozzle Neck Thickness = 24.6100 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 24.6, tr = 11.2, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.52, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-12 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 33, tr = 22.5, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.75, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	12 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 12 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 5.25/15.31 = 0.343

Weld Size Calculations, Description: Manhole

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



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Nozzle Calcs.: Manhole

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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (0.018 - 0.00079 + 2 \cdot 147 \cdot 1 \cdot (1 \cdot 30 - 28.7)) \cdot 116013)$$

$$= 2002506.25 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (0.02 + 0.99958E-04 - 0 \cdot 1) \cdot 116013$$

$$= 2319997.25 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (0.02 + 0 + 0.99958E-04 + (0.0088)) \cdot 116013$$

$$= 3343030.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (0.02 + 0 + 0.99958E-04 + 0 + (0.0088)) \cdot 116013$$

$$= 3343030.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 910 \cdot 10 \cdot 0.49 \cdot 115993$$

$$= 812059. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 381) \cdot (150 - 3) \cdot 0.7 \cdot 115993$$

$$= 14297270. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 910 \cdot (33 - 3) \cdot 0.74 \cdot 116013$$

$$= 3679757. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (812059 + 14297270) = 15109329 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (812059 + 0 + 3679757 + 0) = 4491816 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (812059 + 3679757 + 0) = 4491816 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 15109329N	, must exceed W = 2002506 N	or W1 = 2319997 N
Path 2-2 = 4491816 N	, must exceed W = 2002506 N	or W2 = 3343030 N
Path 3-3 = 4491816 N	, must exceed W = 2002506 N	or W3 = 3343030 N



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Nozzle Calcs.: Manhole

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Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5.49 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 84.9438 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 514.8439 mm

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Nozzle Calcs.: N1- Tube inlet

Nozl: 34 11:13am Apr 10,2026

Input, Nozzle Desc: N1- Tube inlet

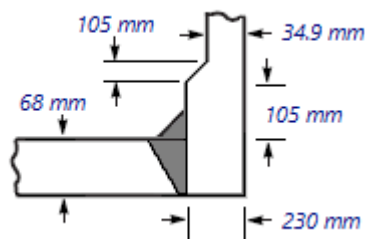
From: 20

Pressure for Reinforcement Calculations	P	5.2510	N/mm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	116013.23	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2520.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1.10	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22 +IT	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115993.27	kPa
Allowable Stress At Ambient	Sna	147991.42	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	34.9250	mm
Hub Height of Integral Nozzle	h	105.0000	mm
Height of Beveled Transition	L'	105.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		230.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N1- Tube inlet



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Nozzle Calcs.: N1- Tube inlet Nozl: 34 11:13am Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 1.375 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5.25 \cdot 1263 / (116013 \cdot 1 - 0.6 \cdot 5.25)$
= 58.7579 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5.25 \cdot 460 / (115993 \cdot 1 - 0.6 \cdot 5.25)$
= 21.4134 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 195/0.5773
= 337.8800 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1840.8000	mm
Parallel to Vessel Wall, opening length	d	920.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	162.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 115993/116013)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 115993/116013)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



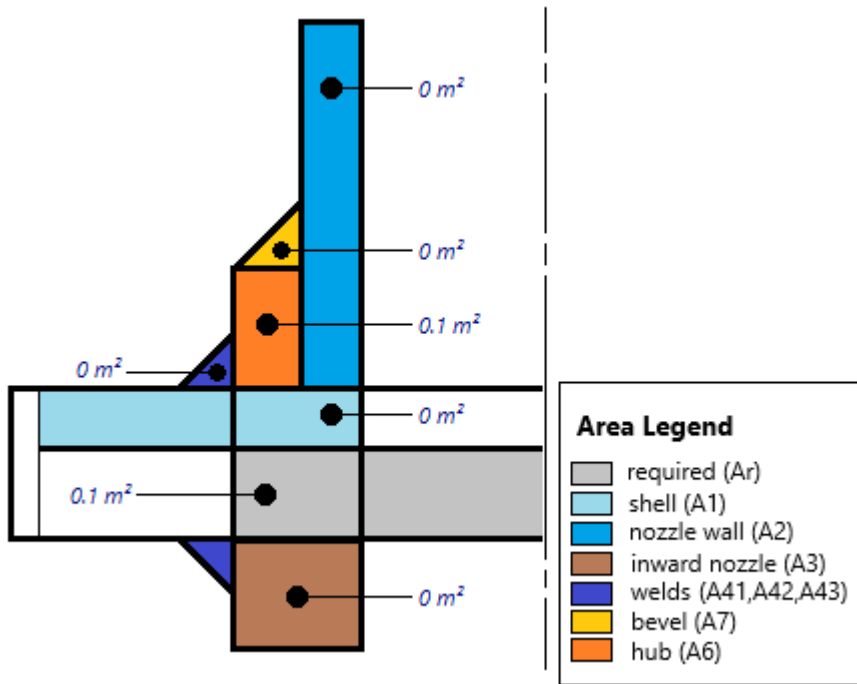
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Nozzle Calcs.: N1- Tube inlet

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.054	NA	NA
Area in Shell	A1	0.006	NA	NA
Area in Nozzle Wall	A2	0.003	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.057	NA	NA
TOTAL AREA AVAILABLE	Atot	0.067	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (920 \cdot 58.8 \cdot 1 + 2 \cdot 227 \cdot 58.8 \cdot 1 \cdot (1 - 1))$$

$$= 0.054 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 920(1 \cdot 65 - 1 \cdot 58.8) - 2 \cdot 227$$

$$(1 \cdot 65 - 1 \cdot 58.8) \cdot (1 - 1)$$

$$= 0.006 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 163)(31.9 - 21.4)1$$

$$= 0.003 \text{ m}^2$$



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Nozzle Calcs.: N1- Tube inlet Nozl: 34 11:13am Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(163, 600, 105)) \cdot (230 - 34.9) \cdot 1$$

$$= 0.057 \text{ m}^2 \text{ Includes Bevel Area of } 0.016 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 24.4134 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 61.7579 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 61.7579 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(61.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(24.4, 11.3)$$

$$= 24.4134 \text{ mm}$$

Available Nozzle Neck Thickness = 34.9250 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48582.3,	Allowable	: 115993.3 kPa	Passed
Expansion	: 0.0,	Allowable	: 281398.5 kPa	Passed
Occasional	: 36575.6,	Allowable	: 154271.0 kPa	Passed
Shear	: 7924.1,	Allowable	: 81195.3 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 68, tr = 45.8, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.7, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	26 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 26 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 195 / 0.5773$$

$$= 337.8800 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]



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Nozzle Calcs.: N1- Tube inlet Nozl: 34 11:13am Apr 10,2026

Parallel to Vessel Wall (Diameter Limit) D1 1380.6000 mm
 Parallel to Vessel Wall, opening length d 690.3000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 162.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (m²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		0.036	0.036
Area Available in Shell A1		0.003	0.003
Area Available in Nozzle Wall A2		0.003	0.003
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.000	0.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		0.057	0.057
TOTAL AREA AVAILABLE Atot		0.064	0.064

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N1- Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (0.054 - 0.0057 + 2 \cdot 227 \cdot 1 \cdot (1 \cdot 65 - 58.8)) \cdot 116013)$
 $= 5936703.50 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (0.061 + 0.00022 - 0 \cdot 1) \cdot 116013$
 $= 7063580.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (0.061 + 0 + 0.00022 + (0.029)) \cdot 116013$
 $= 10486449.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (0.061 + 0 + 0.00022 + 0 + (0.029)) \cdot 116013$
 $= 10486449.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 1374 \cdot 15 \cdot 0.49 \cdot 115993$
 $= 1840524. \text{ N}$



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Nozzle Calcs.: N1- Tube inlet Nozl: 34 11:13am Apr 10,2026

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 574) \cdot (230 - 3) \cdot 0.7 \cdot 115993$$

$$= 33218478. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1374 \cdot (68 - 3) \cdot 0.74 \cdot 116013$$

$$= 12046863. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (1840523 + 33218478) = 35059004 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (1840523 + 0 + 12046863 + 0) = 13887386 N$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (1840523 + 12046863 + 0) = 13887386 N$$

Summary of Failure Path Calculations:

Path 1-1 = 35059004N , must exceed W = 5936703 N or W1 = 7063580 N
 Path 2-2 = 13887386N , must exceed W = 5936703 N or W2 = 10486449 N
 Path 3-3 = 13887386N , must exceed W = 5936703 N or W3 = 10486449 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5.79 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 203.8958 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 871.8958 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 920.40 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N1- Tube inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 116013.23	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	230.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F22	
Nozzle Cold S.I. Allowable	SNmc 147991.42	kPa



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Nozzle Calcs.: N1- Tube inlet

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Nozzle Hot S.I. Allowable SNmh 115993.27 kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	5.25	N/mm ²
Radial Load	P	-94499.98	N
Circumferential Shear	Vc	72000.22	N
Longitudinal Shear	Vl	72000.22	N
Circumferential Moment	Mc	202499.66	N-m
Longitudinal Moment	Ml	202499.66	N-m
Torsional Moment	Mt	243000.12	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.517) is > 0.5
check the Limitations of WRC bulletin 297.
Warning - The ratio, Dn/Tn (5.976) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{R_{mean}(t - ca)}$
= 1374.4 + 2 • 1.65 • $\sqrt{1295.5(68.0 - 3.0)}$
= 2332.011 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2591.000 mm
Nozzle Outside Diameter	(d) =	1374.400 mm
Vessel Thickness used	(T) =	65.000 mm
Nozzle Thickness used	(t) =	227.000 mm
T / t	=	0.286
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.349$$

Nr/P	=	0.056
Mr/P	=	0.095
M0/P	=	0.030
N0/P	=	0.066
MrD/Mc	=	0.261
NrDL/Mc	=	0.093
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.080
NrDL/Ml	=	0.054
M0D/Ml	=	0.023
N0DL/Ml	=	0.075

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1475	1475	1475	1475
Outplane Bending (P)	4047	-4047	4047	-4047
Outplane Membrane (Mc)	0	0	0	0



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Nozzle Calcs.: N1- Tube inlet Nozl: 34 11:13am Apr 10,2026

Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2606	-2606	2606	2606
Outplane Bending (ML)	-4840	4840	4840	-4840
Normal Pressure Stress	99461	104715	99461	104715

Outplane Stress Summary	97537	104377	112430	99909
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1254	1254	1254	1254
Inplane Bending (P)	12810	-12810	12810	-12810
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1889	-1889	1889	1889
Inplane Bending (ML)	-16829	16829	16829	-16829
Inplane Pressure Stress	49730	49730	49730	49730

Inplane Stress Summary	45076	53115	82514	23234
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	510	510	-510	-510
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1254	1254	1254	1254

Shear Stress Summary	1765	1765	744	744
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	97593	104432	112444	99916
------------------------	-------	--------	--------	-------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1475	1475	1475	1475
Outplane Bending (P)	4047	-4047	4047	-4047
Outplane Membrane (Mc)	-2985	-2985	2985	2985
Outplane Bending (Mc)	-16395	16395	16395	-16395
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	49730	49730	49730	49730

Outplane Stress Summary	35872	60569	74634	33749
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1254	1254	1254	1254
Inplane Bending (P)	12810	-12810	12810	-12810
Inplane Membrane (Mc)	-3247	-3247	3247	3247
Inplane Bending (Mc)	-54639	54639	54639	-54639
Inplane Membrane (ML)	0	0	0	0



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Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	99461	104715	99461	104715

Inplane Stress Summary	55639	144552	171413	41767
------------------------	-------	--------	--------	-------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-510	-510	510	510
Torsional Shear (Mt)	1254	1254	1254	1254

Shear Stress Summary	744	744	1765	1765
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	55667	144552	171441	42132
------------------------	-------	--------	--------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	1475	1475	1475	1475
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2606	-2606	2606	2606
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10245	16547	10245	16547

Hoop Stress Summary	9114	15416	14326	20628
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	110	110	110	110
Axial Bending (P)	737	-737	737	-737
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-992	-992	992	992
Axial Bending (ML)	-910	910	910	-910
Axial Pressure Stress	5122	5122	5122	5122

Axial Stress Summary	4067	4412	7873	4578
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	144	144	-144	-144
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	358	358	358	358

Shear Stress Summary	503	503	213	213
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stress Intensities	Outside	Inside	Outside	Inside
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Two * Max Shear Stress	9162	15437	14333	20628
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	1475	1475	1475	1475
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2985	-2985	2985	2985
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10245	16547	10245	16547

Hoop Stress Summary	8735	15037	14706	21007
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	110	110	110	110
Axial Bending (P)	737	-737	737	-737
Axial Membrane (Mc)	-992	-992	992	992
Axial Bending (Mc)	-3674	3674	3674	-3674
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5122	5122	5122	5122

Axial Stress Summary	1303	7177	10638	1813
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-144	-144	144	144
Shear due to Torsion	358	358	358	358

Shear Stress Summary	213	213	503	503
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	8735	15037	14761	21014
------------------------	------	-------	-------	-------

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	99461	104715	99461	104715	99461	104715	99461	104715
Circ. Pl (SUS)	-1131	-1131	4081	4081	-1993	-1993	4501	4501
Circ. Q (SUS)	-793	793	8887	-8887	-41829	41829	67449	-67449
Long. Pm (SUS)	49730	49730	49730	49730	49730	49730	49730	49730
Long. Pl (SUS)	-635	-635	3143	3143	-1510	-1510	4460	4460
Long. Q (SUS)	-4019	4019	29639	-29639	-12348	12348	20442	-20442



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Nozzle Calcs.: N1- Tube inlet

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	510	510	-510	-510	-510	-510	510	510
Shear Q (SUS)	1254	1254	1254	1254	1254	1254	1254	1254
Pm (SUS)	99461	104715	99461	104715	99461	104715	99461	104715
Pm+P1 (SUS)	98335	103588	103547	108800	97473	102726	103967	109220
Pm+P1+Q (Total)	97596	104437	112447	99916	55666	144557	171443	42137

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	104715	116013	Passed
Pm+P1 (SUS)	109220	174019	Passed
Pm+P1+Q (TOTAL)	171443	348039	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10245	16547	10245	16547	10245	16547	10245	16547
Circ. P1 (SUS)	-1131	-1131	4081	4081	-1510	-1510	4460	4460
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5122	5122	5122	5122	5122	5122	5122	5122
Long. P1 (SUS)	-882	-882	1102	1102	-882	-882	1102	1102
Long. Q (SUS)	-173	173	1647	-1647	-2937	2937	4411	-4411
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	144	144	-144	-144	-144	-144	144	144
Shear Q (SUS)	358	358	358	358	358	358	358	358
Pm (SUS)	10245	16547	10245	16547	10245	16547	10245	16547
Pm+P1 (SUS)	9118	15417	14328	20629	8739	15038	14707	21008
Pm+P1+Q (Total)	9163	15438	14333	20630	8741	15042	14766	21020

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16547	115993	Passed
Pm+P1 (SUS)	21008	173989	Passed
Pm+P1+Q (TOTAL)	21020	347979	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Nozzle Calcs.: N2- Tube outlet

Nozl: 35 11:13am Apr 10,2026

Input, Nozzle Desc: N2- Tube outlet

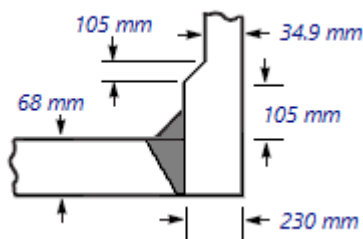
From: 20

Pressure for Reinforcement Calculations	P	5.2514	N/mm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	116013.23	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2520.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1.10	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	107048.78	kPa
Allowable Stress At Ambient	Sna	147988.41	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	34.9250	mm
Hub Height of Integral Nozzle	h	105.0000	mm
Height of Beveled Transition	L'	105.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		230.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2- Tube outlet



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Nozzle Calcs.: N2- Tube outlet Nozl: 35 11:13am Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 1.375 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5.25 \cdot 1263 / (116013 \cdot 1 - 0.6 \cdot 5.25)$
= 58.7631 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5.25 \cdot 460 / (107049 \cdot 1 - 0.6 \cdot 5.25)$
= 23.2590 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 195/0.5773
= 337.8800 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1840.8000	mm
Parallel to Vessel Wall, opening length	d	920.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	162.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 107049/116013)
= 0.923

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 107049/116013)
= 0.923

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.92, 1)
= 0.923



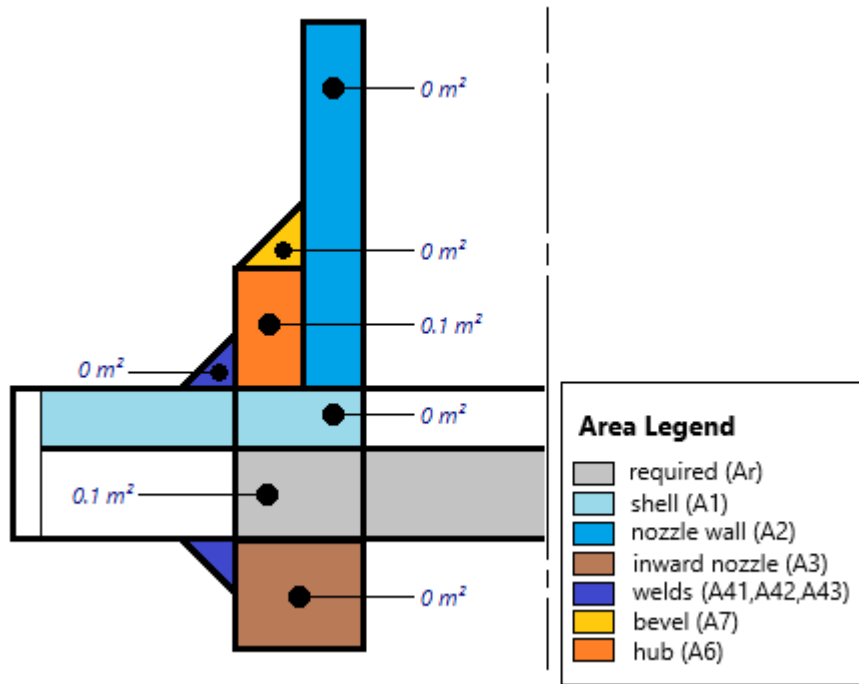
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Nozzle Calcs.: N2- Tube outlet

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.056	NA	NA
Area in Shell	A1	0.006	NA	NA
Area in Nozzle Wall	A2	0.003	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.053	NA	NA
TOTAL AREA AVAILABLE	Atot	0.061	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (920 \cdot 58.8 \cdot 1 + 2 \cdot 227 \cdot 58.8 \cdot 1 \cdot (1 - 0.92))$$

$$= 0.056 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 920(1 \cdot 65 - 1 \cdot 58.8) - 2 \cdot 227$$

$$(1 \cdot 65 - 1 \cdot 58.8) \cdot (1 - 0.92)$$

$$= 0.006 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 163)(31.9 - 23.3)0.92$$

$$= 0.003 \text{ m}^2$$



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Nozzle Calcs.: N2- Tube outlet Nozl: 35 11:13am Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 0.92 + (0)^2 \cdot 0.92$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(163, 600, 105)) \cdot (230 - 34.9) \cdot 0.92$$

$$= 0.053 \text{ m}^2 \text{ Includes Bevel Area of } 0.015 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 26.2590 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 61.7631 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 61.7631 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(61.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(26.3, 11.3)$$

$$= 26.2590 \text{ mm}$$

Available Nozzle Neck Thickness = 34.9250 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 51922.9,	Allowable	: 107048.8 kPa	Passed
Expansion	: 0.0,	Allowable	: 266873.5 kPa	Passed
Occasional	: 36578.7,	Allowable	: 142374.9 kPa	Passed
Shear	: 7924.1,	Allowable	: 74934.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 68, tr = 45.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.7, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	26 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 26 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 195 / 0.5773$$

$$= 337.8800 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]



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Nozzle Calcs.: N2- Tube outlet Nozl: 35 11:13am Apr 10,2026

Parallel to Vessel Wall (Diameter Limit) D1 1380.6000 mm
 Parallel to Vessel Wall, opening length d 690.3000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 162.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (m²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		0.037	0.037
Area Available in Shell A1		0.003	0.003
Area Available in Nozzle Wall A2		0.003	0.003
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.000	0.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		0.053	0.053
TOTAL AREA AVAILABLE Atot		0.058	0.058

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N2- Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (0.056 - 0.0055 + 2 \cdot 227 \cdot 0.92 \cdot (1 \cdot 65 - 58.8)) \cdot 116013)$
 $= 6176167.00 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (0.055 + 0.00021 - 0 \cdot 0.92) \cdot 116013$
 $= 6454686.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (0.055 + 0 + 0.00021 + (0.027)) \cdot 116013$
 $= 9613611.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (0.055 + 0 + 0.00021 + 0 + (0.027)) \cdot 116013$
 $= 9613611.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 1374 \cdot 15 \cdot 0.49 \cdot 107049$
 $= 1698597. \text{ N}$



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Nozzle Calcs.: N2- Tube outlet Nozl: 35 11:13am Apr 10,2026

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 574) \cdot (230 - 3) \cdot 0.7 \cdot 107049$$

$$= 30656930. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1374 \cdot (68 - 3) \cdot 0.74 \cdot 116013$$

$$= 12046863. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (1698597 + 30656930) = 32355528 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (1698597 + 0 + 12046863 + 0) = 13745460 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (1698597 + 12046863 + 0) = 13745460 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 32355528N , must exceed W = 6176167 N or W1 = 6454686 N
 Path 2-2 = 13745460N , must exceed W = 6176167 N or W2 = 9613611 N
 Path 3-3 = 13745460N , must exceed W = 6176167 N or W3 = 9613611 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5.47 N/mm²

The Drop for this Nozzle is : 203.8958 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 871.8958 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 815.60 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 920.40 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2- Tube outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 116013.23	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	230.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.41	kPa
Nozzle Hot S.I. Allowable	SNmh 107048.78	kPa



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Nozzle Calcs.: N2- Tube outlet

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Using 3*S for Discontinuity Allowable (Div. 2, 4.1.6.3) in Creep range.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5.25	N/mm ²
Radial Load	P	94499.98	N
Circumferential Shear	Vc	72000.22	N
Longitudinal Shear	Vl	72000.22	N
Circumferential Moment	Mc	202499.66	N-m
Longitudinal Moment	Ml	243000.12	N-m
Torsional Moment	Mt	243000.12	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.517) is > 0.5
check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (5.976) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1374.4 + 2 \cdot 1.65 \cdot \sqrt{1295.5(68.0 - 3.0)}$$

$$= 2332.011 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2591.000 mm
Nozzle Outside Diameter	(d) =	1374.400 mm
Vessel Thickness used	(T) =	65.000 mm
Nozzle Thickness used	(t) =	227.000 mm
T / t	=	0.286
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.349$$

Nr/P	=	0.056
Mr/P	=	0.095
M0/P	=	0.030
N0/P	=	0.066
MrD/Mc	=	0.261
NrDL/Mc	=	0.093
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.080
NrDL/Ml	=	0.054
M0D/Ml	=	0.023
N0DL/Ml	=	0.075

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1475	-1475	-1475	-1475
Outplane Bending (P)	-4047	4047	-4047	4047
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3130	-3130	3130	3130



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Nozzle Calcs.: N2- Tube outlet Nozl: 35 11:13am Apr 10,2026

Outplane Bending (ML)	-5812	5812	5812	-5812
Normal Pressure Stress	99468	104722	99468	104722

Outplane Stress Summary	85003	109975	102888	104611
-------------------------	-------	--------	--------	--------

Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	-1254	-1254	-1254	-1254
Inplane Bending (P)	-12810	12810	-12810	12810
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2268	-2268	2268	2268
Inplane Bending (ML)	-20194	20194	20194	-20194
Inplane Pressure Stress	49730	49730	49730	49730

Inplane Stress Summary	13203	79212	58128	43360
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	510	510	-510	-510
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1254	1254	1254	1254

Shear Stress Summary	1765	1765	744	744
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	85044	110072	102895	104618
------------------------	-------	--------	--------	--------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-1475	-1475	-1475	-1475
Outplane Bending (P)	-4047	4047	-4047	4047
Outplane Membrane (Mc)	-2985	-2985	2985	2985
Outplane Bending (Mc)	-16395	16395	16395	-16395
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	49730	49730	49730	49730

Outplane Stress Summary	24827	65712	63588	38892
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-1254	-1254	-1254	-1254
Inplane Bending (P)	-12810	12810	-12810	12810
Inplane Membrane (Mc)	-3247	-3247	3247	3247
Inplane Bending (Mc)	-54639	54639	54639	-54639
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	99468	104722	99468	104722



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Nozzle Calcs.: N2- Tube outlet

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Inplane Stress Summary	27516	167669	143290	64885
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-510	-510	510	510
Torsional Shear (Mt)	1254	1254	1254	1254

Shear Stress Summary	744	744	1765	1765
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	27702	167669	143324	65002
------------------------	-------	--------	--------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-1475	-1475	-1475	-1475
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3130	-3130	3130	3130
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10245	16547	10245	16547

Hoop Stress Summary	5639	11941	11900	18201
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	-110	-110	-110	-110
Axial Bending (P)	-737	737	-737	737
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1192	-1192	1192	1192
Axial Bending (ML)	-1096	1096	1096	-1096
Axial Pressure Stress	5122	5122	5122	5122

Axial Stress Summary	1985	5653	6563	5846
----------------------	------	------	------	------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	144	144	-144	-144
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	358	358	358	358

Shear Stress Summary	503	503	213	213
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------



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Nozzle Calcs.: N2- Tube outlet Nozl: 35 11:13am Apr 10,2026

Two * Max Shear Stress 5701 11975 11906 18201

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-1475	-1475	-1475	-1475
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2985	-2985	2985	2985
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10245	16547	10245	16547
Hoop Stress Summary	5784	12086	11755	18056

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	-110	-110	-110	-110
Axial Bending (P)	-737	737	-737	737
Axial Membrane (Mc)	-992	-992	992	992
Axial Bending (Mc)	-3674	3674	3674	-3674
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5122	5122	5122	5122
Axial Stress Summary	-392	8432	8942	3068

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-144	-144	144	144
Shear due to Torsion	358	358	358	358
Shear Stress Summary	213	213	503	503

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	6191	12093	11838	18070

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	99468	104722	99468	104722	99468	104722	99468	104722
Circ. Pl (SUS)	-4605	-4605	1655	1655	-4501	-4501	1993	1993
Circ. Q (SUS)	-9859	9859	1765	-1765	-67449	67449	41829	-41829
Long. Pm (SUS)	49730	49730	49730	49730	49730	49730	49730	49730
Long. Pl (SUS)	-3522	-3522	1014	1014	-4460	-4460	1510	1510
Long. Q (SUS)	-33004	33004	7384	-7384	-20442	20442	12348	-12348
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N2- Tube outlet

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Shear P1 (SUS)	510	510	-510	-510	-510	-510	510	510
Shear Q (SUS)	1254	1254	1254	1254	1254	1254	1254	1254
Pm (SUS)	99468	104722	99468	104722	99468	104722	99468	104722
Pm+P1 (SUS)	94868	100121	101128	106381	94972	100225	101466	106719
Pm+P1+Q (Total)	85047	110076	102900	104621	27710	167675	143329	65005

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	104722	116013	Passed
Pm+P1 (SUS)	106719	174019	Passed
Pm+P1+Q (TOTAL)	167675	348039	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10245	16547	10245	16547	10245	16547	10245	16547
Circ. P1 (SUS)	-4605	-4605	1655	1655	-4460	-4460	1510	1510
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5122	5122	5122	5122	5122	5122	5122	5122
Long. P1 (SUS)	-1302	-1302	1082	1082	-1102	-1102	882	882
Long. Q (SUS)	-1833	1833	359	-359	-4411	4411	2937	-2937
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	144	144	-144	-144	-144	-144	144	144
Shear Q (SUS)	358	358	358	358	358	358	358	358
Pm (SUS)	10245	16547	10245	16547	10245	16547	10245	16547
Pm+P1 (SUS)	5651	11944	11903	18203	5796	12089	11758	18058
Pm+P1+Q (Total)	5707	11981	11908	18205	6190	12099	11841	18073

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16547	107048	Passed
Pm+P1 (SUS)	18203	160573	Passed
Pm+P1+Q (TOTAL)	18205	321146	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell inlet-4

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Input, Nozzle Desc: Shell inlet-4

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6.74	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	18.2600	mm

Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



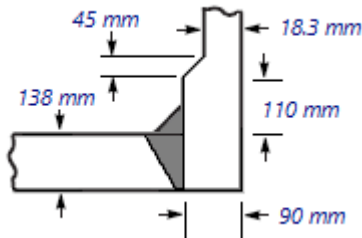
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Nozzle Calcs.: Shell inlet-4

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Reinforcement CALCULATION, Description: Shell inlet-4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 6.000 in.
Actual Thickness Used in Calculation 0.719 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
= 133.8976 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.8 \cdot 79.2 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
= 8.3964 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6836 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 71.7/0.5773
= 124.2574 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	458.9200	mm
Parallel to Vessel Wall	Rn+tn+t	229.4600	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	162.4074	mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n/S_v)
= min(1, 147989/147989)
= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n/S_v)
= min(1, 147989/147989)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(1, 1)
= 1.000



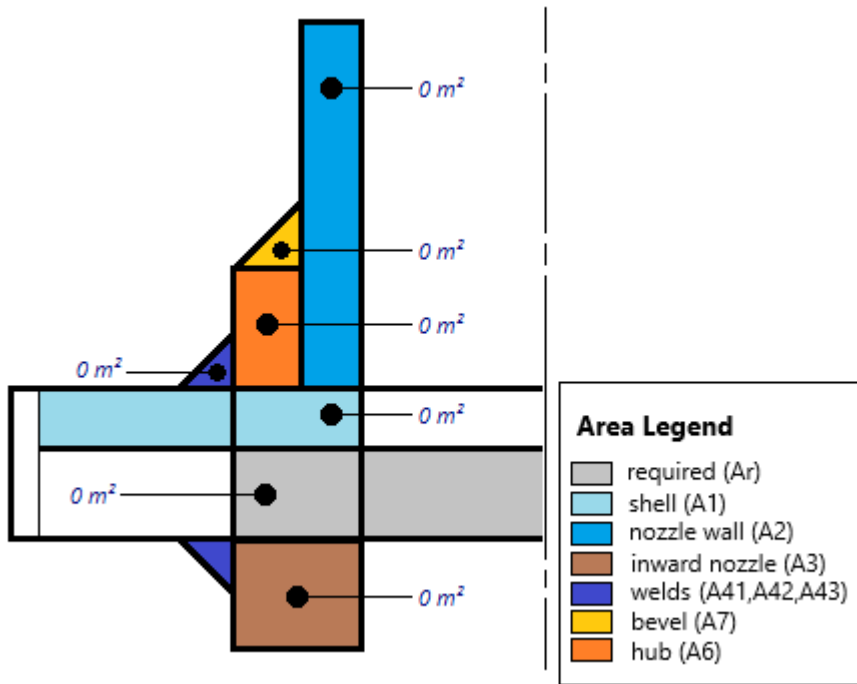
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Nozzle Calcs.: Shell inlet-4

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.021	0.001	NA
Area in Shell	A1	0.000	0.037	NA
Area in Nozzle Wall	A2	0.002	0.005	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.019	0.019	NA
TOTAL AREA AVAILABLE	Atot	0.022	0.061	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (158 \cdot 134 \cdot 1 + 2 \cdot 87 \cdot 134 \cdot 1 \cdot (1 - 1)) \\
 &= 0.021 \text{ m}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 301(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 87 \\
 &\quad (1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1) \\
 &= 0.000 \text{ m}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tnp)(tn - trn)fr2 \\
 &= (2 \cdot 162)(15.3 - 8.4)1 \\
 &= 0.002 \text{ m}^2
 \end{aligned}$$



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Nozzle Calcs.: Shell inlet-4

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(162, 250, 110)) \cdot (90 - 18.3) \cdot 1$$

$$= 0.019 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 11.3964 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.2, \max(137, 5.5)]$$

$$= 10.1600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(11.4, 10.2)$$

$$= 11.3964 \text{ mm}$$

Available Nozzle Neck Thickness = 18.2600 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 67979.9,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 301991.9 kPa	Passed
Occasional	: 34915.0,	Allowable	: 196825.0 kPa	Passed
Shear	: 16806.1,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 90, tr = 8.4, c = 3 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.097, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	47 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: Shell inlet-4

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: Shell inlet-4 Nozl: 36 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.021 - 0.00033 + 2 \cdot 87 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 3118023.25 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / 1.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.021 + 0.99975 \text{E-}04 - 0 \cdot 1) \cdot 147989 \\
 &= 3158066.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.021 + 0 + 0.99975 \text{E-}04 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.021 + 0 + 0.99975 \text{E-}04 + 0 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 332 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 378612. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 123) \cdot (90 - 3) \cdot 0.7 \cdot 147989 \\
 &= 3473990. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 332 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 7719042. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (378612 + 3473990) = 3852602 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (378612 + 0 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (378612 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 3852602 N	, must exceed W = 3118023 N	or W1 = 3158066 N
Path 2-2 = 8097653 N	, must exceed W = 3118023 N	or W2 = 6634228 N
Path 3-3 = 8097653 N	, must exceed W = 3118023 N	or W3 = 6634228 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: Shell inlet-4

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Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 11.0094 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.0094 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 79.38 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 229.46 mm

Input Echo, WRC297 Item 1, Description: Shell inlet-4 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	152.400	mm
Nozzle Thickness	90.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.75	N/mm ²
Radial Load	P	18750.05	N
Circumferential Shear	Vc	15000.04	N
Longitudinal Shear	Vl	15000.04	N
Circumferential Moment	Mc	7849.99	N-m
Longitudinal Moment	Ml	6700.00	N-m
Torsional Moment	Mt	6749.99	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (2.409) is < 5
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.693) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 332.4 + 2 \cdot 1.65 \cdot \sqrt{(1330.5(138.0 - 3.0))} \\
 &= 1730.983 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2661.000 mm



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Nozzle Calcs.: Shell inlet-4

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Nozzle Outside Diameter (d) = 332.400 mm
 Vessel Thickness used (T) = 135.000 mm
 Nozzle Thickness used (t) = 87.000 mm
 T / t = 1.552
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.555$$

Nr/P = 0.181
 Mr/P = 0.217
 M0/P = 0.099
 N0/P = 0.160
 MrD/Mc = 0.254
 NrDL/Mc = 0.174
 M0d/Mc = 0.146
 N0DL/Mc = -0.095
 MrD/Ml = 0.226
 NrDL/Ml = 0.164
 M0D/Ml = 0.129
 N0DL/Ml = 0.026

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-27	-27	27	27
Outplane Bending (ML)	-854	854	854	-854
Normal Pressure Stress	130997	145744	130997	145744
Outplane Stress Summary	129342	147027	131107	145372

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-179	-179	179	179
Inplane Bending (ML)	-1496	1496	1496	-1496
Inplane Pressure Stress	65498	65498	65498	65498
Inplane Stress Summary	62299	67966	65650	65333

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	206	206	-206	-206
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	282	282	282	282
Shear Stress Summary	489	489	75	75

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: Shell inlet-4

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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	129342	147027	131107	145372

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	117	117	-117	-117
Outplane Bending (Mc)	-1130	1130	1130	-1130
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65498	65498	65498	65498
Outplane Stress Summary	63712	67201	65740	64705

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	-220	-220	220	220
Inplane Bending (Mc)	-1971	1971	1971	-1971
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130997	145744	130997	145744
Inplane Stress Summary	127281	148647	131666	145145

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-206	-206	206	206
Torsional Shear (Mt)	282	282	282	282
Shear Stress Summary	75	75	489	489

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	127281	148647	131666	145145

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-27	-27	27	27
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095



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Hoop Stress Summary	10210	27909	10266	27964
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1958	-1958	1958	1958
Axial Bending (ML)	-2302	2302	2302	-2302
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-1213	7142	7308	6453
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	441	441	441	441

Shear Stress Summary	765	765	117	117
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	11520	27936	10266	27964
------------------------	-------	-------	-------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	117	117	-117	-117
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095

Hoop Stress Summary	10355	28054	10121	27819
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	-2289	-2289	2289	2289
Axial Bending (Mc)	-3116	3116	3116	-3116
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-2357	7625	8452	5970
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
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Nozzle Calcs.: Shell inlet-4

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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-324	-324	324	324
Shear due to Torsion	441	441	441	441
Shear Stress Summary	117	117	765	765

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	12713	28054	10417	27840

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Circ. Pl (SUS)	-185	-185	-131	-131	-406	-406	34	34
Circ. Q (SUS)	-1467	1467	241	-241	-3308	3308	634	-634
Long. Pm (SUS)	65498	65498	65498	65498	65498	65498	65498	65498
Long. Pl (SUS)	-365	-365	-7	-7	-41	-41	-275	-275
Long. Q (SUS)	-2833	2833	159	-159	-1743	1743	517	-517
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	206	206	-206	-206	-206	-206	206	206
Shear Q (SUS)	282	282	282	282	282	282	282	282
Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Pm+Pl (SUS)	130812	145559	130866	145613	130591	145338	131031	145778
Pm+Pl+Q (Total)	129348	147029	131107	145372	127283	148646	131668	145146

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145744	147988	Passed
Pm+P1 (SUS)	145778	221983	Passed
Pm+P1+Q (TOTAL)	148646	443966	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was $3 * S_m h$.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

[illegible]



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Nozzle Calcs.: Shell inlet-4

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Long. P1 (SUS)	-2233	-2233	1683	1683	-2564	-2564	2014	2014
Long. Q (SUS)	-4177	4177	427	-427	-4991	4991	1241	-1241
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	324	324	-324	-324	-324	-324	324	324
Shear Q (SUS)	441	441	441	441	441	441	441	441
Pm (SUS)	10397	28095	10397	28095	10397	28095	10397	28095
Pm+P1 (SUS)	10226	27914	10296	27968	10369	28058	10157	27825
Pm+P1+Q (Total)	11526	27938	10270	27964	12715	28054	10419	27846

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28095	147988	Passed
Pm+P1 (SUS)	28058	221983	Passed
Pm+P1+Q (TOTAL)	28054	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell outlet-3

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Input, Nozzle Desc: Shell outlet-3

From: 40

Pressure for Reinforcement Calculations	P	14.7270	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5.74	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	23.0124	mm

Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



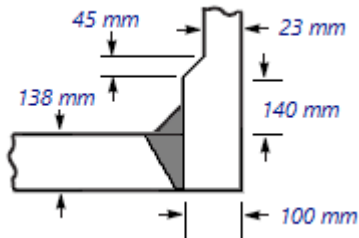
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Nozzle Calcs.: Shell outlet-3

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Reinforcement CALCULATION, Description: Shell outlet-3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.906 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.7 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
 $= 133.6591 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.7 \cdot 105 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
 $= 11.0695 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8049 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 77 / 0.5773$
 $= 133.3465 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	519.2248	mm
Parallel to Vessel Wall	Rn+tn+t	259.6124	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	183.3775	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147989 / 147989)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147989 / 147989)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



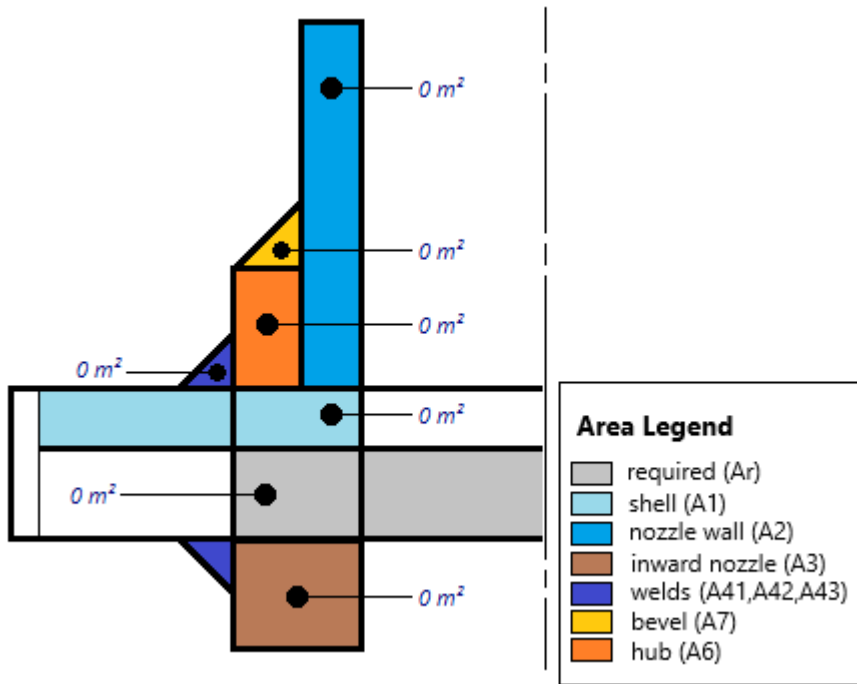
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Nozzle Calcs.: Shell outlet-3

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.028	0.001	NA
Area in Shell	A1	0.000	0.038	NA
Area in Nozzle Wall	A2	0.003	0.007	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	0.025	0.025	NA
TOTAL AREA AVAILABLE	Atot	0.029	0.070	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (209 \cdot 134 \cdot 1 + 2 \cdot 97 \cdot 134 \cdot 1 \cdot (1 - 1))$$

$$= 0.028 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 310(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 97$$

$$(1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1)$$

$$= 0.000 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 183)(20 - 11.1)1$$

$$= 0.003 \text{ m}^2$$



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Nozzle Calcs.: Shell outlet-3

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{inp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(183, 250, 140)) \cdot (100 - 23) \cdot 1$$

$$= 0.025 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 14.0695 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.6591 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.6591 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.1, \max(137, 5.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(14.1, 11.1)$$

$$= 14.0695 \text{ mm}$$

Available Nozzle Neck Thickness = 23.0124 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 59027.4,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 310944.4 kPa	Passed
Occasional	: 35124.8,	Allowable	: 196825.0 kPa	Passed
Shear	: 40737.2,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 100, tr = 11.1, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.11, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: Shell outlet-3

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: Shell outlet-3 Nozl: 37 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.028 - 0.00042 + 2 \cdot 97 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 4114850.50 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.028 + 0.99975E-04 - 0 \cdot 1) \cdot 147989 \\
 &= 4202223.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.028 + 0 + 0.99975E-04 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.028 + 0 + 0.99975E-04 + 0 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 403 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 459255. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 153) \cdot (100 - 3) \cdot 0.7 \cdot 147989 \\
 &= 4832944. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 403 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 9363169. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (459255 + 4832944) = 5292198 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (459255 + 0 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (459255 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 5292198 N	, must exceed W = 4114850 N	or W1 = 4202223 N
Path 2-2 = 9822424 N	, must exceed W = 4114850 N	or W2 = 8077944 N
Path 3-3 = 9822424 N	, must exceed W = 4114850 N	or W3 = 8077944 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: Shell outlet-3

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Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 16.2325 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 404.2325 mm

Input Echo, WRC297 Item 1, Description: Shell outlet-3:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	203.200	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.73	N/mm ²
Radial Load	P	24999.98	N
Circumferential Shear	Vc	20000.03	N
Longitudinal Shear	Vl	20000.33	N
Circumferential Moment	Mc	12000.01	N-m
Longitudinal Moment	Ml	12000.01	N-m
Torsional Moment	Mt	14000.00	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (2.922) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.032) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 403.2 + 2 \cdot 1.65 \cdot \sqrt{1330.5(138.0 - 3.0)}$$

$$= 1801.783 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2661.000 mm
Nozzle Outside Diameter	(d) =	403.200 mm
Vessel Thickness used	(T) =	135.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	1.392



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Nozzle Calcs.: Shell outlet-3

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$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.673$$

Nr/P	=	0.163
Mr/P	=	0.208
MØ/P	=	0.086
NØ/P	=	0.149
MrD/Mc	=	0.262
NrDL/Mc	=	0.166
MØd/Mc	=	0.134
NØDL/Mc	=	-0.082
MrD/Ml	=	0.220
NrDL/Ml	=	0.148
MØD/Ml	=	0.112
NØDL/Ml	=	0.067

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-103	-103	103	103
Outplane Bending (ML)	-1089	1089	1089	-1089
Normal Pressure Stress	130776	145503	130776	145503
Outplane Stress Summary	128680	146992	131066	145021

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-241	-241	241	241
Inplane Bending (ML)	-2151	2151	2151	-2151
Inplane Pressure Stress	65388	65388	65388	65388
Inplane Stress Summary	61072	68780	65857	64960

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	227	227	-227	-227
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	399	399	399	399
Shear Stress Summary	627	627	172	172

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	128680	146992	131066	145021



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Nozzle Calcs.: Shell outlet-3

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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	130	130	-130	-130
Outplane Bending (Mc)	-1309	1309	1309	-1309
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65388	65388	65388	65388
Outplane Stress Summary	63306	67332	65664	64450

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	-268	-268	268	268
Inplane Bending (Mc)	-2564	2564	2564	-2564
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130776	145503	130776	145503
Inplane Stress Summary	126019	149281	131686	144690

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2337	-2337	2337	2337
Torsional Shear (Mt)	399	399	399	399
Shear Stress Summary	-1937	-1937	2737	2737

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	126074	149323	131797	144779

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-103	-103	103	103
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687
Hoop Stress Summary	12713	30384	12920	30591

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: Shell outlet-3

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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2006	-2006	2006	2006
Axial Bending (ML)	-2764	2764	2764	-2764
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-537	9018	9004	7501
----------------------	------	------	------	------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	558	558	558	558

Shear Stress Summary	882	882	234	234
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	13361	30418	12934	30591
------------------------	-------	-------	-------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	130	130	-130	-130
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687

Hoop Stress Summary	12948	30618	12686	30356
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	-2006	-2006	2006	2006
Axial Bending (Mc)	-3399	3399	3399	-3399
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-1172	9652	9638	6867
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3254	-3254	3254	3254



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Shear due to Torsion	558	558	558	558
----------------------	-----	-----	-----	-----

Shear Stress Summary	-2695	-2695	3812	3812
----------------------	-------	-------	------	------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	15112	30956	15264	30956

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Circ. Pl (SUS)	-302	-302	-96	-96	-488	-488	48	48
Circ. Q (SUS)	-1792	1792	386	-386	-4266	4266	862	-862
Long. Pm (SUS)	65388	65388	65388	65388	65388	65388	65388	65388
Long. Pl (SUS)	-461	-461	21	21	-69	-69	-329	-329
Long. Q (SUS)	-3853	3853	449	-449	-2012	2012	606	-606
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	227	227	-227	-227	-2337	-2337	2337	2337
Shear Q (SUS)	399	399	399	399	399	399	399	399
Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Pm+Pl (SUS)	130474	145201	130680	145407	130371	145083	130906	145618
Pm+Pl+Q (Total)	128687	146998	131066	145021	126081	149326	131799	144782

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145503	147988	Passed
Pm+Pl (SUS)	145618	221983	Passed
Pm+Pl+Q (TOTAL)	149326	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Circ. Pl (SUS)	-302	-302	-96	-96	-69	-69	-329	-329
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6508	6508	6508	6508	6508	6508	6508	6508
Long. Pl (SUS)	-2267	-2267	1745	1745	-2267	-2267	1745	1745
Long. Q (SUS)	-4777	4777	751	-751	-5412	5412	1386	-1386
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: Shell outlet-3

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Shear P1 (SUS)	324	324	-324	-324	-3254	-3254	3254	3254
Shear Q (SUS)	558	558	558	558	558	558	558	558
Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Pm+P1 (SUS)	12727	30389	12943	30595	14029	31013	14408	30827
Pm+P1+Q (Total)	13367	30421	12934	30593	15113	30959	15269	30961

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30687	147988	Passed
Pm+P1 (SUS)	31013	221983	Passed
Pm+P1+Q (TOTAL)	30961	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell outlet-4

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Input, Nozzle Desc: Shell outlet-4

From: 40

Pressure for Reinforcement Calculations	P	14.7270	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6.74	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	23.0124	mm

Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



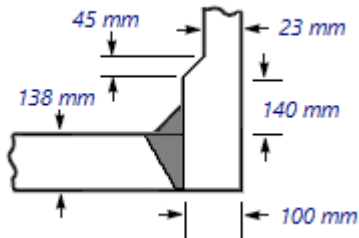
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Nozzle Calcs.: Shell outlet-4

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Reinforcement CALCULATION, Description: Shell outlet-4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 8.000 in.
Actual Thickness Used in Calculation 0.906 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.7 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
= 133.6591 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.7 \cdot 105 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
= 11.0695 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.8049 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $77 / 0.5773$
= 133.3465 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	519.2248	mm
Parallel to Vessel Wall	Rn+tn+t	259.6124	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	183.3775	mm

Weld Strength Reduction Factor [fr1]:

= $\min(1, S_n / S_v)$
= $\min(1, 147989 / 147989)$
= 1.000

Weld Strength Reduction Factor [fr2]:

= $\min(1, S_n / S_v)$
= $\min(1, 147989 / 147989)$
= 1.000

Weld Strength Reduction Factor [fr3]:

= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000



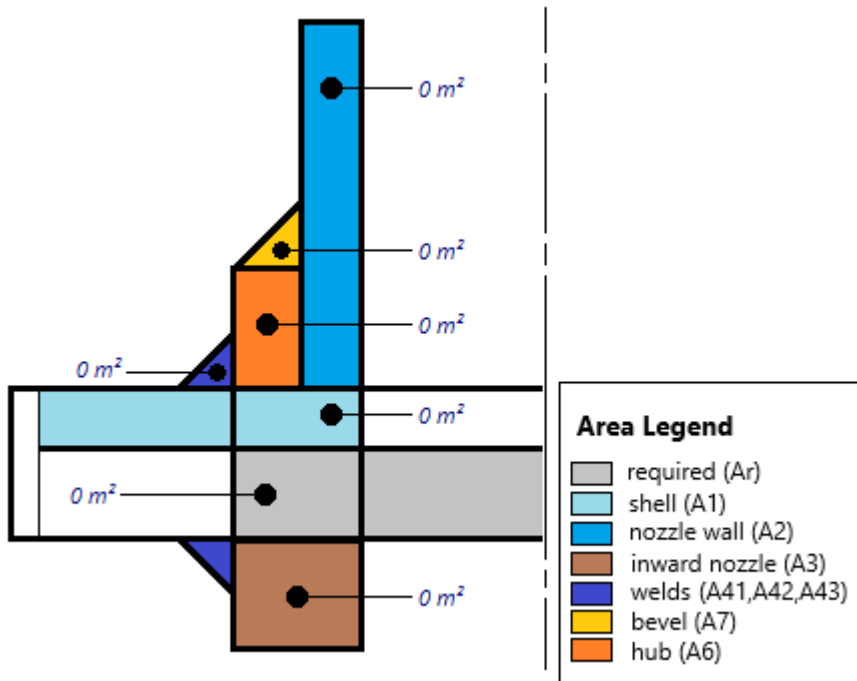
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Nozzle Calcs.: Shell outlet-4

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.028	0.001	NA
Area in Shell	A1	0.000	0.038	NA
Area in Nozzle Wall	A2	0.003	0.007	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	0.025	0.025	NA
TOTAL AREA AVAILABLE	Atot	0.029	0.070	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (209 \cdot 134 \cdot 1 + 2 \cdot 97 \cdot 134 \cdot 1 \cdot (1 - 1))$$

$$= 0.028 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 310(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 97$$

$$(1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1)$$

$$= 0.000 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 183)(20 - 11.1)1$$

$$= 0.003 \text{ m}^2$$



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Nozzle Calcs.: Shell outlet-4

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{inp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(183, 250, 140)) \cdot (100 - 23) \cdot 1$$

$$= 0.025 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 14.0695 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.6591 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.6591 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.1, \max(137, 5.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(14.1, 11.1)$$

$$= 14.0695 \text{ mm}$$

Available Nozzle Neck Thickness = 23.0124 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 59027.4,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 310944.4 kPa	Passed
Occasional	: 35124.8,	Allowable	: 196825.0 kPa	Passed
Shear	: 14473.4,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 100, tr = 11.1, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.11, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: Shell outlet-4

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: Shell outlet-4 Nozl: 38 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.028 - 0.00042 + 2 \cdot 97 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 4114850.50 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.028 + 0.99975E-04 - 0 \cdot 1) \cdot 147989 \\
 &= 4202223.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.028 + 0 + 0.99975E-04 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.028 + 0 + 0.99975E-04 + 0 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 403 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 459255. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 153) \cdot (100 - 3) \cdot 0.7 \cdot 147989 \\
 &= 4832944. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 403 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 9363169. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (459255 + 4832944) = 5292198 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (459255 + 0 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (459255 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 5292198 N	, must exceed W = 4114850 N	or W1 = 4202223 N
Path 2-2 = 9822424 N	, must exceed W = 4114850 N	or W2 = 8077944 N
Path 3-3 = 9822424 N	, must exceed W = 4114850 N	or W3 = 8077944 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: Shell outlet-4

Nozl: 38 11:13am Apr 10,2026

Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 16.2325 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 404.2325 mm

Input Echo, WRC297 Item 1, Description: Shell outlet-4:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	203.200	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.73	N/mm ²
Radial Load	P	24999.98	N
Circumferential Shear	Vc	20000.03	N
Longitudinal Shear	Vl	20000.03	N
Circumferential Moment	Mc	12000.01	N-m
Longitudinal Moment	Ml	12000.01	N-m
Torsional Moment	Mt	14000.00	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (2.922) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.032) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 403.2 + 2 \cdot 1.65 \cdot \sqrt{1330.5(138.0 - 3.0)}$$

$$= 1801.783 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2661.000 mm
Nozzle Outside Diameter	(d) =	403.200 mm
Vessel Thickness used	(T) =	135.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	1.392



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Nozzle Calcs.: Shell outlet-4

Nozl: 38 11:13am Apr 10,2026

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.673$$

Nr/P	=	0.163
Mr/P	=	0.208
MØ/P	=	0.086
NØ/P	=	0.149
MrD/Mc	=	0.262
NrDL/Mc	=	0.166
MØd/Mc	=	0.134
NØDL/Mc	=	-0.082
MrD/Ml	=	0.220
NrDL/Ml	=	0.148
MØD/Ml	=	0.112
NØDL/Ml	=	0.067

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-103	-103	103	103
Outplane Bending (ML)	-1089	1089	1089	-1089
Normal Pressure Stress	130776	145503	130776	145503
Outplane Stress Summary	128680	146992	131066	145021

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-241	-241	241	241
Inplane Bending (ML)	-2151	2151	2151	-2151
Inplane Pressure Stress	65388	65388	65388	65388
Inplane Stress Summary	61072	68780	65857	64960

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	227	227	-227	-227
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	399	399	399	399
Shear Stress Summary	627	627	172	172

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	128680	146992	131066	145021



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Nozzle Calcs.: Shell outlet-4

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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	130	130	-130	-130
Outplane Bending (Mc)	-1309	1309	1309	-1309
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65388	65388	65388	65388
Outplane Stress Summary	63306	67332	65664	64450

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	-268	-268	268	268
Inplane Bending (Mc)	-2564	2564	2564	-2564
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130776	145503	130776	145503
Inplane Stress Summary	126019	149281	131686	144690

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-227	-227	227	227
Torsional Shear (Mt)	399	399	399	399
Shear Stress Summary	172	172	627	627

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	126019	149281	131686	144690

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-103	-103	103	103
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687
Hoop Stress Summary	12713	30384	12920	30591

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: Shell outlet-4

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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2006	-2006	2006	2006
Axial Bending (ML)	-2764	2764	2764	-2764
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-537	9018	9004	7501
----------------------	------	------	------	------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	558	558	558	558

Shear Stress Summary	882	882	234	234
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	13361	30418	12934	30591
------------------------	-------	-------	-------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	130	130	-130	-130
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687

Hoop Stress Summary	12948	30618	12686	30356
---------------------	-------	-------	-------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	-2006	-2006	2006	2006
Axial Bending (Mc)	-3399	3399	3399	-3399
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-1172	9652	9638	6867
----------------------	-------	------	------	------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-324	-324	324	324



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Nozzle Calcs.: Shell outlet-4 Nozl: 38 11:13am Apr 10,2026

Shear due to Torsion	558	558	558	558
----------------------	-----	-----	-----	-----

Shear Stress Summary	234	234	882	882
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14127	30618	12920	30384

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Circ. Pl (SUS)	-302	-302	-96	-96	-488	-488	48	48
Circ. Q (SUS)	-1792	1792	386	-386	-4266	4266	862	-862
Long. Pm (SUS)	65388	65388	65388	65388	65388	65388	65388	65388
Long. Pl (SUS)	-461	-461	21	21	-69	-69	-329	-329
Long. Q (SUS)	-3853	3853	449	-449	-2012	2012	606	-606
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	227	227	-227	-227	-227	-227	227	227
Shear Q (SUS)	399	399	399	399	399	399	399	399
Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Pm+Pl (SUS)	130474	145201	130680	145407	130288	145015	130824	145551
Pm+Pl+Q (Total)	128687	146998	131066	145021	126022	149281	131691	144693

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145503	147988	Passed
Pm+Pl (SUS)	145551	221983	Passed
Pm+Pl+Q (TOTAL)	149281	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Circ. Pl (SUS)	-302	-302	-96	-96	-69	-69	-329	-329
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6508	6508	6508	6508	6508	6508	6508	6508
Long. Pl (SUS)	-2267	-2267	1745	1745	-2267	-2267	1745	1745
Long. Q (SUS)	-4777	4777	751	-751	-5412	5412	1386	-1386
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: Shell outlet-4

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Shear P1 (SUS)	324	324	-324	-324	-324	-324	324	324
Shear Q (SUS)	558	558	558	558	558	558	558	558
Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Pm+P1 (SUS)	12727	30389	12943	30595	12960	30621	12711	30362
Pm+P1+Q (Total)	13367	30421	12934	30593	14126	30620	12924	30391

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30687	147988	Passed
Pm+P1 (SUS)	30621	221983	Passed
Pm+P1+Q (TOTAL)	30620	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell inlet-1

Nozl: 39 11:13am Apr 10,2026

Input, Nozzle Desc: Shell inlet-1

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3.69	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	18.2600	mm

Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



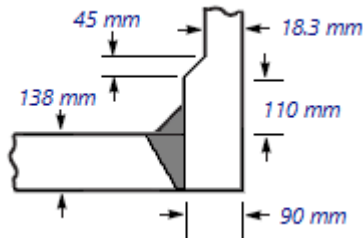
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Nozzle Calcs.: Shell inlet-1

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Reinforcement CALCULATION, Description: Shell inlet-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 6.000 in.
Actual Thickness Used in Calculation 0.719 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
= 133.8976 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.8 \cdot 79.2 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
= 8.3964 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6836 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 71.7/0.5773
= 124.2574 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	458.9200	mm
Parallel to Vessel Wall	Rn+tn+t	229.4600	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	162.4074	mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n/S_v)
= min(1, 147989/147989)
= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n/S_v)
= min(1, 147989/147989)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(1, 1)
= 1.000



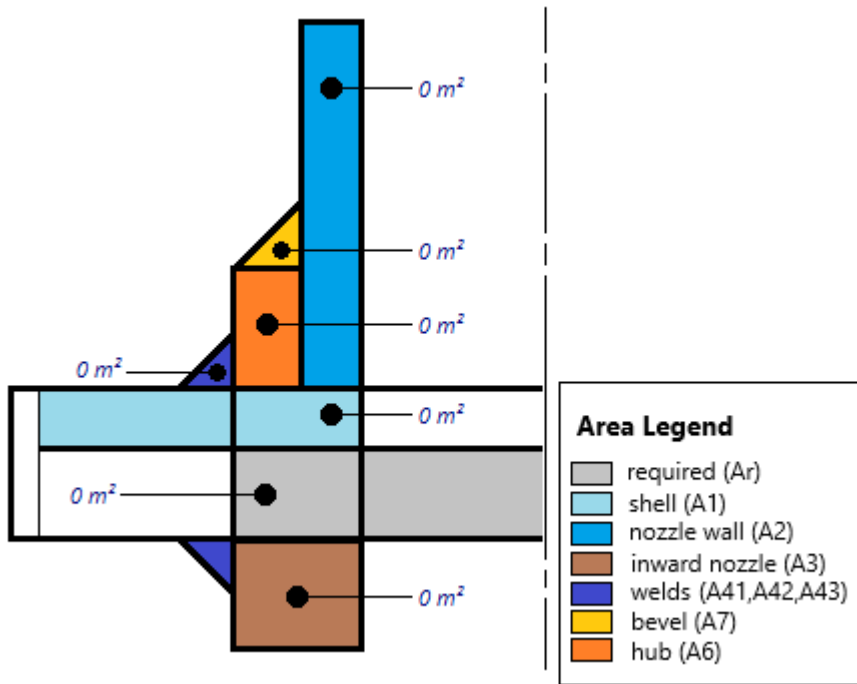
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Nozzle Calcs.: Shell inlet-1

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.021	0.001	NA
Area in Shell	A1	0.000	0.037	NA
Area in Nozzle Wall	A2	0.002	0.005	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.019	0.019	NA
TOTAL AREA AVAILABLE	Atot	0.022	0.061	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (158 \cdot 134 \cdot 1 + 2 \cdot 87 \cdot 134 \cdot 1 \cdot (1 - 1)) \\
 &= 0.021 \text{ m}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 301(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 87 \\
 &\quad (1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1) \\
 &= 0.000 \text{ m}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 162)(15.3 - 8.4)1 \\
 &= 0.002 \text{ m}^2
 \end{aligned}$$



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Nozzle Calcs.: Shell inlet-1 Nozl: 39 11:13am Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(162, 250, 110)) \cdot (90 - 18.3) \cdot 1$$

$$= 0.019 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 11.3964 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.2, \max(137, 5.5)]$$

$$= 10.1600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(11.4, 10.2)$$

$$= 11.3964 \text{ mm}$$

Available Nozzle Neck Thickness = 18.2600 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 65667.4,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 304304.4 kPa	Passed
Occasional	: 34915.0,	Allowable	: 196825.0 kPa	Passed
Shear	: 18676.0,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 90, tr = 8.4, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.097, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	47 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: Shell inlet-1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: Shell inlet-1 Nozl: 39 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.021 - 0.00033 + 2 \cdot 87 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 3118023.25 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.021 + 0.99975E-04 - 0 \cdot 1) \cdot 147989 \\
 &= 3158066.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.021 + 0 + 0.99975E-04 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.021 + 0 + 0.99975E-04 + 0 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 332 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 378612. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 123) \cdot (90 - 3) \cdot 0.7 \cdot 147989 \\
 &= 3473990. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 332 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 7719042. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (378612 + 3473990) = 3852602 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (378612 + 0 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (378612 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 3852602 N	, must exceed W = 3118023 N	or W1 = 3158066 N
Path 2-2 = 8097653 N	, must exceed W = 3118023 N	or W2 = 6634228 N
Path 3-3 = 8097653 N	, must exceed W = 3118023 N	or W3 = 6634228 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: Shell inlet-1

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Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 11.0094 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.0094 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 79.38 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 229.46 mm

Input Echo, WRC297 Item 1, Description: Shell inlet-1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	152.400	mm
Nozzle Thickness	90.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.75	N/mm ²
Radial Load	P	18750.05	N
Circumferential Shear	Vc	15000.04	N
Longitudinal Shear	Vl	15000.04	N
Circumferential Moment	Mc	6749.99	N-m
Longitudinal Moment	Ml	6749.99	N-m
Torsional Moment	Mt	7875.00	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (2.409) is < 5
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.693) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 332.4 + 2 \cdot 1.65 \cdot \sqrt{(1330.5(138.0 - 3.0))} \\
 &= 1730.983 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2661.000 mm



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Nozzle Calcs.: Shell inlet-1

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Nozzle Outside Diameter (d) = 332.400 mm
 Vessel Thickness used (T) = 135.000 mm
 Nozzle Thickness used (t) = 87.000 mm
 T / t = 1.552
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.555$$

Nr/P = 0.181
 Mr/P = 0.217
 M0/P = 0.099
 N0/P = 0.160
 MrD/Mc = 0.254
 NrDL/Mc = 0.174
 M0d/Mc = 0.146
 N0DL/Mc = -0.095
 MrD/Ml = 0.226
 NrDL/Ml = 0.164
 M0D/Ml = 0.129
 N0DL/Ml = 0.026

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-27	-27	27	27
Outplane Bending (ML)	-854	854	854	-854
Normal Pressure Stress	130997	145744	130997	145744
Outplane Stress Summary	129342	147027	131107	145372

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-179	-179	179	179
Inplane Bending (ML)	-1509	1509	1509	-1509
Inplane Pressure Stress	65498	65498	65498	65498
Inplane Stress Summary	62285	67980	65664	65319

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	206	206	-206	-206
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	330	330	330	330
Shear Stress Summary	537	537	124	124

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: Shell inlet-1

Nozl: 39 11:13am Apr 10,2026

	Outside	Inside	Outside	Inside
Two * Max Shear Stress	129342	147027	131107	145372

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	103	103	-103	-103
Outplane Bending (Mc)	-972	972	972	-972
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65498	65498	65498	65498
Outplane Stress Summary	63857	67029	65595	64878

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	-193	-193	193	193
Inplane Bending (Mc)	-1689	1689	1689	-1689
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130997	145744	130997	145744
Inplane Stress Summary	127591	148392	131355	145400

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-206	-206	206	206
Torsional Shear (Mt)	330	330	330	330
Shear Stress Summary	124	124	537	537

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	127591	148392	131355	145400

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-27	-27	27	27
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095



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Nozzle Calcs.: Shell inlet-1 Nozl: 39 11:13am Apr 10,2026

Hoop Stress Summary	10210	27909	10266	27964
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1971	-1971	1971	1971
Axial Bending (ML)	-2323	2323	2323	-2323
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-1247	7149	7342	6446
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	517	517	517	517

Shear Stress Summary	841	841	193	193
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	11576	27936	10272	27964
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	103	103	-103	-103
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095

Hoop Stress Summary	10341	28040	10135	27833
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	-1971	-1971	1971	1971
Axial Bending (Mc)	-2675	2675	2675	-2675
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-1599	7501	7694	6094
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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Nozzle Calcs.: Shell inlet-1

Noz1: 39 11:13am Apr 10, 2026

(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-324	-324	324	324
Shear due to Torsion	517	517	517	517
Shear Stress Summary	193	193	841	841

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11941	28040	10390	27861

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Circ. Pl (SUS)	-185	-185	-131	-131	-379	-379	7	7
Circ. Q (SUS)	-1467	1467	241	-241	-3026	3026	352	-352
Long. Pm (SUS)	65498	65498	65498	65498	65498	65498	65498	65498
Long. Pl (SUS)	-365	-365	-7	-7	-55	-55	-261	-261
Long. Q (SUS)	-2846	2846	172	-172	-1585	1585	359	-359
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	206	206	-206	-206	-206	-206	206	206
Shear Q (SUS)	330	330	330	330	330	330	330	330
Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Pm+Pl (SUS)	130812	145559	130866	145613	130618	145365	131004	145751
Pm+Pl+Q (Total)	129349	147029	131107	145372	127592	148391	131360	145402

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145744	147988	Passed
Pm+P1 (SUS)	145751	221983	Passed
Pm+P1+Q (TOTAL)	148391	443966	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

[illegible]



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Nozzle Calcs.: Shell inlet-1

Nozl: 39 11:13am Apr 10,2026

Long. P1 (SUS)	-2246	-2246	1696	1696	-2246	-2246	1696	1696
Long. Q (SUS)	-4198	4198	448	-448	-4550	4550	800	-800
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	324	324	-324	-324	-324	-324	324	324
Shear Q (SUS)	517	517	517	517	517	517	517	517
Pm (SUS)	10397	28095	10397	28095	10397	28095	10397	28095
Pm+P1 (SUS)	10226	27914	10296	27968	10356	28044	10168	27839
Pm+P1+Q (Total)	11580	27944	10278	27965	11946	28041	10397	27866

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28095	147988	Passed
Pm+P1 (SUS)	28044	221983	Passed
Pm+P1+Q (TOTAL)	28041	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell outlet-1

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Input, Nozzle Desc: Shell outlet-1

From: 40

Pressure for Reinforcement Calculations	P	14.7270	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3.71	m
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	23.0124	mm

Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



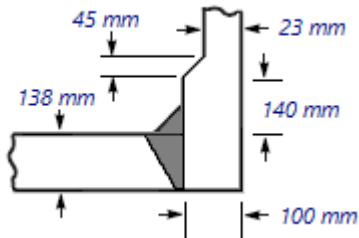
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Nozzle Calcs.: Shell outlet-1

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Reinforcement CALCULATION, Description: Shell outlet-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.906 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.7 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
 $= 133.6591 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.7 \cdot 105 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
 $= 11.0695 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8049 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 77 / 0.5773$
 $= 133.3465 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	519.2248	mm
Parallel to Vessel Wall	Rn+tn+t	259.6124	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	183.3775	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147989 / 147989)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147989 / 147989)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



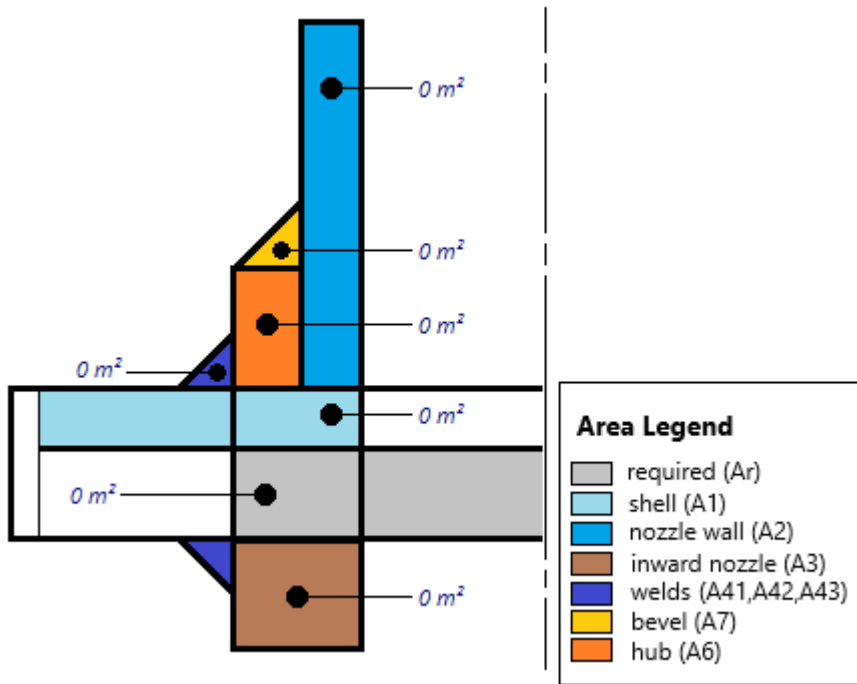
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Nozzle Calcs.: Shell outlet-1

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.028	0.001	NA
Area in Shell	A1	0.000	0.038	NA
Area in Nozzle Wall	A2	0.003	0.007	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	0.025	0.025	NA
TOTAL AREA AVAILABLE	Atot	0.029	0.070	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (209 \cdot 134 \cdot 1 + 2 \cdot 97 \cdot 134 \cdot 1 \cdot (1 - 1))$$

$$= 0.028 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 310(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 97$$

$$(1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1)$$

$$= 0.000 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 183)(20 - 11.1)1$$

$$= 0.003 \text{ m}^2$$



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Nozzle Calcs.: Shell outlet-1

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{inp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(183, 250, 140)) \cdot (100 - 23) \cdot 1$$

$$= 0.025 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 14.0695 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.6591 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.6591 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.1, \max(137, 5.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(14.1, 11.1)$$

$$= 14.0695 \text{ mm}$$

Available Nozzle Neck Thickness = 23.0124 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 59027.4,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 310944.4 kPa	Passed
Occasional	: 35124.8,	Allowable	: 196825.0 kPa	Passed
Shear	: 14473.4,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 100, tr = 11.1, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.11, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: Shell outlet-1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: Shell outlet-1 Nozl: 40 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.028 - 0.00042 + 2 \cdot 97 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 4114850.50 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / 1.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.028 + 0.99975 \text{E-}04 - 0 \cdot 1) \cdot 147989 \\
 &= 4202223.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.028 + 0 + 0.99975 \text{E-}04 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.028 + 0 + 0.99975 \text{E-}04 + 0 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 403 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 459255. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 153) \cdot (100 - 3) \cdot 0.7 \cdot 147989 \\
 &= 4832944. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 403 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 9363169. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (459255 + 4832944) = 5292198 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (459255 + 0 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (459255 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 5292198 N	, must exceed W = 4114850 N	or W1 = 4202223 N
Path 2-2 = 9822424 N	, must exceed W = 4114850 N	or W2 = 8077944 N
Path 3-3 = 9822424 N	, must exceed W = 4114850 N	or W3 = 8077944 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: Shell outlet-1

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Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 16.2325 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 404.2325 mm

Input Echo, WRC297 Item 1, Description: Shell outlet-1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	203.200	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.73	N/mm ²
Radial Load	P	24999.98	N
Circumferential Shear	Vc	20000.03	N
Longitudinal Shear	Vl	20000.03	N
Circumferential Moment	Mc	12000.01	N-m
Longitudinal Moment	Ml	12000.01	N-m
Torsional Moment	Mt	14000.00	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (2.922) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.032) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 403.2 + 2 \cdot 1.65 \cdot \sqrt{1330.5(138.0 - 3.0)}$$

$$= 1801.783 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2661.000	mm
Nozzle Outside Diameter	(d) =	403.200	mm
Vessel Thickness used	(T) =	135.000	mm
Nozzle Thickness used	(t) =	97.000	mm
T / t	=	1.392	



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Nozzle Calcs.: Shell outlet-1

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$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.673$$

Nr/P	=	0.163
Mr/P	=	0.208
MØ/P	=	0.086
NØ/P	=	0.149
MrD/Mc	=	0.262
NrDL/Mc	=	0.166
MØd/Mc	=	0.134
NØDL/Mc	=	-0.082
MrD/Ml	=	0.220
NrDL/Ml	=	0.148
MØD/Ml	=	0.112
NØDL/Ml	=	0.067

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-103	-103	103	103
Outplane Bending (ML)	-1089	1089	1089	-1089
Normal Pressure Stress	130776	145503	130776	145503
Outplane Stress Summary	128680	146992	131066	145021

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-241	-241	241	241
Inplane Bending (ML)	-2151	2151	2151	-2151
Inplane Pressure Stress	65388	65388	65388	65388
Inplane Stress Summary	61072	68780	65857	64960

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	227	227	-227	-227
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	399	399	399	399
Shear Stress Summary	627	627	172	172

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	128680	146992	131066	145021



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Nozzle Calcs.: Shell outlet-1

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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	130	130	-130	-130
Outplane Bending (Mc)	-1309	1309	1309	-1309
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65388	65388	65388	65388
Outplane Stress Summary	63306	67332	65664	64450

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	-268	-268	268	268
Inplane Bending (Mc)	-2564	2564	2564	-2564
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130776	145503	130776	145503
Inplane Stress Summary	126019	149281	131686	144690

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-227	-227	227	227
Torsional Shear (Mt)	399	399	399	399
Shear Stress Summary	172	172	627	627

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	126019	149281	131686	144690

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-103	-103	103	103
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687
Hoop Stress Summary	12713	30384	12920	30591

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: Shell outlet-1

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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2006	-2006	2006	2006
Axial Bending (ML)	-2764	2764	2764	-2764
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-537	9018	9004	7501
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE
(Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	558	558	558	558

Shear Stress Summary	882	882	234	234
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	13361	30418	12934	30591

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stresses in the
hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	130	130	-130	-130
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687

Hoop Stress Summary	12948	30618	12686	30356
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	-2006	-2006	2006	2006
Axial Bending (Mc)	-3399	3399	3399	-3399
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-1172	9652	9638	6867
----------------------	-------	------	------	------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-324	-324	324	324



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Shear due to Torsion	558	558	558	558
----------------------	-----	-----	-----	-----

Shear Stress Summary	234	234	882	882
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14127	30618	12920	30384

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Circ. Pl (SUS)	-302	-302	-96	-96	-488	-488	48	48
Circ. Q (SUS)	-1792	1792	386	-386	-4266	4266	862	-862
Long. Pm (SUS)	65388	65388	65388	65388	65388	65388	65388	65388
Long. Pl (SUS)	-461	-461	21	21	-69	-69	-329	-329
Long. Q (SUS)	-3853	3853	449	-449	-2012	2012	606	-606
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	227	227	-227	-227	-227	-227	227	227
Shear Q (SUS)	399	399	399	399	399	399	399	399
Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Pm+Pl (SUS)	130474	145201	130680	145407	130288	145015	130824	145551
Pm+Pl+Q (Total)	128687	146998	131066	145021	126022	149281	131691	144693

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145503	147988	Passed
Pm+Pl (SUS)	145551	221983	Passed
Pm+Pl+Q (TOTAL)	149281	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Circ. Pl (SUS)	-302	-302	-96	-96	-69	-69	-329	-329
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6508	6508	6508	6508	6508	6508	6508	6508
Long. Pl (SUS)	-2267	-2267	1745	1745	-2267	-2267	1745	1745
Long. Q (SUS)	-4777	4777	751	-751	-5412	5412	1386	-1386
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: Shell outlet-1

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Shear P1 (SUS)	324	324	-324	-324	-324	-324	324	324
Shear Q (SUS)	558	558	558	558	558	558	558	558
Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Pm+P1 (SUS)	12727	30389	12943	30595	12960	30621	12711	30362
Pm+P1+Q (Total)	13367	30421	12934	30593	14126	30620	12924	30391

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30687	147988	Passed
Pm+P1 (SUS)	30621	221983	Passed
Pm+P1+Q (TOTAL)	30620	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell outlet-2

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Input, Nozzle Desc: Shell outlet-2

From: 40

Pressure for Reinforcement Calculations	P	14.7270	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4.73	m
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	23.0124	mm

Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



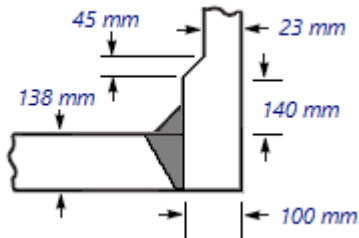
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Nozzle Calcs.: Shell outlet-2

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Reinforcement CALCULATION, Description: Shell outlet-2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 8.000 in.
Actual Thickness Used in Calculation 0.906 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.7 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
= 133.6591 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.7 \cdot 105 / (147989 \cdot 1 - 0.6 \cdot 14.7)$
= 11.0695 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.8049 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 77/0.5773
= 133.3465 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	519.2248	mm
Parallel to Vessel Wall	Rn+tn+t	259.6124	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	183.3775	mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n/S_v)
= min(1, $147989/147989$)
= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n/S_v)
= min(1, $147989/147989$)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(1, 1)
= 1.000



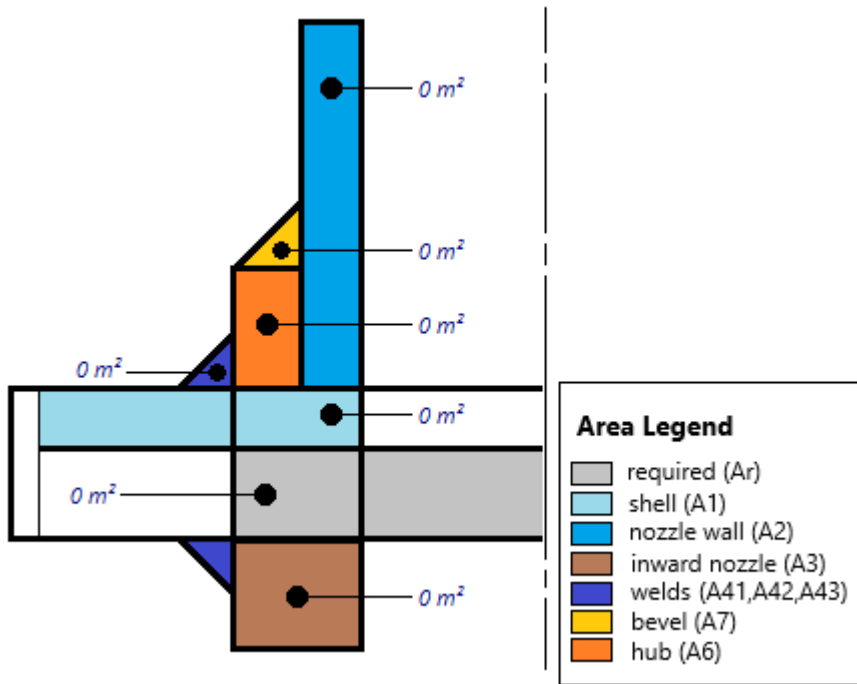
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Nozzle Calcs.: Shell outlet-2

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.028	0.001	NA
Area in Shell	A1	0.000	0.038	NA
Area in Nozzle Wall	A2	0.003	0.007	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	0.025	0.025	NA
TOTAL AREA AVAILABLE	Atot	0.029	0.070	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (209 \cdot 134 \cdot 1 + 2 \cdot 97 \cdot 134 \cdot 1 \cdot (1 - 1))$$

$$= 0.028 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 310(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 97$$

$$(1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1)$$

$$= 0.000 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 183)(20 - 11.1)1$$

$$= 0.003 \text{ m}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{bht})) \cdot (H_{btk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(183, 250, 140)) \cdot (100 - 23) \cdot 1$$

$$= 0.025 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 14.0695 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.6591 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.6591 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.1, \max(137, 5.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(14.1, 11.1)$$

$$= 14.0695 \text{ mm}$$

Available Nozzle Neck Thickness = 23.0124 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 59027.4,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 310944.4 kPa	Passed
Occasional	: 35124.8,	Allowable	: 196825.0 kPa	Passed
Shear	: 14473.4,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 100, tr = 11.1, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.11, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: Shell outlet-2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.028 - 0.00042 + 2 \cdot 97 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 4114850.50 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.028 + 0.99975E-04 - 0 \cdot 1) \cdot 147989 \\
 &= 4202223.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.028 + 0 + 0.99975E-04 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.028 + 0 + 0.99975E-04 + 0 + (0.026)) \cdot 147989 \\
 &= 8077944.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 403 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 459255. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 153) \cdot (100 - 3) \cdot 0.7 \cdot 147989 \\
 &= 4832944. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 403 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 9363169. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (459255 + 4832944) = 5292198 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (459255 + 0 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (459255 + 9363169 + 0) = 9822424 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 5292198 N	, must exceed W = 4114850 N	or W1 = 4202223 N
Path 2-2 = 9822424 N	, must exceed W = 4114850 N	or W2 = 8077944 N
Path 3-3 = 9822424 N	, must exceed W = 4114850 N	or W3 = 8077944 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: Shell outlet-2

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Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 16.2325 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 404.2325 mm

Input Echo, WRC297 Item 1, Description: Shell outlet-2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	203.200	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.73	N/mm ²
Radial Load	P	24999.98	N
Circumferential Shear	Vc	20000.03	N
Longitudinal Shear	Vl	20000.03	N
Circumferential Moment	Mc	12000.01	N-m
Longitudinal Moment	Ml	12000.01	N-m
Torsional Moment	Mt	14000.00	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (2.922) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.032) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 403.2 + 2 \cdot 1.65 \cdot \sqrt{1330.5(138.0 - 3.0)}$$

$$= 1801.783 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2661.000 mm
Nozzle Outside Diameter	(d) =	403.200 mm
Vessel Thickness used	(T) =	135.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	1.392



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Nozzle Calcs.: Shell outlet-2

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$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.673$$

Nr/P	=	0.163
Mr/P	=	0.208
MØ/P	=	0.086
NØ/P	=	0.149
MrD/Mc	=	0.262
NrDL/Mc	=	0.166
MØd/Mc	=	0.134
NØDL/Mc	=	-0.082
MrD/Ml	=	0.220
NrDL/Ml	=	0.148
MØD/Ml	=	0.112
NØDL/Ml	=	0.067

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-103	-103	103	103
Outplane Bending (ML)	-1089	1089	1089	-1089
Normal Pressure Stress	130776	145503	130776	145503
Outplane Stress Summary	128680	146992	131066	145021

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-241	-241	241	241
Inplane Bending (ML)	-2151	2151	2151	-2151
Inplane Pressure Stress	65388	65388	65388	65388
Inplane Stress Summary	61072	68780	65857	64960

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	227	227	-227	-227
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	399	399	399	399
Shear Stress Summary	627	627	172	172

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	128680	146992	131066	145021



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Nozzle Calcs.: Shell outlet-2

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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-199	-199	-199	-199
Outplane Bending (P)	-703	703	-703	703
Outplane Membrane (Mc)	130	130	-130	-130
Outplane Bending (Mc)	-1309	1309	1309	-1309
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65388	65388	65388	65388
Outplane Stress Summary	63306	67332	65664	64450

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-220	-220	-220	-220
Inplane Bending (P)	-1702	1702	-1702	1702
Inplane Membrane (Mc)	-268	-268	268	268
Inplane Bending (Mc)	-2564	2564	2564	-2564
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130776	145503	130776	145503
Inplane Stress Summary	126019	149281	131686	144690

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-227	-227	227	227
Torsional Shear (Mt)	399	399	399	399
Shear Stress Summary	172	172	627	627

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	126019	149281	131686	144690

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-103	-103	103	103
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687
Hoop Stress Summary	12713	30384	12920	30591

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2006	-2006	2006	2006
Axial Bending (ML)	-2764	2764	2764	-2764
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-537	9018	9004	7501
----------------------	------	------	------	------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	558	558	558	558

Shear Stress Summary	882	882	234	234
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	13361	30418	12934	30591
------------------------	-------	-------	-------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-199	-199	-199	-199
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	130	130	-130	-130
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13017	30687	13017	30687

Hoop Stress Summary	12948	30618	12686	30356
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	-261	-261	-261	-261
Axial Bending (P)	-2013	2013	-2013	2013
Axial Membrane (Mc)	-2006	-2006	2006	2006
Axial Bending (Mc)	-3399	3399	3399	-3399
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6508	6508	6508	6508

Axial Stress Summary	-1172	9652	9638	6867
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-324	-324	324	324



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Shear due to Torsion	558	558	558	558
----------------------	-----	-----	-----	-----

Shear Stress Summary	234	234	882	882
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14127	30618	12920	30384

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Circ. Pl (SUS)	-302	-302	-96	-96	-488	-488	48	48
Circ. Q (SUS)	-1792	1792	386	-386	-4266	4266	862	-862
Long. Pm (SUS)	65388	65388	65388	65388	65388	65388	65388	65388
Long. Pl (SUS)	-461	-461	21	21	-69	-69	-329	-329
Long. Q (SUS)	-3853	3853	449	-449	-2012	2012	606	-606
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	227	227	-227	-227	-227	-227	227	227
Shear Q (SUS)	399	399	399	399	399	399	399	399
Pm (SUS)	130776	145503	130776	145503	130776	145503	130776	145503
Pm+Pl (SUS)	130474	145201	130680	145407	130288	145015	130824	145551
Pm+Pl+Q (Total)	128687	146998	131066	145021	126022	149281	131691	144693

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145503	147988	Passed
Pm+Pl (SUS)	145551	221983	Passed
Pm+Pl+Q (TOTAL)	149281	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Circ. Pl (SUS)	-302	-302	-96	-96	-69	-69	-329	-329
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6508	6508	6508	6508	6508	6508	6508	6508
Long. Pl (SUS)	-2267	-2267	1745	1745	-2267	-2267	1745	1745
Long. Q (SUS)	-4777	4777	751	-751	-5412	5412	1386	-1386
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: Shell outlet-2

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Shear P1 (SUS)	324	324	-324	-324	-324	-324	324	324
Shear Q (SUS)	558	558	558	558	558	558	558	558
Pm (SUS)	13017	30687	13017	30687	13017	30687	13017	30687
Pm+P1 (SUS)	12727	30389	12943	30595	12960	30621	12711	30362
Pm+P1+Q (Total)	13367	30421	12934	30593	14126	30620	12924	30391

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30687	147988	Passed
Pm+P1 (SUS)	30621	221983	Passed
Pm+P1+Q (TOTAL)	30620	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: BD1

Nozl: 42 11:13am Apr 10,2026

Input, Nozzle Desc: BD1

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	150.0000	mm
-------------------------------------	------	----------	----

Distance from Bottom/Left Tangent		3.24	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

Flange Material		SA-336 F11	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	190.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



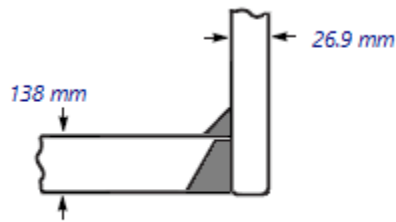
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Nozzle Calcs.: BD1

Nozl: 42 11:13am Apr 10, 2026



Reinforcement CALCULATION, Description: BD1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
 $= 133.8976 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([14.8 / (147989 \cdot 1)] - 1))$
 $= 2.9767 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4325 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.0000 mm
Parallel to Vessel Wall, opening length	d	75.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: BD1.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.9767 mm
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm
Wall Thickness per table UG-45	tb3	= 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(137, 5.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.98, 8.26)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = 26.9240 mm -> OK



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Nozzle Calcs.: BD1

Nozl: 42 11:13am Apr 10,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)**

Govrn. thk, tg = 26.9, tr = 2.98, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: BD1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0869 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 329.5869 mm

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Nozzle Calcs.: BD2

Nozl: 43 11:13am Apr 10,2026

Input, Nozzle Desc: BD2

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	150.0000	mm
-------------------------------------	------	----------	----

Distance from Bottom/Left Tangent		4.59	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	190.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



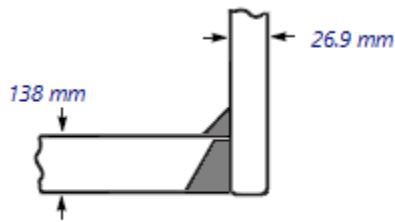
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Nozzle Calcs.: BD2

Nozl: 43 11:13am Apr 10,2026



Reinforcement CALCULATION, Description: BD2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
 $= 133.8976 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([14.8 / (147989 \cdot 1)] - 1))$
 $= 2.9767 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4325 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.0000 mm
Parallel to Vessel Wall, opening length	d	75.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: BD2.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.9767 mm
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm
Wall Thickness per table UG-45	tb3	= 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(137, 5.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.98, 8.26)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = 26.9240 mm -> OK



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Nozzle Calcs.: BD2

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Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)**

Govrn. thk, tg = 26.9, tr = 2.98, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: BD2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0869 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 329.5869 mm

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Nozzle Calcs.: shell inlet-2

Nozl: 44 11:13am Apr 10,2026

Input, Nozzle Desc: shell inlet-2

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4.98	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	18.2600	mm

Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



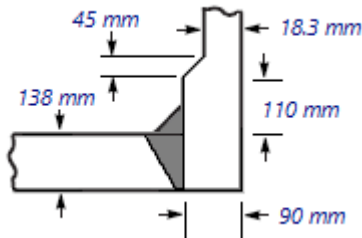
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Nozzle Calcs.: shell inlet-2

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Reinforcement CALCULATION, Description: shell inlet-2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 6.000 in.
Actual Thickness Used in Calculation 0.719 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
= 133.8976 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $14.8 \cdot 79.2 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
= 8.3964 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.6836 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 71.7/0.5773
= 124.2574 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	458.9200	mm
Parallel to Vessel Wall	Rn+tn+t	229.4600	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	162.4074	mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n/S_v)
= min(1, $147989/147989$)
= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n/S_v)
= min(1, $147989/147989$)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(1, 1)
= 1.000



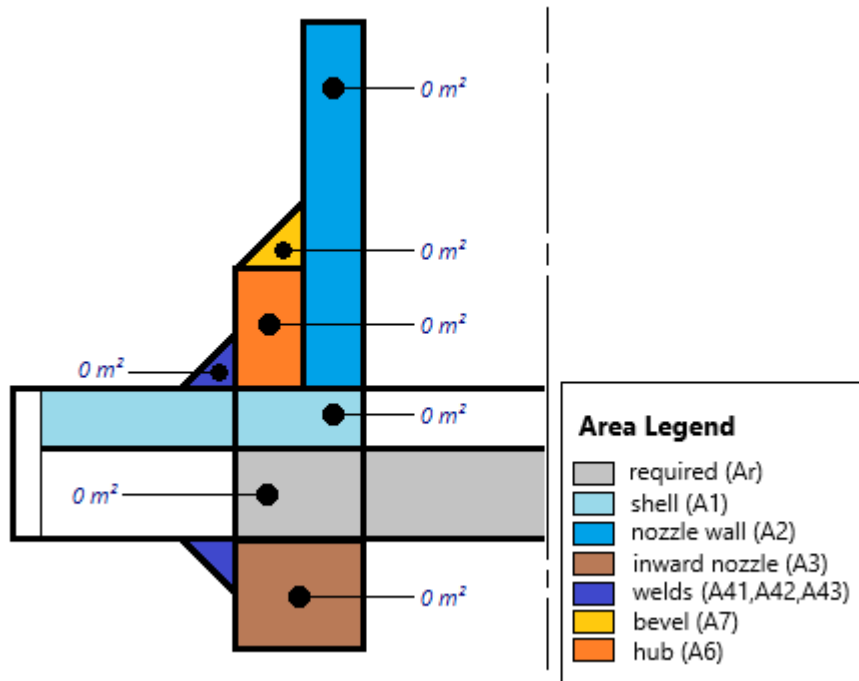
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Nozzle Calcs.: shell inlet-2

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.021	0.001	NA
Area in Shell	A1	0.000	0.037	NA
Area in Nozzle Wall	A2	0.002	0.005	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.019	0.019	NA
TOTAL AREA AVAILABLE	Atot	0.022	0.061	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (158 \cdot 134 \cdot 1 + 2 \cdot 87 \cdot 134 \cdot 1 \cdot (1 - 1))$$

$$= 0.021 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 301(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 87(1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1)$$

$$= 0.000 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 162)(15.3 - 8.4)1$$

$$= 0.002 \text{ m}^2$$



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Nozzle Calcs.: shell inlet-2

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(162, 250, 110)) \cdot (90 - 18.3) \cdot 1$$

$$= 0.019 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 11.3964 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.2, \max(137, 5.5)]$$

$$= 10.1600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(11.4, 10.2)$$

$$= 11.3964 \text{ mm}$$

Available Nozzle Neck Thickness = 18.2600 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 65667.4,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 304304.4 kPa	Passed
Occasional	: 34915.0,	Allowable	: 196825.0 kPa	Passed
Shear	: 18676.0,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 90, tr = 8.4, c = 3 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.097, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	47 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: shell inlet-2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: shell inlet-2 Nozl: 44 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.021 - 0.00033 + 2 \cdot 87 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 3118023.25 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.021 + 0.99975\text{E-}04 - 0 \cdot 1) \cdot 147989 \\
 &= 3158066.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.021 + 0 + 0.99975\text{E-}04 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.021 + 0 + 0.99975\text{E-}04 + 0 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 332 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 378612. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 123) \cdot (90 - 3) \cdot 0.7 \cdot 147989 \\
 &= 3473990. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 332 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 7719042. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (378612 + 3473990) = 3852602 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (378612 + 0 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (378612 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 3852602 N	, must exceed W = 3118023 N	or W1 = 3158066 N
Path 2-2 = 8097653 N	, must exceed W = 3118023 N	or W2 = 6634228 N
Path 3-3 = 8097653 N	, must exceed W = 3118023 N	or W3 = 6634228 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: shell inlet-2

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Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 11.0094 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.0094 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 79.38 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 229.46 mm

Input Echo, WRC297 Item 1, Description: shell inlet-2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	152.400	mm
Nozzle Thickness	90.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.75	N/mm ²
Radial Load	P	18750.05	N
Circumferential Shear	Vc	15000.04	N
Longitudinal Shear	Vl	15000.04	N
Circumferential Moment	Mc	6749.99	N-m
Longitudinal Moment	Ml	6749.99	N-m
Torsional Moment	Mt	7875.00	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (2.409) is < 5
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.693) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 332.4 + 2 \cdot 1.65 \cdot \sqrt{(1330.5(138.0 - 3.0))} \\
 &= 1730.983 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2661.000 mm



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Nozzle Calcs.: shell inlet-2

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Nozzle Outside Diameter (d) = 332.400 mm
 Vessel Thickness used (T) = 135.000 mm
 Nozzle Thickness used (t) = 87.000 mm
 T / t = 1.552
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.555$$

Nr/P = 0.181
 Mr/P = 0.217
 M0/P = 0.099
 N0/P = 0.160
 MrD/Mc = 0.254
 NrDL/Mc = 0.174
 M0d/Mc = 0.146
 N0DL/Mc = -0.095
 MrD/Ml = 0.226
 NrDL/Ml = 0.164
 M0D/Ml = 0.129
 N0DL/Ml = 0.026

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-27	-27	27	27
Outplane Bending (ML)	-854	854	854	-854
Normal Pressure Stress	130997	145744	130997	145744
Outplane Stress Summary	129342	147027	131107	145372

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-179	-179	179	179
Inplane Bending (ML)	-1509	1509	1509	-1509
Inplane Pressure Stress	65498	65498	65498	65498
Inplane Stress Summary	62285	67980	65664	65319

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	206	206	-206	-206
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	330	330	330	330
Shear Stress Summary	537	537	124	124

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: shell inlet-2

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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	129342	147027	131107	145372

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	103	103	-103	-103
Outplane Bending (Mc)	-972	972	972	-972
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65498	65498	65498	65498
Outplane Stress Summary	63857	67029	65595	64878

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	-193	-193	193	193
Inplane Bending (Mc)	-1689	1689	1689	-1689
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130997	145744	130997	145744
Inplane Stress Summary	127591	148392	131355	145400

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-206	-206	206	206
Torsional Shear (Mt)	330	330	330	330
Shear Stress Summary	124	124	537	537

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	127591	148392	131355	145400

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-27	-27	27	27
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095



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Nozzle Calcs.: shell inlet-2

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Hoop Stress Summary	10210	27909	10266	27964
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1971	-1971	1971	1971
Axial Bending (ML)	-2323	2323	2323	-2323
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-1247	7149	7342	6446
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	517	517	517	517

Shear Stress Summary	841	841	193	193
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	11576	27936	10272	27964
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	103	103	-103	-103
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095

Hoop Stress Summary	10341	28040	10135	27833
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	-1971	-1971	1971	1971
Axial Bending (Mc)	-2675	2675	2675	-2675
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-1599	7501	7694	6094
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
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Nozzle Calcs.: shell inlet-2

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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-324	-324	324	324
Shear due to Torsion	517	517	517	517
Shear Stress Summary	193	193	841	841

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11941	28040	10390	27861

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Circ. Pl (SUS)	-185	-185	-131	-131	-379	-379	7	7
Circ. Q (SUS)	-1467	1467	241	-241	-3026	3026	352	-352
Long. Pm (SUS)	65498	65498	65498	65498	65498	65498	65498	65498
Long. Pl (SUS)	-365	-365	-7	-7	-55	-55	-261	-261
Long. Q (SUS)	-2846	2846	172	-172	-1585	1585	359	-359
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	206	206	-206	-206	-206	-206	206	206
Shear Q (SUS)	330	330	330	330	330	330	330	330
Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Pm+Pl (SUS)	130812	145559	130866	145613	130618	145365	131004	145751
Pm+Pl+Q (Total)	129349	147029	131107	145372	127592	148391	131360	145402

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145744	147988	Passed
Pm+P1 (SUS)	145751	221983	Passed
Pm+P1+Q (TOTAL)	148391	443966	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

[illegible]



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Nozzle Calcs.: shell inlet-2

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Long. P1 (SUS)	-2246	-2246	1696	1696	-2246	-2246	1696	1696
Long. Q (SUS)	-4198	4198	448	-448	-4550	4550	800	-800
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	324	324	-324	-324	-324	-324	324	324
Shear Q (SUS)	517	517	517	517	517	517	517	517
Pm (SUS)	10397	28095	10397	28095	10397	28095	10397	28095
Pm+P1 (SUS)	10226	27914	10296	27968	10356	28044	10168	27839
Pm+P1+Q (Total)	11580	27944	10278	27965	11946	28041	10397	27866

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28095	147988	Passed
Pm+P1 (SUS)	28044	221983	Passed
Pm+P1+Q (TOTAL)	28041	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell inlet-3

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Input, Nozzle Desc: Shell inlet-3

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5.71	m
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	18.2600	mm

Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



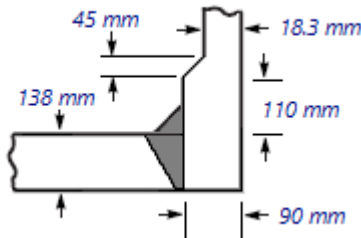
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Nozzle Calcs.: Shell inlet-3

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Reinforcement CALCULATION, Description: Shell inlet-3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	6.000 in.
Actual Thickness Used in Calculation	0.719 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
 $= 133.8976 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.8 \cdot 79.2 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
 $= 8.3964 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6836 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 71.7 / 0.5773$
 $= 124.2574 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	458.9200 mm
Parallel to Vessel Wall	Rn+tn+t	229.4600 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	162.4074 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147989 / 147989)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147989 / 147989)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



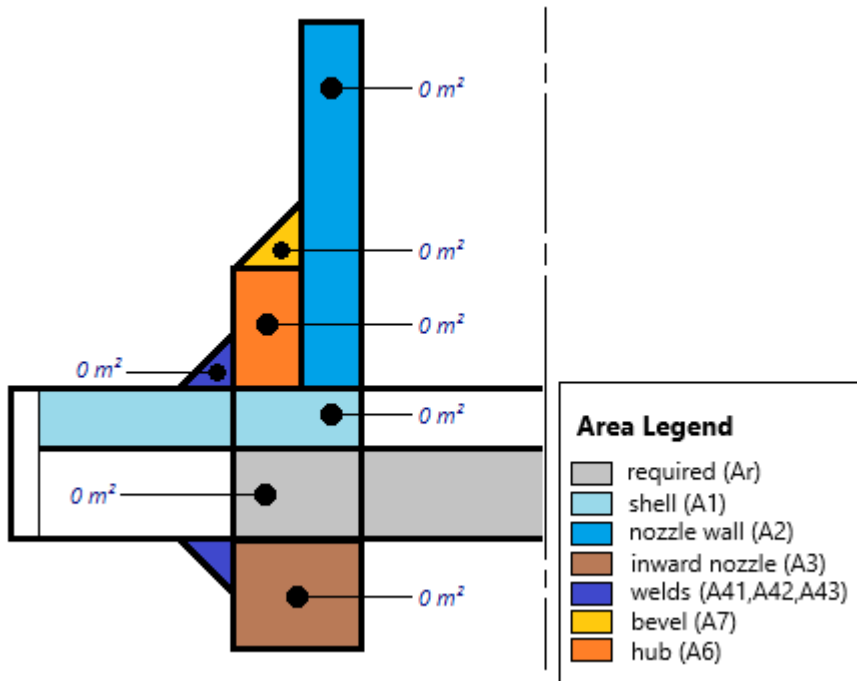
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Nozzle Calcs.: Shell inlet-3

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.021	0.001	NA
Area in Shell	A1	0.000	0.037	NA
Area in Nozzle Wall	A2	0.002	0.005	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.019	0.019	NA
TOTAL AREA AVAILABLE	Atot	0.022	0.061	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (158 \cdot 134 \cdot 1 + 2 \cdot 87 \cdot 134 \cdot 1 \cdot (1 - 1))$$

$$= 0.021 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 301(1 \cdot 135 - 1 \cdot 134) - 2 \cdot 87(1 \cdot 135 - 1 \cdot 134) \cdot (1 - 1)$$

$$= 0.000 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 162)(15.3 - 8.4)1$$

$$= 0.002 \text{ m}^2$$



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Nozzle Calcs.: Shell inlet-3

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(162, 250, 110)) \cdot (90 - 18.3) \cdot 1$$

$$= 0.019 \text{ m}^2 \text{ Includes Bevel Area of } 0.003 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 11.3964 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.2, \max(137, 5.5)]$$

$$= 10.1600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(11.4, 10.2)$$

$$= 11.3964 \text{ mm}$$

Available Nozzle Neck Thickness = 18.2600 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 65667.4,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 304304.4 kPa	Passed
Occasional	: 34915.0,	Allowable	: 196825.0 kPa	Passed
Shear	: 18634.4,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 90, tr = 8.4, c = 3 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.097, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	47 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: Shell inlet-3

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: Shell inlet-3 Nozl: 45 11:13am Apr 10, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (0.021 - 0.00033 + 2 \cdot 87 \cdot 1 \cdot (1 \cdot 135 - 134)) \cdot 147989) \\
 &= 3118023.25 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (0.021 + 0.99975E-04 - 0 \cdot 1) \cdot 147989 \\
 &= 3158066.50 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (0.021 + 0 + 0.99975E-04 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (0.021 + 0 + 0.99975E-04 + 0 + (0.023)) \cdot 147989 \\
 &= 6634228.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 332 \cdot 10 \cdot 0.49 \cdot 147989 \\
 &= 378612. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 123) \cdot (90 - 3) \cdot 0.7 \cdot 147989 \\
 &= 3473990. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 332 \cdot (138 - 3) \cdot 0.74 \cdot 147989 \\
 &= 7719042. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (378612 + 3473990) = 3852602 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (378612 + 0 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (378612 + 7719041 + 0) = 8097653 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 3852602 N	, must exceed W = 3118023 N	or W1 = 3158066 N
Path 2-2 = 8097653 N	, must exceed W = 3118023 N	or W2 = 6634228 N
Path 3-3 = 8097653 N	, must exceed W = 3118023 N	or W3 = 6634228 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.8 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: Shell inlet-3

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Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 11.0094 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.0094 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 79.38 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 229.46 mm

Input Echo, WRC297 Item 1, Description: Shell inlet-3 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2520.000	mm
Shell Thickness	138.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	152.400	mm
Nozzle Thickness	90.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	14.75	N/mm ²
Radial Load	P	18750.05	N
Circumferential Shear	Vc	15000.04	N
Longitudinal Shear	Vl	15000.04	N
Circumferential Moment	Mc	6749.99	N-m
Longitudinal Moment	Ml	6749.99	N-m
Torsional Moment	Mt	7849.99	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (2.409) is < 5
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.693) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 332.4 + 2 \cdot 1.65 \cdot \sqrt{(1330.5(138.0 - 3.0))} \\
 &= 1730.983 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2661.000 mm



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Nozzle Calcs.: Shell inlet-3

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Nozzle Outside Diameter (d) = 332.400 mm
 Vessel Thickness used (T) = 135.000 mm
 Nozzle Thickness used (t) = 87.000 mm
 T / t = 1.552
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.555$$

Nr/P = 0.181
 Mr/P = 0.217
 M0/P = 0.099
 N0/P = 0.160
 MrD/Mc = 0.254
 NrDL/Mc = 0.174
 M0d/Mc = 0.146
 N0DL/Mc = -0.095
 MrD/Ml = 0.226
 NrDL/Ml = 0.164
 M0D/Ml = 0.129
 N0DL/Ml = 0.026

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-27	-27	27	27
Outplane Bending (ML)	-854	854	854	-854
Normal Pressure Stress	130997	145744	130997	145744
Outplane Stress Summary	129342	147027	131107	145372

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-179	-179	179	179
Inplane Bending (ML)	-1509	1509	1509	-1509
Inplane Pressure Stress	65498	65498	65498	65498
Inplane Stress Summary	62285	67980	65664	65319

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	206	206	-206	-206
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	330	330	330	330
Shear Stress Summary	537	537	124	124

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: Shell inlet-3

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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	129342	147027	131107	145372

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-158	-158	-158	-158
Outplane Bending (P)	-613	613	-613	613
Outplane Membrane (Mc)	103	103	-103	-103
Outplane Bending (Mc)	-972	972	972	-972
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65498	65498	65498	65498
Outplane Stress Summary	63857	67029	65595	64878

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-186	-186	-186	-186
Inplane Bending (P)	-1337	1337	-1337	1337
Inplane Membrane (Mc)	-193	-193	193	193
Inplane Bending (Mc)	-1689	1689	1689	-1689
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130997	145744	130997	145744
Inplane Stress Summary	127591	148392	131355	145400

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-206	-206	206	206
Torsional Shear (Mt)	330	330	330	330
Shear Stress Summary	124	124	537	537

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	127591	148392	131355	145400

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-27	-27	27	27
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095



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Nozzle Calcs.: Shell inlet-3 Nozl: 45 11:13am Apr 10,2026

Hoop Stress Summary	10210	27909	10266	27964
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1971	-1971	1971	1971
Axial Bending (ML)	-2323	2323	2323	-2323
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-1247	7149	7342	6446
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	324	324	-324	-324
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	517	517	517	517

Shear Stress Summary	841	841	193	193
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	11576	27936	10272	27964
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-158	-158	-158	-158
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	103	103	-103	-103
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10397	28095	10397	28095

Hoop Stress Summary	10341	28040	10135	27833
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-275	-275	-275	-275
Axial Bending (P)	-1875	1875	-1875	1875
Axial Membrane (Mc)	-1971	-1971	1971	1971
Axial Bending (Mc)	-2675	2675	2675	-2675
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5198	5198	5198	5198

Axial Stress Summary	-1599	7501	7694	6094
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
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Nozzle Calcs.: Shell inlet-3

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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-324	-324	324	324
Shear due to Torsion	517	517	517	517
Shear Stress Summary	193	193	841	841

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11941	28040	10390	27861

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Circ. Pl (SUS)	-185	-185	-131	-131	-379	-379	7	7
Circ. Q (SUS)	-1467	1467	241	-241	-3026	3026	352	-352
Long. Pm (SUS)	65498	65498	65498	65498	65498	65498	65498	65498
Long. Pl (SUS)	-365	-365	-7	-7	-55	-55	-261	-261
Long. Q (SUS)	-2846	2846	172	-172	-1585	1585	359	-359
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	206	206	-206	-206	-206	-206	206	206
Shear Q (SUS)	330	330	330	330	330	330	330	330
Pm (SUS)	130997	145744	130997	145744	130997	145744	130997	145744
Pm+Pl (SUS)	130812	145559	130866	145613	130618	145365	131004	145751
Pm+Pl+Q (Total)	129349	147029	131107	145372	127592	148391	131360	145402

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145744	147988	Passed
Pm+P1 (SUS)	145751	221983	Passed
Pm+P1+Q (TOTAL)	148391	443966	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

[illegible]



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Nozzle Calcs.: Shell inlet-3

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Long. P1 (SUS)	-2246	-2246	1696	1696	-2246	-2246	1696	1696
Long. Q (SUS)	-4198	4198	448	-448	-4550	4550	800	-800
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	324	324	-324	-324	-324	-324	324	324
Shear Q (SUS)	517	517	517	517	517	517	517	517
Pm (SUS)	10397	28095	10397	28095	10397	28095	10397	28095
Pm+P1 (SUS)	10226	27914	10296	27968	10356	28044	10168	27839
Pm+P1+Q (Total)	11580	27944	10278	27965	11946	28041	10397	27866

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28095	147988	Passed
Pm+P1 (SUS)	28044	221983	Passed
Pm+P1+Q (TOTAL)	28041	443966	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: BD3

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Input, Nozzle Desc: BD3

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	150.0000	mm
-------------------------------------	------	----------	----

Distance from Bottom/Left Tangent		5.37	m
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	190.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



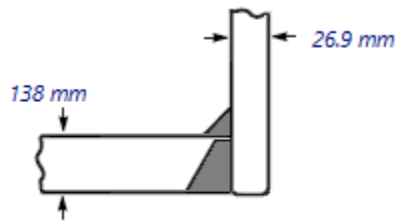
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Nozzle Calcs.: BD3

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Reinforcement CALCULATION, Description: BD3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
 $= 133.8976 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([14.8 / (147989 \cdot 1)] - 1))$
 $= 2.9767 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4325 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.0000 mm
Parallel to Vessel Wall, opening length	d	75.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: BD3.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.9767 mm
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm
Wall Thickness per table UG-45	tb3	= 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(137, 5.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.98, 8.26)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = 26.9240 mm -> OK



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FileName : HTS EFF

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Nozzle Calcs.: BD3

Nozl: 46 11:13am Apr 10,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)**

Govrn. thk, tg = 26.9, tr = 2.98, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: BD3

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0869 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 329.5869 mm

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Nozzle Calcs.: BD5

Nozl: 47 11:13am Apr 10,2026

Input, Nozzle Desc: BD5

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	150.0000	mm
-------------------------------------	------	----------	----

Distance from Bottom/Left Tangent		6.04	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	190.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



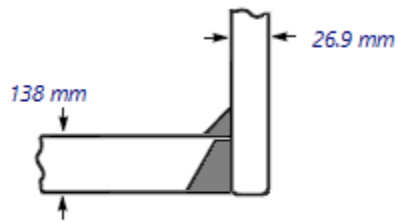
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Nozzle Calcs.: BD5

Nozl: 47 11:13am Apr 10,2026



Reinforcement CALCULATION, Description: BD5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
 $= 133.8976 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([14.8 / (147989 \cdot 1)] - 1))$
 $= 2.9767 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4325 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.0000 mm
Parallel to Vessel Wall, opening length	d	75.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: BD5.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.9767 mm
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm
Wall Thickness per table UG-45	tb3	= 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(137, 5.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.98, 8.26)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = 26.9240 mm -> OK



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FileName : HTS EFF

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Nozzle Calcs.: BD5

Nozl: 47 11:13am Apr 10,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)**

Govrn. thk, tg = 26.9, tr = 2.98, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: BD5

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * Wo$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0869 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 329.5869 mm

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Nozzle Calcs.: BD4

Nozl: 48 11:13am Apr 10,2026

Input, Nozzle Desc: BD4

From: 40

Pressure for Reinforcement Calculations	P	14.7517	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2520.00	mm
Design Length of Section	L	6120.0308	mm
Shell Finished (Minimum) Thickness	t	138.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	150.0000	mm
-------------------------------------	------	----------	----

Distance from Bottom/Left Tangent		7.16	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	190.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	138.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



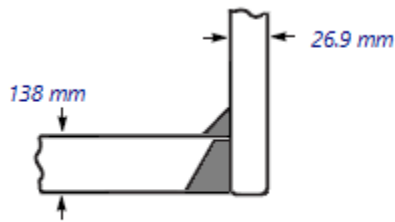
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Nozzle Calcs.: BD4

Nozl: 48 11:13am Apr 10, 2026



Reinforcement CALCULATION, Description: BD4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.8 \cdot 1263 / (147989 \cdot 1 - 0.6 \cdot 14.8)$
 $= 133.8976 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([14.8 / (147989 \cdot 1)] - 1))$
 $= 2.9767 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4325 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.0000 mm
Parallel to Vessel Wall, opening length	d	75.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: BD4.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.9767 mm
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 136.8976 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 136.8976 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.8649 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm
Wall Thickness per table UG-45	tb3	= 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(137, 5.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.98, 8.26)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = 26.9240 mm -> OK



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Nozzle Calcs.: BD4

Nozl: 48 11:13am Apr 10,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)**

Govrn. thk, tg = 26.9, tr = 2.98, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: BD4

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0869 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 329.5869 mm

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ASME TS Calc: DESIGN CASE-

Case: 1 11:13a Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	14.28 N/mm ²
Shell Maximum Operating Pressure	Psox,max	14.28 N/mm ²
Shell Minimum Operating Pressure	Psox,min	12.55 N/mm ²
Shell Thickness	ts	138.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2520.000 mm
Shell Temperature for Internal Pressure	Ts	343.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	147988.73 kPa
Shell Allowable Stress at Ambient		147988.73 kPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4.95 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	4.95 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	4.44 N/mm ²
Channel Thickness	tc	68.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2520.000 mm
Channel Design Temperature	TEMPC	475.00 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	116013.23 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	4952
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	4800.00 mm
Straight Tube Length (bet. inner tubsht faces) L		4260.00 mm
Design Temperature of the Tubes		425.00 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		113991.31 kPa
Tube Allowable Stress At Ambient		117991.02 kPa
Tube Yield Stress At design Temperature	Syt	183987.41 kPa
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1215.220 mm
Largest Center-to-Center Tube Distance	U1	101.6000 mm



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ASME TS Calc: DESIGN CASE-

Case: 1 11:13a Apr 10,2026

Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	4.4 m ²
Perimeter of Tube Layout	Cp	7805.0 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	475.00 °C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

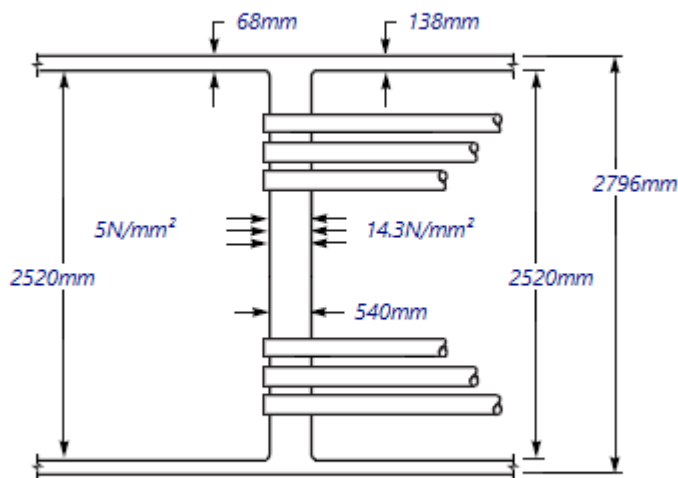
Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	116013.27 kPa
Tubesheet Allowable Stress at Ambient	Tt	147988.41 kPa
Thickness of Tubesheet	h	540.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2796.000 mm

Area of the Untubed Lanes	AL	0.192 m ²
---------------------------	----	----------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025**Elasticity/Expansion Material Properties :**

Shell - TM-1 Material Group C



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ASME TS Calc: DESIGN CASE-

Case: 1 11:13a Apr 10, 2026

Elastic Mod. at Design Temperature	343.0 °C	0.18309E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Channel - TM-1 Material Group D

Elastic Mod. at Design Temperature	475.0 °C	0.17714E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tubes - TM-1 Material Group D

Elastic Mod. at Tubsht. Design Temp.	475.0 °C	0.17714E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	475.0 °C	0.17714E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21099E+06 N/mm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4.95 \cdot (25.4/2 - 0)) / (113991 \cdot 1 + 0.4 \cdot 4.95)$$

$$= 0.5421 + 0.0000 = 0.5421 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 5475 / \sqrt{(12.7 \cdot 2.11)}$$

$$= 1057.690$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)0.94, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (25.4/2.11)$$

$$= 0.0457$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.046 \cdot 181486 (2.11/25.4)$$

$$= 1102044.250 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 1102044.2/183987.4$$

$$= 5.990$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 183987.406 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [183987] \geq F_y [183987], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y/FS$$

$$= 183987.406/1.667$$

$$= 110370.367 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$

$$= 2 \cdot 110 (2.11/25.4)$$

$$= 18.338 \text{ N/mm}^2$$

Summary of Tube Required Thickness Results:



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ASME TS Calc: DESIGN CASE-

Case: 1 11:13a Apr 10, 2026

Total Required Thickness including Corrosion all.	1.6432 mm
Allowable Internal Pressure at Corroded thickness	20.29 N/mm ²
Required Internal Design Pressure	4.95 N/mm ²
Allowable External Pressure at Corroded thickness	18.34 N/mm ²
Required External Design Pressure	14.28 N/mm ²
Required Thickness due to Shell Side pressure	1.6432 mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1215 + 25.4$$

$$= 2455.840 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2526 / 2456$$

$$= 1.0286$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2526 / 2456$$

$$= 1.0286$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 50 / 524$$

$$= 0.095 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t((Et/E) \cdot (StT/S) \cdot (\rho)); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 2.11 (177143 / 177143) \cdot (0.095); 25.4 - 2 \cdot 2.11)$$

$$= 25.0529 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.19; 4 \cdot 2456 \cdot 0.095) / (\pi \cdot 2456^2)) }$$

$$= 32.4138 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 25.1) / 32.4$$

$$= 0.2271$$

$h/p = 16.503937$; $\mu^* = 0.227093$; $E^* = 33304 \text{ N/mm}^2$

$E^*/E = 0.188009$; $v^* = 0.422233$



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Note: As h/p (16.504) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= \left(\left(12(1 - \nu_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2526 + 135)135} \right)$$

$$= 0.0030 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 183094 \cdot 135^3 / (6(1 - 0.3^2))$$

$$= 250235248.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2526 + 135) \cdot 250235248 / 524^3 \cdot (1 + 524 \cdot 0.003 + (524 \cdot 0.003)^2 / 2)$$

$$= 106968.3 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2526^2 / (4 \cdot 183094 \cdot 135) \cdot (1 - 2526 \cdot 0.3 / (2(2526 + 135)))$$

$$= 0.55350\text{E-}04 \text{ mm/kPa}$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 250235248 \cdot 0.003 \cdot 0.55350\text{E-}04 (1 + 524 \cdot 0.003)$$

$$= 111870.2656 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2526 + 65)65} \right)$$

$$= 0.0044 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0044 \cdot 177143 \cdot 65^3 / (6(1 - 0.3^2))$$

$$= 39467212.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2526 + 65 / \cos(0)) \cdot 39467212 / 524^3 \cdot (1 + 524 \cdot 0.0044 \cdot (1 + 1) / 2 + (524 \cdot 0.0044)^2 / 2 \cdot 1)$$

$$= 25650.1 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2526^2 / (4 \cdot 177143 \cdot 65) \cdot (1 - 2526 \cdot 0.3 / (2(2526 + 65)))$$

$$= 0.11829\text{E-}03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 39467212 [0.000118286 (1 + 0.0044 \cdot 1 \cdot 524) - 0.000 (1/524 + 1 \cdot 0.0044/2)]$$

$$= 70635.9609 \text{ mm}^2$$



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ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2796/2456$$

$$= 1.1385$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 33304 \cdot (106968 + 25650 + 177143 \cdot \ln(1.14))$$

$$= 2.6993$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2456^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 0 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 4.95]$$

$$= -109707.2734 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -109707 + 70636 \cdot 4.95 - 111870 \cdot 0$$

$$= 239961.0312 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F)$$

$$= (239961) - 2456^2 / 32 \cdot 2.7 (0 - 4.95) / (1 + 2.7)$$

$$= 745655.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 745656 + 2456^2 / 64 \cdot (3 + 0.42) \cdot (0 - 4.95)$$

$$= -850812.3750 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|745656|; |-850812|)$$

$$= 850812.3750 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 850812 / (0.23 (524 - 0)^2)$$

$$= 81864.1875 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 116013$$

$$= 232026.53 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 835969 / (0.23 (303 - 0)^2)$$

$$= 231990.7344 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 303 + 3 + 13$$

$$= 319.3886 \text{ mm}$$



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Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS}) \\ = \max(319; 167) \\ = 319.3886 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | \text{Ps} - \text{Pt} | \\ = | 0 - 4.95 | \\ = 4.950 \text{ N/mm}^2$$

Shear Stress check [τlimit]:

$$= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot \text{S}; 0.533 \cdot \text{Sy}) \\ = 4 \cdot 0.2 \cdot 524 / 2456 \cdot \min(0.8 \cdot 116013; 0.533 \cdot 224994) \\ = 15842.323 \text{ kPa}$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot \text{S}; 0.533 \cdot \text{Sy})) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\ = 1 / (4 \cdot 0.2 \cdot 92811) \cdot (4 \cdot 4.41 / 7805) \cdot | 0 - 4.95 | + 16 \\ = 166.6563 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 135.000mm

Min. Shell len. adjacent to the tubesheet is: 1051.129 mm

Axial Shell Membrane Stress [σs,m]:

$$= \text{Ds}^2 / (4 \cdot \text{ts} (\text{Ds} + \text{ts})) \cdot \text{Ps} \\ = 2526^2 / (4 \cdot 135 (2526 + 135)) \cdot 0 \\ = 0.0000 \text{ kPa}$$

Axial Shell Bending Stress [σs,b]:

$$= 6 \cdot \text{ks} / \text{ts}^2 \cdot [\beta \text{S} \cdot \text{P's} + 6(1 - \nu^*) / \text{E}^* \cdot \text{Do} / \text{h}^3 \cdot (1 + \text{h} \cdot \beta \text{S} / 2) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\ = 6 \cdot 56253144 / 135^2 \cdot [0.003 \cdot 0 + 6(1 - 0.42) / 33304 \cdot 2456 / 524^3 \cdot (1 + 524 \cdot 0.003 / 2) \cdot (745656 + 2456^2 / 32 \cdot (0 - 4.95))] \\ = -49206.875 \text{ kPa}$$

Shell Membrane + Bending Stress [σs]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot \text{Ss)} \\ = 49206.875 \text{ kPa (Should be } \leq 221983.11)$$

Note: For a given Channel thickness of: 65.000 mm

Min. Channel len. adjacent to the tubesheet is: 729.367mm

Axial Channel Membrane Stress [σc,m]:

$$= \text{Dc}^2 / (4 \cdot \text{tc} (\text{Dc} + \text{tc})) \cdot \cos(\text{theta}) \cdot \text{Pt} \\ = 2526^2 / (4 \cdot 65 (2526 + 65)) \cdot \cos(0) \cdot 4.95 \\ = 46884.695 \text{ kPa}$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot \text{kc} / \text{tc}^2 \cdot [(\delta_c \cdot \text{Cm} \Delta \cdot \beta_c - \zeta_c \cdot \text{Cm} \Theta) \cdot \text{Pt} - 6((1 - \nu^*) / \text{E}^*) \cdot \text{Do} / \text{h}^3 \cdot (\text{Cm} \Theta + \text{h} \cdot \beta_c / 2 \cdot \text{Cm} \Delta) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\ = 6 \cdot 8872270 / 65^2 \cdot [(0.000118286 \cdot 1 \cdot 0.0044 - 0.000 \cdot 1) \cdot 4.95 - 6((1 - 0.42) / 33304) \cdot 2456 / 524^3 \cdot (1 + 0.0044 \cdot 524 / 2 \cdot 1) \cdot (745656 + 2456^2 / 32 \cdot (0 - 4.95))] \\ = 185661.469 \text{ kPa}$$

Channel Membrane + Bending Stress [σc]:



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$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c \text{)}$$

$$= 232546.156 \text{ kPa (Should be } \leq 174019.86 \text{)}$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 177143 \cdot \sqrt{(1.5 \cdot 116013 / 232546)}$$

$$= 153238.7 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 853903 / (0.23 \cdot (524 - 0)^2)$$

$$= 82161.6 \text{ kPa (Should be } \leq 232026.5 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 303 + 3 + 13 = 319 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS}) = \max(319; 167) = 319 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) \cdot 2.11$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) \cdot 113991$$

$$= 17597.941 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) \cdot 113991$$

$$= 17597.9414 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 113991 / 113991$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) \cdot 113991$$

$$= 16195.8193 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \cdot \sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d) - 0.75 \cdot d_o}$$

$$= 2 \cdot \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1 \cdot 1) - 0.75 \cdot 25.4}$$



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= 2.667 mm

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 17598 is >= tube strength 17598 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	82161.6	<= 232026.5 kPa	Ok
Tubesheet shear stress	26684.2	<= 92810.6 kPa	Ok
Stress in Shell at Tubesheet	49206.9	<= 221983.1 kPa	Ok
Stress in Channel at Tubesheet	232546.2	<= 450009.0 kPa	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	319.389	540.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	3.000	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1215 + 25.4$$

$$= 2455.840 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2526 / 2456$$

$$= 1.0286$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2526 / 2456$$

$$= 1.0286$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 50 / 524$$



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$$= 0.095 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 2.11 (177143/177143) \cdot \\ &\quad (100014/116013) \cdot (0.095); 25.4 - 2 \cdot 2.11) \\ &= 25.0529 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.19 ; 4 \cdot 2456 \cdot 31.8) / (\pi \cdot 2456^2)) } \\ &= 32.4138 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.4 - 25.1) / 32.4 \\ &= 0.2271 \end{aligned}$$

$$h/p = 16.503937 ; \mu^* = 0.227093 ; E^* = 33304 \text{ N/mm}^2$$

$$E^*/E = 0.188009 ; \nu^* = 0.422233$$

Note: As h/p (16.504) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2526 + 135) 135) } \\ &= 0.0030 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\ &= 0.003 \cdot 183094 \cdot 135^3 / (6(1 - 0.3^2)) \\ &= 250235248.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned} &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\ &= 6(2526 + 135) \cdot 250235248 / 524^3 \cdot (1 + 524 \cdot 0.003 + (524 \cdot 0.003)^2 / 2) \\ &= 106968.3 \text{ N/mm}^2 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned} &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\ &= 2526^2 / (4 \cdot 183094 \cdot 135) \cdot (1 - 2526 \cdot 0.3 / (2(2526 + 135))) \\ &= 0.55350E-04 \text{ mm/kPa} \end{aligned}$$

Parameter [w_s]:

$$\begin{aligned} &= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ &= 1.03 \cdot 250235248 \cdot 0.003 \cdot 0.55350E-04 (1 + 524 \cdot 0.003) \\ &= 111870.2656 \text{ mm}^2 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned} &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2526 + 65) 65) } \\ &= 0.0044 \text{ 1/mm} \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2)) \\ &= 0.0044 \cdot 177143 \cdot 65^3 / (6(1 - 0.3^2)) \\ &= 39467212.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta +$$



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$$\begin{aligned}
 & h \cdot \beta c \cdot (Cm\Delta + Cq\Theta) / 2 + (h \cdot \beta C)^2 / 2 \cdot Cq\Delta \\
 & = 6(2526 + 65 / \cos(0)) \cdot 39467212 / 524^3 \cdot (1 + \\
 & \quad 524 \cdot 0.0044 \cdot (1 + 1) / 2 + (524 \cdot 0.0044)^2 / 2 \cdot 1) \\
 & = 25650.1 \text{ N/mm}^2
 \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned}
 & = Dc^2 / (4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc / (2(Dc + tc))) \\
 & = 2526^2 / (4 \cdot 177143 \cdot 65) \cdot (1 - 2526 \cdot 0.3 / (2(2526 + 65))) \\
 & = 0.11829E-03 \text{ mm/kPa}
 \end{aligned}$$

Channel coefficient [zeta α]:

$$zeta\alpha = 0$$

Parameter [ωc]:

$$\begin{aligned}
 & = \rho c \cdot kc [\delta C (Cm\Theta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\
 & \quad \zeta c (Cm\Theta + h \cdot Cq\Theta \cdot h \cdot \beta C / 2)] \\
 & = 1.03 \cdot 39467212 [0.000118286 (1 + 0.0044 \cdot 1 \cdot 524) - \\
 & \quad 0.000 (1/524 + 1 \cdot 0.0044/2)] \\
 & = 70635.9609 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 & = A / Do \\
 & = 2796 / 2456 \\
 & = 1.1385
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 & = (1 - \nu^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K)) \\
 & = (1 - 0.42) / 33304 \cdot (106968 + 25650 + \\
 & \quad 177143 \cdot \ln(1.14)) \\
 & = 2.6993
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 & = Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt] \\
 & = 2456^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 14.3 - \\
 & \quad (1.03 - 1) \cdot (1.03^2 + 1) \cdot 0] \\
 & = 316488.9062 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 & = Mts + \omega c \cdot Pt - \omega s \cdot Ps \\
 & = 316489 + 70636 \cdot 0 - 111870 \cdot 14.3 \\
 & = -1281111.2500 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 & = (M^*) - Do^2 / 32 \cdot F (Ps - Pt) / (1 + F) \\
 & = (-1281111) - 2456^2 / 32 \cdot 2.7 (14.3 - 0) / (1 + 2.7) \\
 & = -2310284.2500 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 & = Mp + Do^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\
 & = -2310284 + 2456^2 / 64 \cdot (3 + 0.42) \cdot (14.3 - 0) \\
 & = 2295284.0000 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 & = \max(|Mp|; |Mo|) \\
 & = \max(|-2310284|; |2295284|) \\
 & = 2310284.2500 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:



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Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 2310284 / (0.23 \cdot (524 - 0)^2)$$

$$= 222292.9062 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 116013$$

$$= 232026.53 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 2314049 / (0.23 \cdot (513 - 0)^2)$$

$$= 232024.1094 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + Cats + Catc$$

$$= 513 + 3 + 13$$

$$= 529.0536 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(529; 451)$$

$$= 529.0536 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |14.3 - 0|$$

$$= 14.281 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 524 / 2456 \cdot \min(0.8 \cdot 116013; 0.533 \cdot 224994)$$

$$= 15842.323 \text{ kPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 92811) \cdot (4 \cdot 4.41 / 7805) \cdot |14.3 - 0| + 16$$

$$= 450.6206 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 135.000mm

Min. Shell len. adjacent to the tubesheet is: 1051.129 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2526^2 / (4 \cdot 135 (2526 + 135)) \cdot 14.3$$

$$= 63409.7852 \text{ kPa}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot$$

$$D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 56253144 / 135^2 \cdot [0.003 \cdot 0.78 + 6(1 - 0.42) / 33304 \cdot$$

$$2456 / 524^3 \cdot (1 + 524 \cdot 0.003 / 2) \cdot (-2310284 + 2456^2 / 32 \cdot$$

$$(14.3 - 0))]$$

$$= 297621.031 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$



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= 361030.812 kPa (Should be ≤ 221983.11)

Note: For a given Channel thickness of: 65.000 mm
Min. Channel len. adjacent to the tubesheet is: 729.367mm

Axial Channel Membrane Stress [σ_{c,m}]:
= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
= $2526^2 / (4 \cdot 65 (2526 + 65) \cdot \cos(0)) \cdot 0$
= 0.000 kPa

Axial Channel Bending Stress [σ_{c,b}]:
= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
= $6 \cdot 8872270 / 65^2 \cdot [(0.000118286 \cdot 1 \cdot 0.0044 - 0.000 \cdot 1) \cdot 0 - 6((1 - 0.42) / 33304) \cdot 2456 / 524^3 \cdot (1 + 0.0044 \cdot 524 / 2 \cdot 1) \cdot (-2310284 + 2456^2 / 32 \cdot (14.3 - 0))]$
= -82024.688 kPa

Channel Membrane + Bending Stress [σ_c]:
= |σ_{c,m}| + |σ_{c,b}| (Should be ≤ 1.5·S_c)
= 82024.688 kPa (Should be ≤ 174019.86)

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.
Performing a simplified Elastic-Plastic calculation
(option 3) to reduce the overstress condition.
Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:
= $E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$
= $183094 \cdot \sqrt{(1.5 \cdot 147989 / 361031)}$
= 143569.2 N/mm²

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:
= $6 \cdot M / (\mu^* (h - h'g)^2)$
= $6 \cdot 2341584 / (0.23 (524 - 0)^2)$
= 225304.6 kPa (Should be ≤ 232026.5)



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:
= h + C_{ats} + C_{atc} = 516 + 3 + 13 = 532 mm

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:
= max(H_{reqB}; H_{reqS}) = max(532 ; 451) = 532 mm

Tube Metal Area [A_t]:
= $\pi (d_o - t) t$
= $\pi (25.4 - 2.11) 2.11$
= 0.000 m²

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:
= $\pi \cdot t (d_o - t) S$
= $\pi \cdot 2.11 (25.4 - 2.11) 113991$
= 17597.941 N

Groove Weld Strength [F_g]:
= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > F_t)



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$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 17597.9414 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 113991/113991$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 16195.8193 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 17598 is >= tube strength 17598 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	225304.6 <=	232026.5 kPa	Ok
Tubesheet shear stress	76979.8 <=	92810.6 kPa	Ok
Stress in Shell at Tubesheet	361030.8 <=	478235.3 kPa	Ok
Stress in Channel at Tubesheet	82024.7 <=	174019.9 kPa	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	532.354	540.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	3.000	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 1215 + 25.4$$

$$= 2455.840 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$



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$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2526 / 2456$$

$$= 1.0286$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2526 / 2456$$

$$= 1.0286$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 50 / 524$$

$$= 0.095 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (177143/177143) \cdot (100014/116013) \cdot (0.095); 25.4 - 2 \cdot 2.11)$$

$$= 25.0529 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 31.8 / \sqrt{ 1 - 4 \cdot \min(0.19; 4 \cdot 2456 \cdot 31.8) / (\pi \cdot 2456^2) }$$

$$= 32.4138 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 25.1) / 32.4$$

$$= 0.2271$$

$$h/p = 16.503937 ; \mu^* = 0.227093 ; E^* = 33304 \text{ N/mm}^2$$

$$E^*/E = 0.188009 ; \nu^* = 0.422233$$

Note: As h/p (16.504) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2526 + 135) 135 }$$

$$= 0.0030 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 183094 \cdot 135^3 / (6(1 - 0.3^2))$$

$$= 250235248.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2526 + 135) \cdot 250235248 / 524^3 \cdot (1 + 524 \cdot 0.003 + (524 \cdot 0.003)^2 / 2)$$

$$= 106968.3 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2526^2 / (4 \cdot 183094 \cdot 135) \cdot (1 - 2526 \cdot 0.3 / (2(2526 + 135)))$$

$$= 0.55350\text{E-}04 \text{ mm/kPa}$$



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Parameter [ws]:

$$\begin{aligned}
 &= ps \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\
 &= 1.03 \cdot 250235248 \cdot 0.003 \cdot 0.55350E-04 (1 + 524 \cdot 0.003) \\
 &= 111870.2656 \text{ mm}^2
 \end{aligned}$$

Channel coefficient [\beta_c]:

$$\begin{aligned}
 &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)} \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2526 + 65)65} \\
 &= 0.0044 \text{ 1/mm}
 \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned}
 &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\
 &= 0.0044 \cdot 177143 \cdot 65^3 / (6(1 - 0.3^2)) \\
 &= 39467212.0000 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [\lambda_c]:

$$\begin{aligned}
 &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\
 &\quad h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta) \\
 &= 6(2526 + 65 / \cos(0)) \cdot 39467212 / 524^3 \cdot (1 + \\
 &\quad 524 \cdot 0.0044 \cdot (1 + 1) / 2 + (524 \cdot 0.0044)^2 / 2 \cdot 1) \\
 &= 25650.1 \text{ N/mm}^2
 \end{aligned}$$

Channel coefficient [\delta_c]:

$$\begin{aligned}
 &= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\
 &= 2526^2 / (4 \cdot 177143 \cdot 65) \cdot (1 - 2526 \cdot 0.3 / (2(2526 + 65))) \\
 &= 0.11829E-03 \text{ mm/kPa}
 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [\omega_c]:

$$\begin{aligned}
 &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \\
 &\quad \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\
 &= 1.03 \cdot 39467212 [0.000118286 (1 + 0.0044 \cdot 1 \cdot 524) - \\
 &\quad 0.000 (1/524 + 1 \cdot 0.0044/2)] \\
 &= 70635.9609 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 2796/2456 \\
 &= 1.1385
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.42) / 33304 \cdot (106968 + 25650 + \\
 &\quad 177143 \cdot \ln(1.14)) \\
 &= 2.6993
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\
 &= 2456^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 14.3 - \\
 &\quad (1.03 - 1) \cdot (1.03^2 + 1) \cdot 4.95] \\
 &= 206781.6250 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]

$$\begin{aligned}
 &= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s \\
 &= 206782 + 70636 \cdot 4.95 - 111870 \cdot 14.3 \\
 &= -1041150.1875 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$



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ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M_p) - Do^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((-1041150) - 2456^2/32 \cdot 2.7 (14.3 - 4.95)) / (1 + 2.7)$$

$$= -1564628.8750 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + Do^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= -1564628 + 2456^2/64 \cdot (3 + 0.42) \cdot (14.3 - 4.95)$$

$$= 1444471.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| -1564628 | ; | 1444471 |)$$

$$= 1564628.8750 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 1564628 / (0.23 (524 - 0)^2)$$

$$= 150546.7969 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 116013$$

$$= 232026.53 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 1581656 / (0.23 (422 - 0)^2)$$

$$= 231910.7969 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 422 + 3 + 13$$

$$= 438.0833 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB} ; H_{reqS})$$

$$= \max(438 ; 300)$$

$$= 438.0833 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 14.3 - 4.95 |$$

$$= 9.331 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 524 / 2456 \cdot \min(0.8 \cdot 116013 ; 0.533 \cdot 224994)$$

$$= 15842.323 \text{ kPa}$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 92811) \cdot (4 \cdot 4.41 / 7805) \cdot | 14.3 - 4.95 | + 16$$

$$= 299.9644 \text{ mm}$$



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ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 135.000mm

Min. Shell len. adjacent to the tubesheet is: 1051.129 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2526^2 / (4 \cdot 135 (2526 + 135)) \cdot 14.3$$

$$= 63409.7852 \text{ kPa}$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta_s \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta_s / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 56253144 / 135^2 \cdot [0.003 \cdot 0.78 + 6(1 - 0.42) / 33304 \cdot$$

$$2456 / 524^3 \cdot (1 + 524 \cdot 0.003 / 2) \cdot (-1564628 + 2456^2 / 32 \cdot$$

$$(14.3 - 4.95))]$$

$$= 248414.156 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s, m} | + | \sigma_{s, b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 311823.969 \text{ kPa (Should be } \leq 221983.11)$$

Note: For a given Channel thickness of: 65.000 mm

Min. Channel len. adjacent to the tubesheet is: 729.367mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2526^2 / (4 \cdot 65 (2526 + 65) \cdot \cos(0)) \cdot 4.95$$

$$= 46884.695 \text{ kPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 8872270 / 65^2 \cdot [(0.000118286 \cdot 1 \cdot 0.0044 - 0.000 \cdot 1) \cdot 4.95 -$$

$$6((1 - 0.42) / 33304) \cdot 2456 / 524^3 \cdot (1 +$$

$$0.0044 \cdot 524 / 2 \cdot 1) \cdot (-1564628 + 2456^2 / 32 \cdot (14.3 - 4.95))]$$

$$= 103636.781 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c, m} | + | \sigma_{c, b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 150521.484 \text{ kPa (Should be } \leq 174019.86)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$$

$$= 183094 \cdot \sqrt{(1.5 \cdot 147989 / 311824)}$$

$$= 154482.3 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 1548134 / (0.23 (524 - 0)^2)$$

$$= 148959.7 \text{ kPa (Should be } \leq 232026.5)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:



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$$= h + \text{Cats} + \text{Catc} = 422 + 3 + 13 = 438 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS}) = \max(438 ; 300) = 438 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 113991$$

$$= 17597.941 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 17597.9414 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 113991 / 113991$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 113991$$

$$= 16195.8193 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 17598 is $\geq$ tube strength 17598 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	148959.7	$\leq$ 232026.5 kPa	Ok
Tubesheet shear stress	50295.6	$\leq$ 92810.6 kPa	Ok
Stress in Shell at Tubesheet	311824.0	$\leq$ 478235.3 kPa	Ok
Stress in Channel at Tubesheet	150521.5	$\leq$ 174019.9 kPa	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	438.083	540.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	3.000	Ok



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Tube-Tubesheet Groove Weld Leg :	1.333	3.000	Ok
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U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	312.091	...	79174	232027	25894	92811	Ps+Pt D1	Ok
D2uc	523.506	...	216468	232027	74699	92811	Ps+Pt D2	Ok
D3uc	431.474	...	146180	232027	48805	92811	Ps+Pt D3	Ok
D1c	319.389	...	82162	232027	26684	92811	Ps+Pt-c D1	Ok
D2c	532.354	...	225305	232027	76980	92811	Ps+Pt-c D2	Ok
D3c	438.083	...	148960	232027	50296	92811	Ps+Pt-c D3	Ok
Max:	532.3539	... mm		0.971		0.829	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
	Shell	Channel					
D1uc	...	...	44347	221983	219716	450009	Ok
D2uc	...	...	343172	478235	73292	174020	Ok
D3uc	...	...	298825	478235	146423	174020	Ok
D1c	...	...	49207	221983	232546	450009	Ok
D2c	...	...	361031	478235	82025	174020	Ok
D3c	...	...	311824	478235	150521	174020	Ok
Max	...	...		0.755		0.865	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	532.354	540.000	Ok
Tube thickness :	1.643	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	3.000	Ok
Tube-tubesheet groove weld leg :	1.333	3.000	Ok

Given Shell Thickness: 138.0000 mm

Given Channel Thickness: 68.0000 mm



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Min Shell length of thk, (138.000) adj. to tubesheet: 1061.481 mm

Min Channel length of thk, (68.000) adj. to tubesheet: 745.122 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	14.030	1.000	14.681	0.999
Tubesheet Shear Stress	17.218	1.000	17.218	1.000
Shell Stress (Axial, Junction)	17.218	0.771	17.218	1.179
Tubesheet-Channel Junction	9.578	1.000	17.218	1.179
Tube Pressure Stress	20.288	1.000	18.338	1.000
Minimum MAWP	9.578		14.681	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18.368	1.000	19.154	1.000
Tubesheet Shear Stress	22.634	1.000	22.634	1.000
Shell Stress (Axial, Junction)	22.634	0.884	22.634	1.189
Tubesheet-Channel Junction	13.886	1.000	22.634	1.189
Tube Pressure Stress	21.000	1.000	20.431	1.000
Minimum MAPnc	13.886		19.154	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(540/4, \min(540, 138)) \\
 &= 138.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(14.3/19.2, 1) \\
 &= 0.746
 \end{aligned}$$

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(540/4, \min(540, 68)) \\
 &= 135.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\
 &= \min(4.95/13.9, 1)
 \end{aligned}$$



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= 0.356

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

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Problems/Failures Summary:

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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